



**Board's
Activity Sheet
of July 2025
Included**

**Solution on
QR Code**





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**An excellent book for the
Preparation of the Board's
Examination**

**SCIENCE & TECHNOLOGY
(PART – 2)
STANDARD X**

This book contains ...

1. Format of the Activity Sheet at the beginning of the book
2. Study Tips for Students
3. **Section 1** : Board's July 2024 Activity Sheet with full Solution
4. **Section 2** : Question Sets for Question-wise study and Revision :
 - (a) Guidance on how to attempt the question/activity
 - (b) Solved questions
5. **Section 3** : Assignment – Question for Practice
6. **Section 4** : Board's July 2025 Activity Sheet for Practice
(with Answers on QR Code)

With 21 MLQ... Success awaits you!

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CTP
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Activity Sheet Paper Pattern

Time : 2 Hours]

[Max. Marks : 40

Important points and instructions :

- (1) All questions are compulsory.
- (2) Draw scientifically correct, labelled diagrams wherever necessary.
- (3) Start each main question on a new page.
- (4) The numbers to the right of the questions indicate full marks.
- (5) For each MCQ [Q. 1 (A)], evaluation will be done for the first attempt only.
- (6) For each MCQ, write only the alphabet corresponding to the correct answer, e.g. (i) (A); (ii) (C) ...

		Marks	Marks with option
Q.1(A)	5 Multiple Choice questions carrying 1 mark each	5	5
Q.1(B)	5 Questions carrying 1 mark each (No fill in the blanks questions) Each question will be of different type.	5	5
Q.2(A)	3 Scientific reason questions carrying 2 marks each (Attempt any 2) (Minimum 1 question from Biology and 1 from EVS)	4	6
Q.2(B)	5 Questions carrying 2 marks each (Attempt any 3) (Minimum 2 from Biology and 2 from EVS)	6	10
Q.3	8 Questions carrying 3 marks each (Attempt any 5) (4 from Biology and 4 from EVS)	15	24
Q.4	2 Questions carrying 5 marks (Attempt any 1) (1 from Biology and 1 from EVS)	5	10
Total		40	60

Study Tips

Dear Students,

The SSC examination is an important milestone in your life. If you take appropriate steps in revising Science and Technology (Part 2), you can secure maximum marks.

For this, what you need is 'Smart Study'. Read all the ten chapters of Science and Technology (Part – 2) carefully with full concentration. In Navneet Digest, various types of important questions and their model answers are given. You must have used the Digest throughout the year. Now prepare for the final examination using '**Navneet 21 Most Likely Question Sets**'.

Let's understand the format of activity sheet.

- In **Q. 1**, there will be two sub-questions (A) and (B). They are objective questions, carrying a total of **10 marks**. In **Q. 1 (A)**, there will be 5 multiple choice questions based on textual matter. **Q. 1 (B)** will bear all 5 different objective question types. It includes Find the odd one out, Find correlation, Match the column, True or false, Name the following, Give the examples, Identify figure etc. The question type will not be repeated. Daily read one question set from **Q. Set 1 to Q. Set 6** and revise these objective questions. If you regularly revise these questions, surely you will secure 10 out of 10 marks in the final examination.
- In **Q. 2 (A)**, three 'give scientific reasons' type of question carrying **2 marks each** will be asked. **Q. Set 7** consists of all important 'give reasons' questions. You are expected to write four correct and valid points to obtain 2 marks, by attempting any 2 of these.
- In **Q. 2 (B)**, five questions will be asked carrying **2 marks each**. Students have to attempt any three of these. You must write any four correct points in the answer. Likely questions which can be asked under **Q. 2 (B)** are included in **Q. Set 8 to Q. Set 11** and also in **Q. Set 13, 14, 17 and 18**.
- In **Q. 3**, there will be 8 sub-questions carrying **3 marks each**. Students have to attempt any five out of these eight sub-questions.

Q. Set 9 to Q. Set 17 and Q. Set 19 consists of such questions. The difficulty level of these questions can be little higher. Each answer should ideally have 6 points in order to fetch 3 marks for each question. The diagrams are necessary in the answers.

- In **Q. 4**, there will be two questions carrying **5 marks** each. Students are expected to solve any one out of these. Prepare well for such questions. In **Q. Set 13, 14, 16 and 19**, such questions are included.

In **Q. 4**, any sub-question of 5 marks can be divided and asked for **2 and 3 marks** as well. That is why it will be easier to solve Q. 4 if you also practice the question types from Q. Set which is given for 2 and 3 marks.

When you take into account the complete pattern of the Activity Sheet, you can also understand the importance of regular and proper revision schedule. This will certainly help you to procure maximum marks in the final examination. Further, if you solve the practice activity sheet in stipulated time, that will help you to practise time management during examination.

Guidelines for Time Management

Question	Sub-question	Time to complete the answers
Read the entire activity sheet carefully for first 5 minutes		
Q. 1 (A) and (B)	10 objective questions	20 minutes
Q. 2 (A) and (B)	5 questions of 2 marks each (5 minutes each)	25 minutes
Q. 3	5 questions of 3 marks each (8 minutes each)	40 minutes
Q. 4	1 question of 5 marks	15 minutes
Use the last 15 minutes to check the answer paper.		

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Time : 2.00 Hours]**[Total Marks : 40****Note :** (i) All questions are compulsory.

(ii) Use of calculator is not allowed.

(iii) The numbers to the right of the questions indicate full marks.

(iv) In case of MCQs [Q. No. 1 (A)], only the first attempt will be evaluated and will be given credit.

(v) For each MCQ, the correct alternative (A), (B), (C) or (D) with sub-question number is to be written as an answer.

For e.g. : (i) (A), (ii) (B), (iii) (C).

(vi) Scientifically correct, labelled diagrams should be drawn wherever necessary.

9. 1 A Choose correct option and write its number for the following questions : **5**

Note : This question contains five multiple choice questions which will be based on textual matter. Each one will carry 1 mark. Students are expected to write only alternative number and the respective answer.

(i) Evolution means

(A) Internal change (B) Gradual change

(C) External change (D) Quick change

(ii) One of the rare species is

(A) Lesser florican (B) Red panda

(C) Tiger (D) Shekhru

(iii) The machine which converts the kinetic energy of wind to electrical energy is called

(A) Solar cell (B) Thermal

(C) Generator (D) Wind turbine

(iv) Chemically, vinegar is

- (A) Acetic acid (B) Amino acid
(C) Hydrochloric acid (D) Gluconic acid

(v) We get kcal energy per gram of proteins.

- (A) 4 (B) 3 (C) 2 (D) 1

ANSWERS

- (i) (B) (ii) (B) (iii) (D) (iv) (A) (v) (A).

(1 mark each for the correct answer.)

9. 1 B Solve the following questions :

5

Note :

☛ This question will contain five different types of objective questions of 1 mark each.

☛ Different types of questions will be as follows :

- (1) Find the odd one out
- (2) Correlation
- (3) Match the columns/Make pairs
- (4) True or False
- (5) Give the names/flow charts to be completed

(i) Find odd man out : Starfish, Snail, Frog, Amoeba

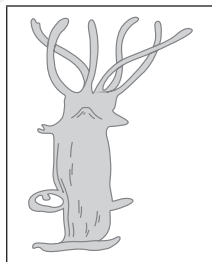
(ii) Match the correct pair :

Column 'A'	Column 'B'
Biotic components	(a) Soil (b) Plant (c) Sunlight

(iii) Write the correlation :

White revolution : Dairy : : Blue revolution :

(iv) Observe the diagram and identify the phylum :



(v) State, whether given statement is right or wrong :

Peer group influence is stronger in case of adolescents.

ANSWERS

(i) Amoeba (All others are multicellular animals.)

OR

Frog (All others are invertebrates.) *(Two answers possible)*

(ii) Biotic component : Plant

(iii) Fishery/Aquaculture

(iv) Coelenterata/Cnidaria

(v) True/Right **(1 mark each for the correct answer.)**

9. 2 A Give scientific reasons : (Any Two)

4

Note : Three scientific reasons will be asked in this question. Students will solve any two of these. Each bears 2 marks. Minimum one question from Biology and one question from EVS will be asked. However, all the answers are given here for guidance to students.

(i) Fibres are one of the important nutrients.

(ii) Though tortoise lives on land as well as in water, it cannot be included in class–Amphibia.

(iii) Hydroelectric energy, solar energy and wind energy are called renewable energies.

ANSWERS

(i) (1) Fibres are indigestible substance.

(2) They are thrown out along with other useless and undigested matter.

(3) This aids in egestion. Some fibres also help in digestion of other substances.

(4) Green leafy vegetables, fruits, cereals, etc. are considered as important in diet as they supply nutritious fibres.

Thus, fibres are considered as one of the important nutrients.

(4 correct points, ½ mark each.)

- (ii) (1) When tortoise lives on the land, it respire with the help of lungs.
- (2) When in water, it puts out its nares (nasal openings) out of the water and breathes air.
- (3) It cannot take up oxygen dissolved in water. In both the habitats, it respire with the help of lungs. In case of true amphibians, this is not the case.
- (4) They can breathe in water with the help of skin and on land with the help of lungs. Tortoise also has exoskeleton which is lacking in Amphibia.

Therefore, tortoise cannot be included in class–Amphibia.

(4 correct points, $\frac{1}{2}$ mark each.)

- (iii) (1) Hydroelectric energy, solar energy and wind energy is obtained respectively from flowing water, solar radiations and flowing wind.
- (2) These sources, i.e. water reservoirs, sun and the wind are inexhaustible and sustainable. They will not be finished.
- (3) On the contrary, the conventional energy sources such as coal and fossil fuels have limited reserves.
- (4) They cannot be renewed and may get exhausted in future. Hydroelectric energy, solar energy and wind energy can be replenished and hence they are called renewable.

(4 correct points, $\frac{1}{2}$ mark each.)

Q. 2 B Answer the following questions : (Any Three)

6

Note :

- This question will contain five sub-questions of 2 marks each. You have to attempt any 3 questions.
- Minimum 2 questions from Biology and 2 from EVS will be asked.
- Different types of questions are as follows :
 - (1) Laws/definitions/principles. (2) Short notes.
 - (3) Complete the flowchart. (4) Give difference between.
 - (5) Write properties/characteristics/uses/advantages/effects.
 - (6) Give examples based on concept, a process and component.
- For students' guidance all the answers are given.

- (i) Write down four benefits of Biotechnology.
- (ii) Distinguish between Sexual reproduction and Asexual reproduction.
- (iii) Observe the following diagram and complete the table :



'A'

Disaster

Precaution

- (1)
 (2)



'B'

Snake Bite

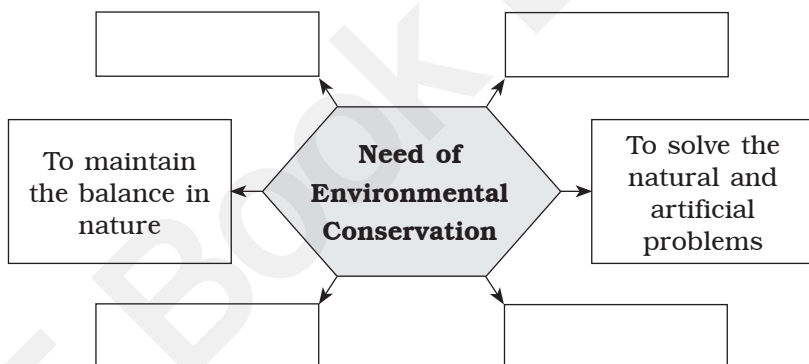


Primary Stage

- (1)
 (2)

(iv) Write down any two features of Industrial Microbiology.

(v) Complete the following flow chart :



- (i) (1) Animal husbandry is now using biotechnological techniques for artificial insemination and embryo transfer by which the breeds of cattle are improved.
- (2) To improve and to manage the human health, diagnosis and treatment of diseases have to be focused. Diagnosis of diabetes, heart diseases and infectious diseases such as AIDS and dengue can be done rapidly due to biotechnology.
- (3) The treatment and prevention of diseases need hormones, interferons, antibiotics and different vaccines which are now manufactured through biotechnology. Gene therapy is also used to treat hereditary disorders.
- (4) Industrial products and clean technology to combat environmental pollution uses biotechnology practices.
- (5) DNA fingerprinting has revolutionised the profession of forensic science.

(Any 4 correct benefits, $\frac{1}{2}$ mark each.)

(ii)	Sexual reproduction	Asexual reproduction
	1. Reproduction that occurs due to fertilisation of gametes is called sexual reproduction. 2. Male and female parent are necessary for sexual reproduction. 3. This reproduction occurs with the help of both mitosis and meiosis. 4. New individual formed by this method is genetically different from parents.	1. Reproduction that occurs with the help of somatic cells is called asexual reproduction. 2. For asexual reproduction, only one parent is necessary. 3. This reproduction occurs with the help of mitosis only. 4. New individual formed by this method is genetically identical with parents.

5. Sexual reproduction occurs in two steps : First formation of haploid gametes by meiosis and then fertilization of these haploid gametes to form diploid zygote. There are no subtypes in the sexual reproduction.

5. Asexual reproduction occurs in different individuals by various methods like binary fission, multiple fission, budding, fragmentation, regeneration, vegetative propagation, spore production, etc.

(4 correct points of differentiation, $\frac{1}{2}$ mark each.)

(iii)

'A'

Disaster



Earthquake :
Natural disaster



Precaution



- (1) Immediately come out of the house and stand in open ground.
- (2) Take care to avoid head injury.

'B'

Snake Bite



Primary Stage



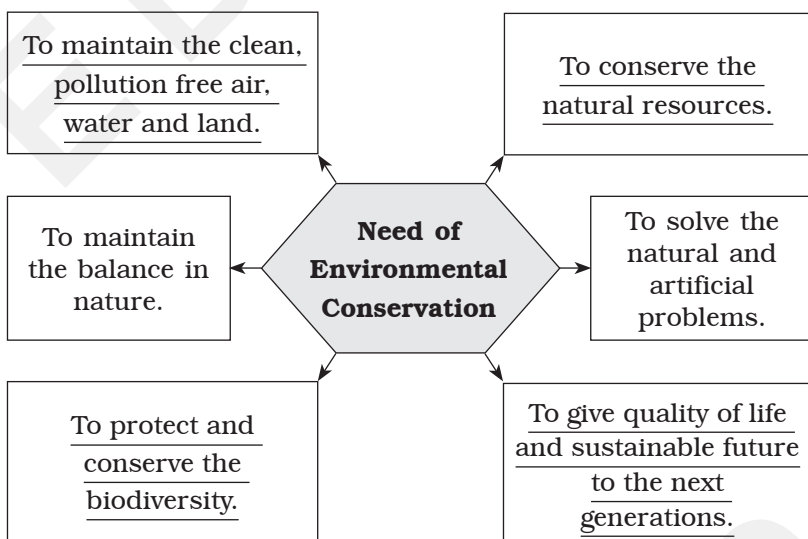
- (1) Small incision should be made near bite wound so that blood can flow out.
- (2) Injection of antivenin should be given.

(Correct disaster : 1 mark, 1 correct precaution : $\frac{1}{2}$ mark, 1 correct treatment : $\frac{1}{2}$ mark.)

(iv) (1) Products like bread, cheese, wine, etc. can be made with the help of fermentation process. Raw materials for chemicals, enzymes, nutrients and medicines can be manufactured.

(2) Microbes can be used for waste management and pollution control. **(2 correct features, 1 mark each.)**

(v) (Answers are underlined.)



(4 correct points, $\frac{1}{2}$ mark each.)

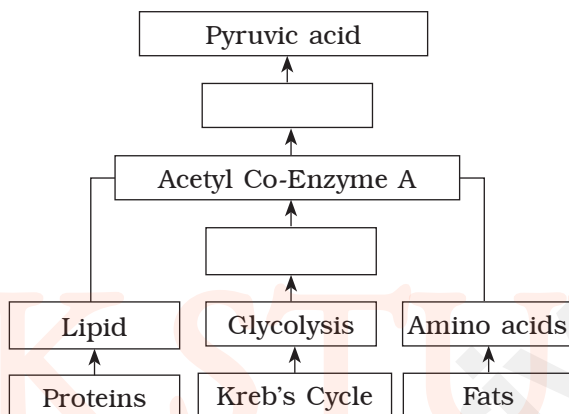
9. ③ Solve the following questions : (Any Five)

15

Note :

- ☛ This question will contain 8 sub-questions of 3 marks each. You have to attempt any 5 questions.
- ☛ 4 questions from Biology and 4 from EVS will be asked.
- ☛ Different types of questions are as follows :
 - (1) Give explanation using given statements.
 - (2) Suggest remedies/measures.
 - (3) Explanation of diagrams.
 - (4) Complete the label/chart.
 - (5) Explain with the help of examples.
 - (6) Complete the diagram.
 - (7) Answer the questions based on figures.
 - (8) Write the answers with explanations.
 - (9) Write laws, theories and explain.
 - (10) Complete the paragraph by filling the gaps.
- ☛ For guidance of students all answers are given here.

- (i) Write any three characteristic features of class 'Aves'.
- (ii) (a) What are vestigial organs?
 (b) Write two names of some vestigial organs in human body.
 (c) Write the names of those animals in whom same organs are used? Give two examples.
- (iii) How energy is formed from oxidation of carbohydrates, fats and proteins? Correct the diagram given below :



- (iv) What is the meaning of the following symbols? Write your role accordingly :



A

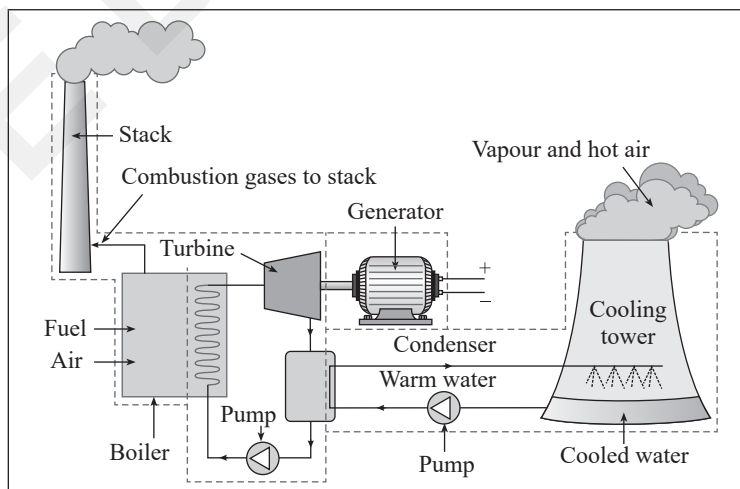


B



C

(v) Observe the diagram and answer the following questions :



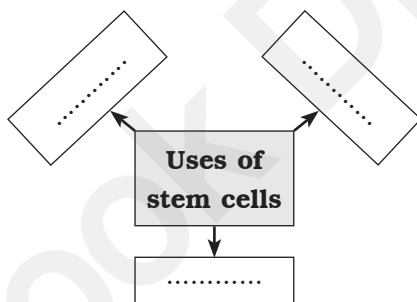
(a) Which type of energy is created in the above diagram?

(b) Which type of fuel is used in power plant?

(c) State one problem created in such type of power plant.

(vi) Explain the importance of Biopesticides in organic farming.

(vii) Complete the given conceptual picture and write its uses in treatment of various diseases :



(viii) Which are three objectives of disaster management?

ANSWERS

(i) (1) Aves are totally adapted for the aerial mode of life. Body is spindle shaped. Body is divisible into head, neck and trunk.

(2) There are two pairs of limbs. The forelimbs are modified to form wings for flight.

- (3) Digits have scales and claws. Jaws are modified into a beak.
- (4) The exoskeleton is in the form of feathers.

(Any 3 characteristics : 1 mark each.)

- (ii)** (a) Vestigial organs are degenerated or underdeveloped organs of organisms which do not perform any function.

(Correct answer, 1 mark.)

- (b) Names of vestigial organs in human body – Appendix, tail-bone or coccyx, wisdom teeth and body hair.

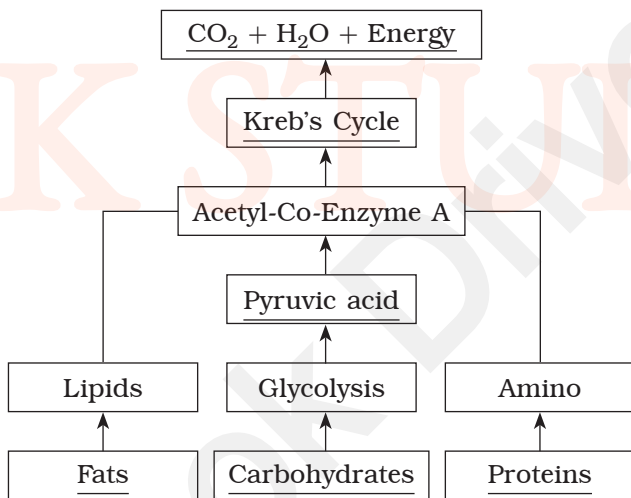
(Any two vestigial organs, 1 mark each.)

- (c) (1) Appendix is vestigial for humans, it does not perform any function but in ruminant animals it is concerned with digestion.
- (2) Ear muscles are vestigial for us but in monkeys and cattle they are functional.

(2 correct examples, $\frac{1}{2}$ mark each.)

- (iii)** (1) First of all the dietary carbohydrates are digested in the digestive system with the help of various enzymes and converted into glucose. Similarly, proteins are converted into amino acids and fats are broken down into fatty acid and glycerol (alcohol).
- (2) Oxidation of carbohydrates takes place during cellular respiration. Glucose is oxidised by three steps during aerobic respiration, viz. glycolysis, tricarboxylic acid cycle or Krebs's cycle and electron transfer chain.
- (3) From one molecule of glucose, two molecules of each pyruvic acid, ATP, NADH_2 and water are formed during glycolysis. Pyruvic acid which is formed in this process is converted into Acetyl-Coenzyme-A along with release of two molecules each of NADH_2 and CO_2 .
- (4) In the next step, i.e. in TCA cycle, molecules of Acetyl-Co-A enter the mitochondria and a cyclic chain of reactions take place. Acetyl part of Acetyl-Co-A is completely oxidized through this cyclical process. The molecules CO_2 , H_2O , NADH_2 , FADH_2 are released in this process.

- (5) In third step, i.e. in ETC reaction, NADH_2 and FADH_2 formed during first two steps are used for obtaining ATP molecules. 3 molecules of ATP are obtained from each NADH_2 molecule and 2 molecules of ATP from each FADH_2 .
- (6) Thus, one molecule of glucose upon complete oxidation in presence of oxygen yields 38 molecules of ATP. This is how from carbohydrates, energy is obtained.
- (7) If carbohydrates are insufficient in diet, then proteins or lipids are used for energy production. Fatty acids derived from fats and amino acids derived from proteins are converted into Acetyl-Co-A. Acetyl-Co-A once again can yield energy through TCA cycle.



**(Correct process of energy formation : 2 marks,
corrected diagram : 1 mark.)**

- (iv) (A) This symbol tells us to keep our wastes carefully. The garbage should not be strewn anywhere. But it should be properly managed. Waste if managed properly can be a wealth.
- (B) This symbol tells us to save electricity. If electricity is carefully used, we can save our natural resources. This message is given through this picture.

- (C) Use of bicycle means use of green energy. By riding on a bicycle we save on fuel and use our own muscular energy. It is the best ecofriendly, non-polluting vehicle.

(3 correct meaning of symbol and role, 1 mark each.)

- (v) (a) Thermal power is created in the diagram.

(Correct answer : 1 mark.)

- (b) Coal is used in the power plant.

(Correct answer : 1 mark.)

- (c) Toxic gases such as carbon dioxide, sulphur oxide and nitrogen oxide are emitted which causes air pollution.

(Correct answer : 1 mark.)

- (vi) (1) By using biopesticides, soil pollution is minimised. Otherwise by using chemical pesticides and fertilizers, there is large scale soil pollution.

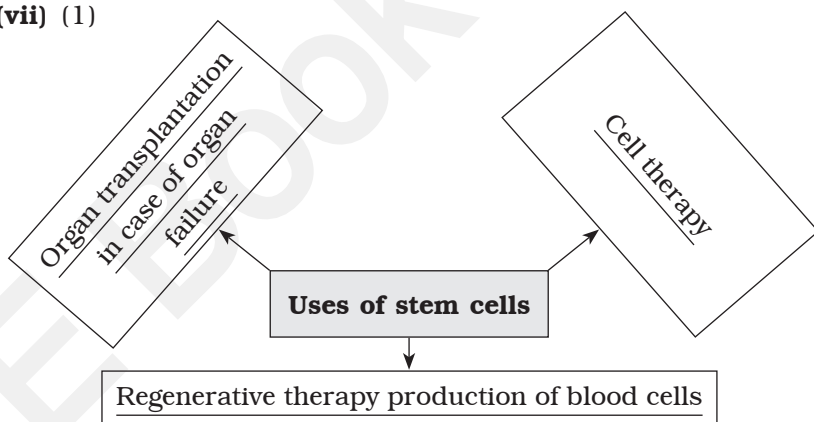
- (2) When chemical pesticides are used in agriculture, there is contamination of soil by fluoroacetamide-like chemicals.

- (3) These are harmful to other plants, animals as well as for human beings. They may cause skin diseases in humans.

- (4) By using bacterial and fungal toxins, the pests and pathogens can be destroyed. Such toxins are directly incorporated in the plant materials.

(Any 3 correct points of importance, 1 mark each.)

- (vii) (1)



(3 correct uses : 1/2 mark each.)

- (2) Regenerative therapy is used to replace dead cells in conditions like diabetes, myocardial infarction, Alzheimer's disease, Parkinson's disease, etc. Blood cells are also produced for conditions of anaemia, leukemia, thalassemia, etc.

(Correct uses in treatment of diseases : 1½ marks.)

(viii) Objectives of disaster management :

- (1) To save human life from disasters. To help them for moving away from the place of disasters by rapid action.
- (2) To supply essential commodities to the affected people. This helps to reduce the gravity of disaster. People are given grains, water and clothes and other basic necessities under this objective.
- (3) To bring back the conditions of affected people to normalcy.
- (4) To rehabilitate the affected and displaced victims.
- (5) To think and execute the protective measures in order to develop capability to face the disasters in future.

(3 correct objectives : 1 mark each.)

9. 4 Solve the following questions : (Any One)

5

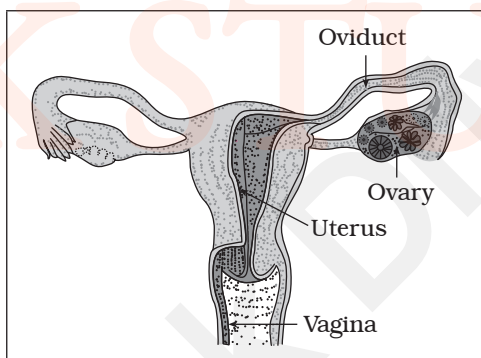
Note :

- ☛ This question will contain two sub-questions of 5 marks each. You have to attempt any one question.
- ☛ One question from Biology and one from EVS will be asked.
- ☛ This question will contain following types of questions :
 - (1) Draw a figure and give explanation.
 - (2) Correct an incorrect diagram.
 - (3) Classify with detailed explanation.
 - (4) Read the given paragraph and answer the questions.
 - (5) Complete the given incomplete table/chart and give explanation.
 - (6) Answer the question in detail.
 - (7) Make a concept diagram based on any component or technology.

- (i) (a) Draw well labelled diagram of female reproductive system.
 (b) Which type of hormones are secreted in female reproductive system?
 (c) When does menopause period stop?
 (d) How many ovaries are there in female foetus at the time of birth?
- (ii) Give two examples of each :
- (a) Hobbies to reduce stress.
 (b) Diseases endangering the social health.
 (c) Physical problems arising due to excessive use of mobile phones.
 (d) Activities under the jurisdiction of cyber crime laws.
 (e) Write problems due to alcoholism.

ANSWERS

(i) (a)



Human female reproductive system

(Correct diagram : 1 mark, any two correct labels : $\frac{1}{2}$ mark each.)

- (b) Estrogen and progesterone are secreted by female reproductive system.

(2 correct hormones : $\frac{1}{2}$ mark each.)

- (c) Menopause takes place between 45 to 50 years of age in females.

(Correct answer : 1 mark.)

- (d) 2 – 4 million immature oocytes are present in ovaries of female foetus at the time of birth.

(Correct answer : 1 mark.)

(ii) (a) Hobbies to reduce stress : (1) To listen to music (2) Bird watching and nature trails (3) Reading good books.

(Any 2 hobbies : $\frac{1}{2}$ mark each.)

(b) Diseases endangering the social health : (1) AIDS (2) Tuberculosis (3) Leprosy.

(Any 2 diseases : $\frac{1}{2}$ mark each.)

(c) Physical problems arising due to excessive use of mobile phones : (1) Headache (2) Vision problems (3) Joint pains.

(Any 2 problems : $\frac{1}{2}$ mark each.)

(d) Activities under the jurisdiction of Cyber Crime Laws : (1) To do bank transactions by procuring PIN number of somebody. (2) Misuse of written material of someone or illegal sale of the same. (3) Hacking the information of government institutes and companies.

(Any 1 activity : 1 mark.)

(e) Write problems due to alcoholism : (1) Vital organs like liver and kidney are affected. (2) Nervous system and brain is affected, causing memory loss and lack of concentration.

(Any 1 problem : 1 mark.)



Question
Set
1MULTIPLE CHOICE QUESTIONS
BASED ON TEXTBOOK

[Q. 1 (A) : 1 Mark each]

Remember :

- ☛ In question 1 (A), 5 sub-questions of Multiple Choice bearing 1 mark each will be asked.
- ☛ These sub-questions are based on textual matter.
- ☛ While answering these questions, you are expected to write only the alphabet of correct choice.
- ☛ The alphabet that you write in the initial attempt will be considered for marking. Thus, be sure when you enter the answer.

Q. Write the correct alphabet of the answer :**CHAPTER 1 : HEREDITY AND EVOLUTION**

- (1) is pioneer of the modern genetics.
(A) Charles Darwin (B) Gregor Johann Mendel
(C) Hargobind Khurana (D) Walter Sutton
- (2) Any nucleotide of the gene suddenly changes its position that causes a minor change which is nothing but the
(A) changes (B) mutation
(C) disability (D) translocation **(March '25)**
- (3) Walter Sutton observed the paired chromosomes in the cells of
(A) monkey (B) cockroach
(C) grasshopper (D) housefly

- (4) Genes are carried via
 (A) cells (B) chromosomes
 (C) mitochondria (D) Golgi body
- (5) The proof for the fact that protein synthesis occurs through was given by George Beadle and Edward Tatum.
 (A) genes (B) cells (C) tissues (D) amino acid
- (6) Transfer of information from molecule of DNA to mRNA is called process.
 (A) translocation (B) translation
 (C) transcription (D) mutation **(March '20)**
- (7) A minor change occurs due to change in position of any nucleotide is termed as
 (A) transcription (B) translocation
 (C) mutation (D) translation **(March '24)**
- (8) is a connecting link between Annelida and Arthropoda.
 (A) Duck-billed platypus (B) Peripatus
 (C) Lungfish (D) Whale **(March '19)**
- (9) Similarities in the initial stages of living beings indicate the evidence.
 (A) connecting links (B) anatomical
 (C) embryological (D) palaentological
- (10) About 10 thousand years ago, started to practise agriculture.
 (A) Gorilla (B) wise man
 (C) Ramapithecus (D) Australopithecus
- (11) can be considered as the first example of wise man.
 (A) Australopithecus (B) Ramapithecus
 (C) Cro Magnon (D) Neanderthal man



(12) Genetic disorder like sickle cell anemia may be caused due to

(A) mutation

(B) translation

(C) translocation

(D) transcription

(July '23)

Ans. (1) (B), (2) (B), (3) (C), (4) (B), (5) (A), (6) (C), (7) (C), (8) (B), (9) (C), (10) (B), (11) (D), (12) (A).

CHAPTER 2 : LIFE PROCESSES IN LIVING ORGANISMS : PART – 1

(1) We get of energy from one gram of lipid.

(A) 9 cal

(B) 4 kcal

(C) 4 cal

(D) 9 kcal

(July '23)

(2) A molecule of glucose is completely oxidised in aerobic respiration and molecules of H_2O and are produced along with energy.

(A) CO_2

(B) O_2

(C) NaOH

(D) HNO_3

(March '25)

(3) Glycolysis and are two steps of anaerobic respiration.

(A) Krebs cycle

(B) Electron transfer chain reaction

(C) Fermentation

(D) Reduction

(July '22)

(4) Bones contain amino acids.

(A) Melanin

(B) Haemoglobin

(C) Ossein

(D) Insulin

(March '22)

(5) After complete oxidation of a glucose molecule, number of ATP molecules are formed.

(A) 14

(B) 28

(C) 29

(D) 38

(6) Which of the following vitamins is necessary for synthesis of $NADH_2$?

(A) Vitamin B_1

(B) Vitamin B_3

(C) Vitamin C

(D) Vitamin K

(7) Somatic and stem cells undergo division.

(A) meiosis

(B) mitosis

(C) budding

(D) cloning

(March '19)

(8) In mitotic division, nuclear membrane completely disappears in phase.

(A) telophase

(B) prophase

(C) metaphase

(D) anaphase

(Nov. '20)

(9) All chromosomes are arranged parallel to equatorial plane of cell in phase of mitosis.

(A) prophase

(B) metaphase

(C) anaphase

(D) telophase

(Sept. '21)

(10) The spindle fibres start appearing from stage of karyokinesis.

(A) prophase

(B) metaphase

(C) anaphase

(D) telophase

(March '20)

Ans. (1) (D), (2) (A), (3) (C), (4) (C), (5) (D), (6) (B),

(7) (B), (8) (C), (9) (B), (10) (B).

CHAPTER 3 : LIFE PROCESSES IN LIVING ORGANISMS : PART - 2

(1) Human males have numbers of chromosomes.

(A) $44 + XX$

(B) $44 + XY$

(C) $22 + XY$

(D) $22 + XX$

(July '23)

(2) Body breaks up into several fragments and each fragment begins to live as a new individual. This is type of reproduction.

(A) regeneration

(B) fragmentation

(C) binary fission

(D) budding

(March '20)

(3) Regeneration occurs in

(A) amoeba

(B) paramoecium

(C) euglena

(D) planaria

(March '24)

(4) Type of sexual reproduction is

(A) fragmentation

(B) regeneration

(C) gamete formation

(D) budding

(March '23)

(5) Pollen grains are formed by division in locules of anthers.

(A) meiosis

(B) mitosis

(C) amitosis

(D) binary



(6) In humans, sperm production occurs in the organ

- (A) *vas deferens* (B) *ejaculator duct*
(C) *testes* (D) *urinogenital duct*

(Nov. '20; March '25)

(7) The length of a sperm is about micrometres.

- (A) 400 (B) 5 (C) 60 (D) 600 (July '19)

(8) Pregnant mother supplies nourishment to her foetus through

- (A) *breasts* (B) *uterus*
(C) *placenta* (D) *ovaries*

(9) Progesterone is secreted from

- (A) *corpus albicans* (B) *corpus luteum*
(C) *testis* (D) *uterus*

(10) is modern remedial reproductive technique used if there is a problem in implantation of embryo in the uterus.

- (A) *surrogacy* (B) *sperm bank*
(C) *in vitro fertilization* (D) *adoption*

(11) Pranav and Pritee are twins in your class. They belong to twins type.

- (A) *monozygotic* (B) *dizygotic*
(C) *Siamese* (D) *none of the above*

(March '19)

Ans. (1) (B), (2) (B), (3) (D), (4) (C), (5) (A), (6) (C), (7) (C),

(8) (C), (9) (B), (10) (A), (11) (C).

CHAPTER 4 : ENVIRONMENTAL MANAGEMENT

(1) In a food chain, autotrophic plants are present at the level.

- (A) *tertiary nutrition* (B) *secondary nutrition*
(C) *producer* (D) *apex* (March '19)

(2) The apex consumer is

- (A) *eagle* (B) *frog*
(C) *snake* (D) *grasshopper* (March '23)

(3) Basic functional unit to study the ecology is termed as

- (A) *environment* (B) *niche*
(C) *ecosystem* (D) *food chain*

- (4) Genetic changes occur due to pollution.
 (A) radioactive (B) air
 (C) water (D) soil **(Nov. '20)**
- (5) is world's largest organisation engaged in environmental activities.
 (A) Green peace (B) Hariyali
 (C) B.N.H.S. (D) I.I.T.
- (6) Which sanctuary of Assam is under threat due to dams and indiscriminate use of water?
 (A) Tadoba (B) Kaziranga
 (C) Manas (D) Sundarbans **(Nov. '20)**
- (7) from Manas sanctuary in Assam is under threat.
 (A) One-horned rhino (B) Lion
 (C) Musk deer (D) Giant squirrel/Shekhru
- (8) World Biodiversity Day is celebrated on every year.
 (A) 22nd April (B) 5th June
 (C) 16th September (D) 22nd May
- (9) Giant squirrel is an example of threatened species.
 (A) endangered (B) rare
 (C) vulnerable (D) indeterminate **(March '22)**

Ans. (1) (C), (2) (A), (3) (C), (4) (A), (5) (A), (6) (C), (7) (A), (8) (D), (9) (D).

CHAPTER 5 : TOWARDS GREEN ENERGY

- (1) Solar photovoltaic cell converts the solar radiation energy directly into energy.
 (A) chemical (B) solar
 (C) mechanical (D) electrical **(March '25)**
- (2) In thermal power plants, the energy in the coal is converted into electrical energy through several steps.
 (A) solar (B) nuclear
 (C) thermal (D) chemical **(Sept. '21)**
- (3) is used in thermal power plant.
 (A) Coal (B) Uranium
 (C) Thermal (D) Water **(July '22)**



- (4) Burning of coal may cause serious health problems related to system.
 (A) digestive (B) respiratory
 (C) nervous (D) excretory
- (5) Atomic power plant in Maharashtra is
 (A) Chandrapur (B) Koyana
 (C) Tarapur (D) Anjanwel **(March '24)**
- (6) Incomplete combustion of fuels leads to formation of
 (A) carbon monoxide (B) carbon dioxide
 (C) chlorofluorocarbon (D) hydrogen sulphide **(March '20)**
- (7) Water reservoir possesses energy.
 (A) nuclear (B) thermal
 (C) kinetic (D) potential **(Nov. '20)**
- (8) The hydroelectric capacity of Koyna dam is
 (A) 2400 MW (B) 1960 MW
 (C) 1900 MW (D) 1961 MW **(March '23)**
- (9) Solar cells are made of a special type of material called semiconductor such as
 (A) silicon (B) uranium
 (C) borosilicate (D) hydrogen

Ans. (1) (D), (2) (D), (3) (A), (4) (B), (5) (C), (6) (A), (7) (D),
 (8) (B), (9) (A).

CHAPTER 6 : ANIMAL CLASSIFICATION

- (1) Artificial method of animal classification was proposed by
 (A) Aristotle (B) Darwin
 (C) Lamarck (D) Whittaker
- (2) Animal with bilateral symmetry is
 (A) Star fish (B) human
 (C) Paramoecium (D) Amoeba **(March '23)**
- (3) Body cavity between the body and internal organs is called
 (A) gastrocoel (B) enteron (C) coelom (D) cave

- (4) Planet Earth has highest number of animals from phylum.
 (A) *Mollusca* (B) *Porifera*
 (C) *Arthropoda* (D) *Platyhelminthes*
- (5) Animals attached to substratum are called animals.
 (A) *sessile* (B) *sedentary*
 (C) *lame* (D) *motionless*
- (6) Which of the following is not in phylum Cnidaria?
 (A) *Hydra* (B) *Sea anemone*
 (C) *Star fish* (D) *Physalia*
- (7) Example of phylum Arthropoda is
 (A) *Scorpion* (B) *Star fish*
 (C) *Earthworm* (D) *Hydra* **(March '24)**
- (8) Due to which similar characteristics honey bee and cockroach are included in the same phylum?
 (A) *Wings* (B) *Three pair of legs*
 (C) *Jointed appendages* (D) *Antenna*
- (9) is second largest phylum in animal kingdom.
 (A) *Mollusca* (B) *Arthropoda*
 (C) *Porifera* (D) *Platyhelminthes*
- (10) Calcareous spines are present on the body of animal.
 (A) *fish* (B) *snail* (C) *sponge* (D) *star fish* **(July '19)**
- (11) is a cold blooded animal.
 (A) *Bat* (B) *Snake* (C) *Rabbit* (D) *Elephant* **(March '19)**
- (12) Human being belongs to class.
 (A) *Mammalia* (B) *Amphibia*
 (C) *Reptilia* (D) *Cyclostomata* **(March '22)**
- Ans.** (1) (A), (2) (B), (3) (C), (4) (C), (5) (B), (6) (D), (7) (A), (8) (C), (9) (A), (10) (D), (11) (B), (12) (A).

CHAPTER 7 : INTRODUCTION TO MICROBIOLOGY

- (1) The bacteria are useful for cleaning the hydrocarbons and oil pollutants from soil and water.
 (A) *Lactobacilli* (B) *Acidobacillus*
 (C) *Pseudomonas* (D) *Streptococcus* **(July '23)**



- (2) Due to fermentation of carbohydrates by yeast, sugar is converted into carbon dioxide and
 (A) methanol (B) ethanol
 (C) butanol (D) none of these
- (3) is produced by fermentation of the mixture of flour of wheat or rice and soybean with the help of the fungus *Aspergillus oryzae*.
 (A) Soya sauce (B) Curds
 (C) Cheese (D) Shrikand
- (4) *Loctobacillus brevis* gives us the beverage
 (A) Cocoa (B) Coffee (C) Wine (D) Cider
- (5) beverage is obtained by fermentation of apple juice.
 (A) Cider (B) Wine (C) Coffee (D) Cocoa **(July '19)**
- (6) Venegar contains acid.
 (A) acetic (B) lactic
 (C) tartaric (D) hydrochloric **(March '25)**
- (7) gas is considered to be the fuel of future.
 (A) Hydrogen (B) Nitrogen (C) Methane (D) Butane
- (8) Bacteria used to clear the oil spills are called bacteria.
 (A) phenol oxidising (B) electrolytic
 (C) hydrocarbonoclastic (D) decomposing

Ans. (1) (C), (2) (B), (3) (A), (4) (B), (5) (A), (6) (A), (7) (A), (8) (C).

CHAPTER 8 : CELL BIOLOGY AND BIOTECHNOLOGY

- (1) Protein like interleukin produced by biotechnology is used against disease.
 (A) anaemia (B) cancer
 (C) diabetes (D) haemophilia **(July '23)**
- (2) is the revolutionary event in biotechnology after cloning.
 (A) Human genome project (B) DNA discovery
 (C) Stem cell research (D) All the above
- (3) The property of stem cells is called
 (A) diversity (B) equality
 (C) differentiation (D) pluripotency

- (4) Organs like can be donated after death.
 (A) skin (B) heart (C) lungs (D) bones **(March '22)**
- (5) Genetically modified crop has capacity to destroy bollworms.
 (A) BT brinjal (B) Golden rice
 (C) BT cotton (D) Maize **(Sept. '21)**
- (6) Due to use of biofertilizers, solubilisation abilities of the plants are improved.
 (A) sulphates (B) oxides
 (C) nitrates (D) phosphates
- (7) The disease related with the synthesis of insulin is
 (A) cancer (B) diabetes
 (C) viral infection (D) dwarfness **(Sept. '21)**
- (8) Transgenic raw potatoes generate the immunity against disease.
 (A) plague (B) cholera (C) leprosy (D) TB

Ans. (1) (B), (2) (C), (3) (D), (4) (B), (5) (C), (6) (D), (7) (B), (8) (B).

CHAPTER 9 : SOCIAL HEALTH

- (1) IT Act 2000 is to control the
 (A) house breaking (B) cybercrimes
 (C) cheating (D) pickpocketing
- (2) Laughter club is a remedy to drive away
 (A) stress (B) addictions (C) lethargy (D) epidemics
- (3) Alcohol consumption mainly affects system.
 (A) Respiratory (B) Nervous
 (C) Digestive (D) Circulatory **(July '22)**
- (4) Persons continuously using computers and the internet become
 (A) courageous (B) timid (C) solitary (D) criminal
- (5) has been newly launched in Police Department.
 (A) Cybercrime unit (B) Women protection unit
 (C) Senior citizen care unit (D) Forensic unit



(6) helps to improve concentration in the studies.

(A) Meditation

(B) Music

(C) Dancing

(D) Laughing

(Sept. '21)

Ans. (1) (B), (2) (A), (3) (B), (4) (C), (5) (A), (6) (A).

CHAPTER 10 : DISASTER MANAGEMENT

(1) After the subsidence of any type of disaster, rehabilitation work is started in phase.

(A) later

(B) transitional

(C) terminal

(D) ultimate

(2) Which of the following is man-made disaster?

(A) Earthquake

(B) Flood

(C) Meteor

(D) Leakage of toxic gases

(March '19; July '19)

(3) Volcano is a disaster.

(A) social

(B) political

(C) biological

(D) geological

(March '22)

(4) There had been a huge in the village Malin, Tal. Ambegaon in 2014.

(A) earthquake

(B) storm

(C) landslide

(D) change

(5) The atomic energy plant of Chernobyl was used only for generating

(A) electricity

(B) solar power

(C) atomic energy

(D) wind energy

Ans. (1) (B), (2) (D), (3) (D), (4) (C), (5) (A).

**Question
Set
2**

FIND THE ODD ONE OUT

[Q. 1 (B) : 1 Mark each]

Remember :

- ☛ 'Find the odd one out' type questions are given in this question set.
- ☛ Four to five terms will be given. Take into consideration the relationship among them and find out the one which does not fit into this group. You are expected to write the justification about why this term is odd among the group.
- ☛ Write this explanation in one sentence only.

Q. Find the odd one out :

CHAPTER 1 : HEREDITY AND EVOLUTION

- (1) Venation, Shape of seeds, Leaf petiole, Leaf shape.
- (2) Transcription, Translation, Mutation, Translocation.
- (3) Duck-billed Platypus, Pomfret, Lung fish, Peripatus.

(March '24)

- (4) Lung fish, Pomfret, Cat fish, Bombay duck.
- (5) Egyptopithecus, Ramapithecus, Dryopithecus, Cro-Magnon.

Ans. (1) Shape of seeds. (All others are morphological evidences in plants.)

(2) **Mutation.** (All others are stages of protein synthesis.)

(3) **Pomfret.** (All others are connecting links.)

(4) **Lung fish.** (All others are fishes which are not connecting links.)

(5) **Cro-Magnon.** (All others are ape like stages of human evolution.)

CHAPTER 2 : LIFE PROCESSES IN LIVING ORGANISMS : PART - 1

- (1) Alcohol, Vinegar, Pyruvic acid, Lactic acid.
- (2) Actin, Ossein, Myosin, Aldosterone. **(July '22)**
- (3) Progesterone, Estrogen, Melanin, Testosterone. **(Sept. '21)**
- (4) Vitamin A, Vitamin B, Vitamin D, Vitamin K.



- Ans.** (1) **Pyruvic acid.** (All others are chemical substances formed by the process of fermentation.)
(2) **Aldosterone.** (All others are proteins concerned with movement.)
(3) **Melanin.** (All others are hormones related to reproductive system.)
(4) **Vitamin B.** (All others are fat soluble vitamins.)

CHAPTER 3 : LIFE PROCESSES IN LIVING ORGANISMS : PART – 2

- (1) Fragmentation, Regeneration, Budding, Fertilization.

(March '25)

- (2) Circulation, Excretion, Sensation, Reproduction.
(3) Carrot, Potato, Radish, Sweet Potato.
(4) Stigma, Style, Pollen, Ovary.

- Ans.** (1) **Fertilization.** (All others are processes of asexual reproduction in multicellular organisms.)
(2) **Reproduction.** (All others are the processes responsible for the survival of the individual.)
(3) **Potato.** (All others are edible roots.)
(4) **Pollen.** (All others are parts of gynoecium.)

CHAPTER 4 : ENVIRONMENTAL MANAGEMENT

- (1) Lion tailed monkey, White rats, Musk deer, Tiger.
(2) Conservation, Regulation, Pollution, Prohibition.
(3) Manas sanctuary, Sunderbans sanctuary, The Western Ghats, Tadoba National Park.

- Ans.** (1) **White rats.** (All others are threatened species.)
(2) **Pollution.** (All others are ways of environmental protection.)
(3) **Tadoba National Park.** (All others are endangered heritage places of India.)

CHAPTER 5 : TOWARDS GREEN ENERGY

- (1) Kudankulam, Tarapur, Ravatabhata, Anjanvel.
(2) Atomic energy, Hydroelectric energy, Wind energy, Solar energy.

- Ans.** (1) **Anjanvel.** (All others are places having nuclear power plants.)
(2) **Atomic energy.** (All others are green energies.)

CHAPTER 6 : ANIMAL CLASSIFICATION

- (1) Starfish, Sea-urchin, Nereis, Sea-cucumber.
- (2) Planaria, Liverfluke, Filarial worm, Tape worm.
- (3) Tube feet, Setae, Parapodia, Suckers.
- (4) Frog, Tortoise, Toad, Salamander.

Ans. (1) **Nereis.** (Nereis belongs to Annelida, all the remaining are Echinoderm animals.)

(2) **Filarial worm.** (Filarial worm belongs to Aschelminthes, remaining are platyhelminthes.)

(3) **Tube feet.** (Tube feet are locomotory organs of Echinoderms, the remaining are locomotory organs of Annelids.)

(4) **Tortoise.** (Tortoise is a reptile, the remaining are amphibians.)

CHAPTER 7 : INTRODUCTION TO MICROBIOLOGY

(1) β -carotene, lycopenes, xylitol, lutein. (**July '23**)

(2) Actinomycetes, Streptomyces, Nocardia, Yeast.

Ans. (1) **Xylitol.** (All others are edible colours.)

(2) **Yeast.** (All others have ability of decomposing rubber from garbage.)

CHAPTER 8 : CELL BIOLOGY AND BIOTECHNOLOGY

(1) DDT, malathion, chloropyriphos, Humus.

(2) Diabetes, Anaemia, Leukaemia, Thalassemia. (**July '19**)

(3) Drying, Salting, Cooking, Soaking with sugar. (**March '20**)

Ans. (1) **Humus.** (All others are chemical insecticides.)

(2) **Diabetes.** (All others are diseases concerned with blood cells.)

(3) **Cooking.** (All others are food preservation methods.)

CHAPTER 9 : SOCIAL HEALTH

(1) Tobacco, Laughter club, Alcoholism, Drug abuse.

(2) Aadhaar card, PAN card, Greeting card, Credit card.

Ans. (1) **Laughter club.** (All others are addictions.)

(2) **Greeting card.** (All others are important cards of personal use.)



- (1) Earthquake, Flooding, Tsunami, War. (**March '22**)
- (2) Drought, Bomb explosion, Forest fire, Floods.
- (3) Preparation, Redemption, Preparedness, Resurgence.
- (4) Forest fire, Snow storms, Drought, Meteorite.

Ans. (1) **War.** (All others are natural disasters.)

(2) **Bomb explosion.** (All others are natural disasters.)

(3) **Resurgence.** (All others are pre-disaster management techniques.)

(4) **Forest fire.** (All others are atmospheric disasters.)

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**Question
Set
3**

FIND OUT THE CORRELATION

[Q. 1 (B) : 1 Mark each]

Remember :

- ☛ 'Find the correlation' type questions are given in this question set.
- ☛ Taking into consideration the relation between first two terms or group of words, you have to write appropriate fourth term or group of words that bears similar correlation with the third term.

Q. Complete the correct correlation :

CHAPTER 1 : HEREDITY AND EVOLUTION

- (1) Survival of fittest : Darwin : : Acquired characters :
(2) Father of Evolution : Darwin : : Father of Genetic :
(3) m-RNA : Transcription : : t-RNA :
(4) Wisdom teeth : Vestigial organs : : Lung fish :

(Sept. '21)

- (5) Peripatus : Connecting link : : Appendix :

Ans. (1) Lamarck (2) Mendel (3) Translation (4) Connecting link
(5) Vestigial organ.

CHAPTER 2 : LIFE PROCESSES IN LIVING ORGANISMS : PART - 1

- (1) Skin : Melanin : : Pancreas : **(March '24)**
(2) Breakdown of glucose molecule : Glycolysis : : Formation of glucose from proteins :
(3) Condensation of chromosomes : Prophase : : Formation of spindle fibres :
(4) Division of nucleus : Karyokinesis : :
Division of cytoplasm :

Ans. (1) Insulin or trypsin (2) Gluconeogenesis (3) Metaphase
(4) Cytokinesis.

CHAPTER 3 : LIFE PROCESSES IN LIVING ORGANISMS : PART – 2

- (1) Ovary : Estrogen : : Testes : (July '23)
(2) Transverse binary fission : Paramecium : : Longitudinal binary fission : (July '19)
(3) Spirogyra : Fragmentations : : Planaria : (Nov. '20)
(4) Amoeba : Fission : : Hydra :
(5) FSH : Development of oocyte : : LH :
(6) Female hormone : Estrogen : : Male hormone :

Ans. (1) Testosterone (2) Euglena (3) Regeneration (4) Budding
(5) Ovulation (6) Testosterone.

CHAPTER 4 : ENVIRONMENTAL MANAGEMENT

- (1) Western Ghat : Asiatic lion : : Sundarbans sanctuary : (March '22)
(2) Manas : One-horned rhino : : Gir :
(3) Red panda : Rare species : : Giant Squirrel :

Ans. (1) Tiger (2) Asiatic Lion (3) Indeterminate species.

CHAPTER 5 : TOWARDS GREEN ENERGY

- (1) Steam engine : Thermal energy : : Solar panels :
(2) Uranium : Nuclear energy : : Coal :
(3) Electrical energy generated through solar cells : DC : :
Electrical energy required for domestic appliances :

Ans. (1) Solar energy (2) Thermal energy (3) AC.

CHAPTER 6 : ANIMAL CLASSIFICATION

- (1) Acoelomate : Porifera : : Pseudocoelomate :
(2) Annelida : Earthworm : : Platyhelminthes : (March '20)
(3) Proboscis : Hemichordata : : Suctorial mouth :
(4) Horse : Mammal : : Seahorse :
(5) Dog fish : Cartilaginous : : Cat fish :

Ans. (1) Aschelminthes (2) Planaria (3) Cyclostomata (4) Pisces
(5) Bony.

CHAPTER 7 : INTRODUCTION TO MICROBIOLOGY

- (1) Oil slick : *Alcanovorax* : : Rubber from garbage :
- (2) Coffee : *Coffea arabica* : : Cocoa :
- (3) Bread : Baker's yeast : : Soya sauce :

Ans. (1) *Actinomyces* (2) *Theobroma cacao* (3) *Aspergillus oryzae*.

CHAPTER 8 : CELL BIOLOGY AND BIOTECHNOLOGY

- (1) Diabetes : Insulin : : Cancer : (March '25)
- (2) Insulin : Diabetes : : Interferon : (March '23)
- (3) : Dwarfness : : Factor VIII : Haemophilia.
- (4) White revolution : Increase in dairy production : : Green revolution : (March '19)

Ans. (1) Interleukin (2) Viral infection (3) Somatostatin
(4) Increase in agricultural production or crop yield.

CHAPTER 9 : SOCIAL HEALTH

- (1) Addictive substances : Drugs : : Carcinogenic (Cancer causing) substances :
- (2) Radiations from cell phones : Headache : : : Hindrance to the brain development.
- (3) Movement against tobacco : Tata trust : : Education of slum children :

Ans. (1) Tobacco (2) Alcoholism (3) Salaam Mumbai Foundation.

CHAPTER 10 : DISASTER MANAGEMENT

- (1) Pre-disaster management : Preparation and warning : : Post-disaster management :
- (2) Floods : Geophysical disaster : : Covid-19 pandemic :
- (3) Bringing normalcy to : Public life : Recovery phase : : Restoring and building house :
- (4) Toxic gas leakage : Accidental disaster : : War :

Ans. (1) Resurgence and restoration (2) Biological disaster
(3) Reconstruction phase (4) Intentional type of disaster.



**Question
Set
4**

MATCH THE COLUMNS

[Q. 1 (B) : 1 Mark each for correct pair]

Remember :

- ☛ Match the pairs/match the columns will have one word or more words in the first column and two or three words in the second column.
- ☛ Since there would be more words in the second column, a wise selection of the terms is needed here.

Q. Match the columns :

CHAPTER 1 : HEREDITY AND EVOLUTION

(1) Scientist	Discovery
Hugo de Vries	(a) Pioneer of the modern genetics (b) Theory of natural selection (c) Mutational theory

Ans. Hugo de Vries – Mutational theory.

(2) Theory	Scientist
Survival of fittest	(a) Lamarck (b) Walter Sutton (c) Charles Darwin

Ans. Survival of fittest – Charles Darwin.

CHAPTER 2 : LIFE PROCESSES IN LIVING ORGANISMS : PART – 1

(1) Protein	Part of the body (July '19)
(1) Haemoglobin (2) Ossein	(a) skin (b) bones (c) blood (d) muscles

Ans. (1) Haemoglobin – blood (2) Ossein – bones.

(2) Protein	Part of the body
Keratin	(a) Muscles (b) Skin (c) Bones

Ans. Keratin – Skin.

CHAPTER 3 : LIFE PROCESSES IN LIVING ORGANISMS : PART – 2

(1) Example	Types of Reproduction
Mucor	(a) Fragmentation (b) Spore formation (c) Binary fission

Ans. Mucor – Spore formation.

(2) Column 'A'	Column 'B' (March '24)
Male	(a) 44 + XX (b) 44 + XY (c) 44 + YY

Ans. Male – 44 + XY.

CHAPTER 4 : ENVIRONMENTAL MANAGEMENT

(1) Column 'A'	Column 'B' (March '23)
Forest Conservation Act	(a) 1986 (b) 1980 (c) 1970

Ans. Forest Conservation Act – 1980.

(2) Species	Examples
Rare Species	(a) Red panda, Musk deer (b) Tiger, Lion (c) Lion-tailed monkey, Lesser florican

Ans. Rare Species – Red panda, Musk deer.



CHAPTER 5 : TOWARDS GREEN ENERGY

(1) Redraw the table taking into account the relation between entries in the three columns. (July '22)

Column I	Column II	Column III
Coal	Kinetic energy	Nuclear Power plant
Uranium	Thermal energy	Wind electricity plant
Wind	Nuclear energy	Thermal plant

Ans.

Column I	Column II	Column III
Coal	Thermal energy	Thermal plant
Uranium	Nuclear energy	Nuclear Power plant
Wind	Kinetic energy	Wind electricity plant

(2) Column I	Column II
Eco-friendly energy	(a) Thermal energy (b) Wind energy (c) Nuclear energy

Ans. Eco-friendly energy – Wind energy.

CHAPTER 6 : ANIMAL CLASSIFICATION

(1) Phylum	Characteristics
Cnidaria	(a) Tunic (b) Collar cells (c) Tentacles with Cnidoblasts

Ans. Cnidaria – Tentacles with Cnidoblasts.

(2) Phylum	Characteristics
Hemichordata	(a) Collar cells (b) Mantle (c) Proboscis

Ans. Hemichordata – Proboscis.



CHAPTER 7 : INTRODUCTION TO MICROBIOLOGY

(1)	Column 'A'	Column 'B'
Protease		(a) Molasses (b) Production of cheese (c) Ice creams and puddings

Ans. Protease – Production of cheese.

(2)	Column 'A'	Column 'B'
Lycopene		(a) Colouring (b) Sweetening (c) Giving Acidity

Ans. Lycopene – Colouring.

CHAPTER 8 : CELL BIOLOGY AND BIOTECHNOLOGY

(1)	Organism	Substance that is absorbed
Factor VIII		(a) Haemophilia (b) Viral infection (c) Cancer

Ans. Factor VIII – Haemophilia.

(2)	Organism	Substance that is absorbed
Pseudomonas		(a) Radiations (b) Hydrocarbons (c) Arsenic

Ans. Pseudomonas – Hydrocarbons.

CHAPTER 9 : SOCIAL HEALTH

(1)	Column 'A'	Column 'B'
Cyber crime		(a) Bank frauds (b) Drug peddling (c) Burglary by breaking into house

Ans. Cyber crime – Bank frauds.

(2) Column 'A'	Column 'B'
Laughter club	(a) Stress Management (b) Self-centred attitude (c) Alcoholism

Ans. Laughter club – Stress Management.

CHAPTER 10 : DISASTER MANAGEMENT

(1) Column A : Disaster	Column B : Types of Disaster
Snow fall	(a) Geological (b) Atmospheric (c) Animal origin

Ans. Snow fall – Atmospheric.

(2) Column A : Example	Column B : Disaster
Bomb explosion	(a) Geological (b) Atmospheric (c) Intentional

Ans. Bomb explosion – Intentional.

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**Question
Set
5**

TRUE/FALSE
[Q. 1 (B) : 1 Mark each]

Remember :

- ☛ The given statement may be true or false. Identify and write only the answer.
- ☛ Correcting the statement and rewriting it is not expected.

Q. State whether True/False :

CHAPTER 1 : HEREDITY AND EVOLUTION

- (1) Changes acquired during life time are transferred to next generation.
- (2) It seems that invertebrates have been slowly originated from vertebrates.
- (3) The decaying process of C-12 occurs continuously from the dead remains of living organisms.
- (4) Mesozoic era was dominated by variety of mammals.
- (5) Dr. Har Govind Khorana was awarded Nobel prize for his invention and publication in the journal Radio Carbon.
- (6) Study of fossils is an important aspect of the study of evolution. **(March '25)**

Ans. (1) False (2) False (3) False (4) False (5) False (6) True.

CHAPTER 2 : LIFE PROCESSES IN LIVING ORGANISMS : PART - 1

- (1) In aerobic respiration, glucose is oxidised in three steps.
- (2) Oxidation of proteins is carried out in aerobic respiration.
(March '22)
- (3) Molecules of pyruvic acid formed in this glycolysis are converted into molecules of Acetyl-coenzyme A.
- (4) For formation of plasma membrane, phospholipid molecules are necessary.
- (5) Proteins of animal origin are called 'first class' proteins.

(6) The disease related with the deficient synthesis of insulin is heart disease.

Ans. (1) True (2) False (3) True (4) True (5) True (6) False.

CHAPTER 3 : LIFE PROCESSES IN LIVING ORGANISMS : PART – 2

- (1) If oocyte is fertilized, secretion of estrogen and progesterone stops completely.
- (2) During unfavourable conditions, binary fission is performed by amoeba.
- (3) The pollen grains from anthers fall on stigma of carpel.
- (4) There is glucose sugar in semen.
- (5) There is need of rest along with special personal hygiene during phase of menstruation.

Ans. (1) False (2) False (3) True (4) False (5) True.

CHAPTER 4 : ENVIRONMENTAL MANAGEMENT

- (1) Environmental balance is not maintained due to various food chains of ecosystem.
- (2) Flow of nutrients in an ecosystem is unidirectional.
- (3) Environmental pollution is unnecessary and never acceptable change in the surrounding environment.
- (4) There are clusters of thick forests only in Western Ghats of India.
- (5) 34 highly sensitive biodiversity spots are reported all over the world.

Ans. (1) False (2) False (3) True (4) False (5) True.

CHAPTER 5 : TOWARDS GREEN ENERGY

- (1) Turbine is used to rotate the magnet in a generator.
- (2) In thermal power plants, the turbines work on solar energy.
(Sept. '21)
- (3) In nuclear power plants, the chain reactions cannot be controlled.
- (4) The potential difference available from a solar cell is independent of its area.
- (5) The power available from the solar cell is AC. **(July '22)**

Ans. (1) True (2) False (3) False (4) True (5) False.

CHAPTER 6 : ANIMAL CLASSIFICATION

- (1) Planet Earth has highest number of animals from phylum Mollusca.
- (2) Animals belonging to phylum Echinodermata perform locomotion with the help of tube-feet.
- (3) Herdmania has notochord in only tail region and hence it is called Urochordate.

Ans. (1) False (2) True (3) True.

CHAPTER 7 : INTRODUCTION TO MICROBIOLOGY

- (1) Rennet obtained from alimentary canal of human beings is used in production of cheese.
- (2) Due to taking probiotics in diet, useful bacteria in the intestine are destroyed.
- (3) Soya sauce is produced with the help of fungus *Aspergillus oryzae*.

Ans. (1) False (2) False (3) True.

CHAPTER 8 : CELL BIOLOGY AND BIOTECHNOLOGY

- (1) Manufacturing alcohol by using transgenic yeast on sugarcane molasses is one of the examples of white revolution.
- (2) Malaria arises due to genetic changes in hepatocytes.
- (3) Various essential elements like N, P, K can be available to crops due to earthworms and fungi.
- (4) DNA fingerprinting is mainly useful in forensic sciences.

(Nov. '20)

- (5) Production of various useful aquatic organisms with the help of water is called blue revolution. **(March '24)**

Ans. (1) False (2) False (3) True (4) True (5) True.

CHAPTER 9 : SOCIAL HEALTH

- (1) Communication media like mobile phone, television, internet should be used often. **(July '23)**
- (2) Tobacco containing substances cannot cause cancer of mouth and lungs. **(March '20)**



(3) Children who watch cartoon films may imitate the characters of those films.

(4) To take selfie on the road is the invitation for accident.

(March '23)

Ans. (1) False (2) False (3) True (4) True.

CHAPTER 10 : DISASTER MANAGEMENT

(1) Toxic gas leakage is an unknown man-made disaster.

(2) One must keep the accident victim in same condition till medical help arrives.

(3) Mock drill is done for virtual situation of disaster.

Ans. (1) True (2) False (3) True.

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**Question
Set
6**

**NAME THE FOLLOWING/
GIVE THE EXAMPLES/FUNCTIONS**

[Q. 1 (B) : 1 Mark each and

Q. 2 (B) : 2 Marks each]

Remember :

- ☛ If a question based on 'Giving suitable examples or writing the names' is asked under Q. 1 (B), then write any two examples.
- ☛ If a similar question is asked under Q. 2 (B) for 2 marks, write any four suitable examples.
- ☛ For such type of questions descriptive answer is not expected.

Q. Name the following :

CHAPTER 1 : HEREDITY AND EVOLUTION

(1) Genetic disorder caused due to mutation.

Ans. Sickle cell anaemia.

(2) Being a fish, I perform respiration with the help of lungs.

Who am I? (March '25)

Ans. Lungfish.

(3) Father of modern genetics.

Ans. Johann Gregor Mendel.

(4) Vestigial organs in human beings.

Ans. Appendix, tail-bone or coccyx, wisdom teeth and body hair.

(5) I am connecting link between Reptilia and mammals.

What is my name? (March '20)

Ans. Duck-billed Platypus.

(6) Examples of connecting links.

Ans. (1) Peripatus (2) Duck-billed Platypus (3) Lungfish.

(7) Scientific name of human being. (July '22)

Ans. *Homo sapiens*.

CHAPTER 2 : LIFE PROCESSES IN LIVING ORGANISMS : PART - 1

(1) Steps of anaerobic respiration.

Ans. Glycolysis and fermentation.

(2) Two co-enzymes involved in cellular respiration.

Ans. NAD → Nicotinamide Adenine Dinucleotide and FAD → Flavin Adenine Dinucleotide.

(3) Scientist who discovered the TCA cycle.

Ans. Sir Hans Krebs.

(4) Most abundantly found protein in nature.

Ans. RUBISCO.

(5) Molecules forming plasma membrane.

OR

Which molecules are necessary for the formation of plasma membrane? (Nov. '20)

Ans. Phospholipids.

CHAPTER 3 : LIFE PROCESSES IN LIVING ORGANISMS : PART - 2

(1) Different glands associated with male reproductive system.

Ans. Seminal vesicles, Prostate gland, Cowper's or bulbourethral glands.

(2) Hormones secreted by ovary of female reproductive system.

Ans. Estrogen and progesteron.

(3) Types of twins.

Ans. Monozygotic twins, Siamese twins and Dizygotic twins.

(4) Methods of (things used for) family planning.

Ans. Copper T, condoms, oral contraceptive pills.

CHAPTER 4 : ENVIRONMENTAL MANAGEMENT

(1) Colour of the pages in Red List of Endangered Species.

Ans. Pink and Green.

(2) Endangered Heritage Places of the country.

Ans. Manas sanctuary, Western ghat, Sunderbans sanctuary.

(3) Factors responsible for creating environmental pollution.

Ans. Industrialisation, Deforestation, Indiscriminate use of natural resources, Unplanned urbanisation.

(4) Levels of biodiversity.

Ans. Genetic diversity, Species diversity, Ecosystem diversity.



CHAPTER 5 : TOWARDS GREEN ENERGY

(1) Which type of fuel is used in thermal power plant?

(March '22)

Ans. Coal.

(2) Major thermal power plants in India.

Ans. Vindhyannagar, Mundra, Tamnar, Chandrapur.

(3) Important nuclear power plants in India.

Ans. Kudankulam, Tarapur, Rawatbhata, Kaiga.

(4) Natural gas-based power plants in India.

Ans. Samaralkota, Anjanvel, Bavanna, Kondapalli.

(5) Major hydroelectric plants in India.

Ans. Tehari, Koyana, Srishailam, Nathpa zakri.

(6) Sources of green energy or environment-friendly energy.

Ans. Wind energy, solar energy, Hydroelectric energy.

CHAPTER 6 : ANIMAL CLASSIFICATION

(1) I have chitinous exoskeleton, I have four pairs of walking appendages. I can sting you. Who am I?

Ans. Scorpion.

(2) Though I am multicellular, there are no tissues in my body. What is the name of my phylum?

Ans. Name of my phylum is Porifera.

(3) I am diploblastic and acoelomate. Which phylum do I belong to?

Ans. I am from phylum Cnidaria or Coelenterata.

(4) I live in your small intestine. Pseudocoelom is present in my thread-like body. In which phylum will you include me?

Ans. I am Ascaris. I am included in Aschelminthes.

(5) I am metamerically segmented, blood sucking, ectoparasite. I have suckers. Who am I? (July '19)

Ans. I am Leech. Classified as Annelidan.

(6) My body is radially symmetrical. Water vascular system is present in my body. I am referred as fish though I am not. What is my name?

Ans. I am Starfish. I am from Echinodermata phylum.



(7) To which class does Bat belong? (March '23)

Ans. Mammals/Mammalia.

CHAPTER 7 : INTRODUCTION TO MICROBIOLOGY

(1) Emulsifiers.

Ans. Polysaccharides and glycolipids.

(2) Microbe used in baking industry.

Ans. *Saccharomyces cerevisiae*.

(3) Molecular formula for 4% Acetic acid (Vinegar). (Nov. '20)

Ans. CH_3COOH .

(4) Microbial enzymes.

Ans. Oxidoreductases, Transferases, Hydrolases, Lyases, Isomerases, Ligases.

(5) Effective antibiotic against tuberculosis.

Ans. Rifamycin.

CHAPTER 8 : CELL BIOLOGY AND BIOTECHNOLOGY

(1) Sources of stem cells.

Ans. (1) Umbilical cord (2) Embryonic cells (3) Red bone marrow
(4) Adipose connective tissue and blood of adult human being.

(2) Organs that can be used in transplantation.

Ans. Heart, Kidney, Liver, Eyes, Skin.

(3) Genetically modified crops.

Ans. BT Cotton, BT Brinjal, Golden rice.

(4) Two main methods used in animal husbandry.

Ans. (1) Artificial insemination (2) Embryo transfer.

(5) Place where DNA fingerprinting research is done in India.

Ans. Centre for DNA fingerprinting and Diagnostics, Hyderabad.

(6) Organisms used as biofertilizers.

Ans. Rhizobium, Azotobacter, Nostoc, Anabaena, Azolla.

CHAPTER 9 : SOCIAL HEALTH

(1) Hobbies to reduce stress.

Ans. (1) To listen to music (2) Bird watching and nature trails
(3) Reading good books.



(2) Diseases endangering the social health.

Ans. (1) AIDS (2) Tuberculosis (3) Leprosy.

(3) Full form of WHO. (March '24)

OR

Write full form of WHO.

Ans. World Health Organisation.

CHAPTER 10 : DISASTER MANAGEMENT

(1) Main aspects of disaster management cycle.

Ans. Preparation, Redemption, Preparedness, Impact of disaster, Response, Resurgence, Restoration.

(2) Write any two examples of man-made disaster. (Sept. '21)

Ans. War, Bomb explosion.

(3) Natural disasters. (July '22)

Ans. Floods, Drought, Earthquake, Volcano.

(4) Geological disasters.

Ans. Earthquake, Volcano, Tsunami, Landslide, Erosion, Salinisation.

(5) Intentional man-made disasters.

Ans. War, Fire (setting purposely), Terrorism, Riots, Bomb blast, Rapes, Child labour.

Q. Give functions :

CHAPTER 1 : HEREDITY AND EVOLUTION

(1) t-RNA.

Ans. t-RNA is present in the cytoplasm. According to the message of the m-RNA, it carries the specific amino acid up to the ribosomes.

(2) Gene.

Ans. Genes in the form of DNA participate in process of protein synthesis. They help in the functioning of cell and maintain the structure of a cell.

(3) m-RNA.

Ans. m-RNA is a messenger RNA molecule which transcribes the information present on the DNA to cytoplasm for protein synthesis.



CHAPTER 2 : LIFE PROCESSES IN LIVING ORGANISMS : PART – I

(1) Mitochondria.

Ans. Mitochondria participates in cellular respiration. It helps to complete electron transfer chain reaction.

(2) Cell division.

Ans. A new organism is formed from pre-existing one. Growth is possible due to cell division. Repair and restoration of emaciated body is done through cell division.

CHAPTER 3 : LIFE PROCESSES IN LIVING ORGANISMS : PART – 2

(1) Corolla.

Ans. Attracting insects for pollination. Protecting inner whorls.

(2) Testis. (March '20)

Ans. Production of sperms and male hormone–testosterone.

(3) Ovary. (July '22)

Ans. Production of oocytes and female hormones–estrogen and progesterone.

(4) Uterus. (July '22)

Ans. Growth and development of foetus during pregnancy. Helping in parturition (childbirth) by contractions.

(5) Fallopian tubes/ducts. (July '22)

Ans. Transporting the released oocyte after ovulation to the uterus. Providing space for fertilization of oocyte by sperm. Conception is possible only when sperm and oocyte meet in the fallopian tube.

(6) Placenta.

Ans. Supplying nourishment to the growing foetus.

CHAPTER 4 : ENVIRONMENTAL MANAGEMENT

(1) Biogeochemical cycles.

Ans. Biogeochemical cycles help the circular flow of nutrients for the development and growth of other living organisms.

(2) Decomposers.

Ans. Decomposers help in conversion of organic matter back to inorganic ions by the process of decomposition.

CHAPTER 5 : TOWARDS GREEN ENERGY

(1) Turbine.

Ans. Turbine is used for rotating the magnet in the generator.

(2) Electric generator.

Ans. The potential difference is created across the conducting wire and thus electricity is generated by the electric generator.

CHAPTER 6 : ANIMAL CLASSIFICATION

(1) Cnidoblast.

Ans. Cnidoblasts are used to inject the toxins in the body of prey. Their function is characteristic in the animals belonging to phylum Cnidaria.

(2) Setae.

Ans. Setae are locomotory organs in some Annelids. Setae help in creeping locomotion.

(3) Tubefeet.

Ans. Tubefeet help in locomotion in case of animals belonging to phylum Echinodermata.

(4) Gills.

Ans. Respiration occurs with gills.

CHAPTER 7 : INTRODUCTION TO MICROBIOLOGY

(1) Probiotics.

Ans. (1) Probiotics improve resistance to fight against disease.

(2) The useful colonies of bacteria are produced in the alimentary canal of human beings due to the probiotics.

(3) The ill-effects of harmful substances formed during metabolic activities are reduced by the probiotics.

(2) Microbial enzymes.

Ans. (1) Microbial enzymes carry out specific processes even at low temperature, pH and pressure.

(2) Unnecessary byproducts are not formed which reduce expenses on purification.

(3) They save energy.

(4) They reduce the necessity of erosion-free instruments.



(3) Antibiotics.

Ans. To control the bacterial diseases. Eradicating both Gram positive and Gram negative bacteria and thereby control diseases and their epidemics.

CHAPTER 8 : CELL BIOLOGY AND BIOTECHNOLOGY

(1) Gene therapy.

Ans. Treatment of genetic disorders present in the somatic cells.

(2) Cloning.

Ans. Producing exact replica of any cell or organ or entire organism.

CHAPTER 9 : SOCIAL HEALTH

• Counsellor.

Ans. To empathetically listen to persons who have some psychological or emotional problems. To understand problems of them and give remedial measures by talking to them.

CHAPTER 10 : DISASTER MANAGEMENT

• Functions of Disaster Management Authority.

Ans. (1) Giving the necessary instructions.

(2) Assessing whether the instructions are followed or not.

(3) Reviewing the entire system.

(4) Controlling the actions.

(5) Planning and executing the plans.

(6) Designing schemes for each district, separately for each type of disaster and getting those sanctioned from state-level authorities.

**Question
Set
7**

GIVE SCIENTIFIC REASONS

[Q. 2 (A) : 2 Marks each]

Remember :

- ☛ In this question set, 'Give scientific reasons' questions are included. This will form Q. 2 (A).
- ☛ Three questions will be asked and you are expected to attempt any two of them.
- ☛ Since each question carries 2 marks, write minimum four correct explanatory points in the answer.
- ☛ Out of the three questions, minimum one will be on portion of Biology and one will be set on portion of Environmental Science.

Q. Give scientific reasons :

CHAPTER 1 : HEREDITY AND EVOLUTION

(1) Some of the characters of parents are seen in their offspring.

Ans. (1) The parental genes are transferred to their progeny through male and female gametes. (2) These genes carry hereditary characters. (3) Since they are transmitted from the parents to their offspring, one can see the parental characters in their offspring.

(2) Duck-billed platypus is a connecting link. (July '23)

Ans. (1) Duck-billed platypus is a connecting link between reptiles and mammals. (2) The reptilian characters are egg laying habit. (3) Mammalian character is possession of mammary glands and hairy skin. (4) Since it shows characters of both these classes, it is called a connecting link.

CHAPTER 2 : LIFE PROCESSES IN LIVING ORGANISMS : PART - 1

(1) Krebs cycle is also known as citric acid cycle.

Ans. (1) Sir Hans Krebs proposed this cycle and hence it is called Krebs cycle. (2) These are series of cyclic chain reactions which begins with Acetyl Coenzyme-A molecules which act with molecules

of oxaloacetic acid. (3) The reactions are catalysed with the help of specific enzymes. (4) The first molecule formed in this reaction is called citric acid. Therefore, Krebs cycle is also called citric acid cycle.

(2) Sometimes higher plants and animals too perform anaerobic respiration. (March '23 – '25)

Ans. (1) When there is deficiency of oxygen in the surrounding, the aerobic respiration is not possible. (2) In such case, to survive, higher plants switch over to anaerobic respiration. (3) In some animal tissues, in case of oxygen deficiency, cells perform anaerobic respiration.

(3) We feel exhausted after exercising. (March '20 – '24)

Ans. (1) When we undertake constant exercises, there may be shortage of oxygen for the cells. (2) Therefore, our muscles and other tissues perform anaerobic respiration in such condition. (3) In this process, lactic acid is formed. (4) Molecules of ATP produced in oxidation of food are also much less. (5) Thus, there is less energy in the body and accumulation of lactic acid too. All this brings about a feeling of exhaustion.

(4) Cell division is one of the important properties of cells and organisms. (Sept. '21; March '22)

Ans. (1) Cell division is very essential for all the living organisms. (2) The growth and development is possible only due to cell division. (3) The emaciated body can be restored only through the cell division which adds new cells. (4) In asexual reproduction, mitosis helps to give rise to new generation, while in sexual reproduction, meiosis helps to form haploid gametes. (5) All such functions show that cell division is one of the important properties of cells and organisms.

CHAPTER 3 : LIFE PROCESSES IN LIVING ORGANISMS : PART – 2

(1) Gender of child is determined by the male partner of couple.

OR

It is not right to consider the mother responsible, for giving birth to a girl child. (March '25)

Ans. (1) In males, XY sex-chromosomes are present and in females, XX sex-chromosomes are present. (2) At the time of

fertilization, if X-chromosome comes from male, the child will be a girl and if Y-chromosome comes from male, the child will be a boy. (3) Therefore, it is not right to consider the mother responsible, for giving birth to a girl child.

(2) Flower is the structural unit of sexual reproduction in plant. (March '23)

Ans. (1) Flower produces male and female gametes. (2) For this purpose there are essential whorls of androecium and gynoecium. (3) The double fertilization also takes place in flower. (4) Therefore, flower is called the structural unit of sexual reproduction in plants.

(3) Indians should follow family planning for controlling the population. (March '20)

Ans. (1) There is severe population explosion in India. It has almost reached to 121 crores. (2) This results into unemployment, decreasing per capita income and increasing loan, stress on natural resources, etc. (3) Only by controlling population, the quality of life can be restored. (4) Therefore, Indians should follow family planning for controlling the population.

CHAPTER 4 : ENVIRONMENTAL MANAGEMENT

(1) There is no definite information about indeterminate species.

Ans. (1) Indeterminate species do not have substantial information about them. (2) The organisms belonging to such species appear to be endangered due to their some behavioural habits. (3) They are shy and do not come in open so that they can be observed keenly. (4) For example, animals like Giant squirrel also do not provide such information.

(2) Humans have a crucial role in maintaining environmental balance. (July '23)

Ans. (1) Humans have exploited all the natural resources and degraded most of the ecosystems. (2) He created pollution and degradation due to his urbanisation and industrialisation. (3) He has not protected other species on the earth. (4) But due to his intelligence and will to conserve the natural resources, he can restore the lost environment balance.



(1) The energy generated from nuclear fuels is not environment friendly. (March '25)

Ans. (1) Radioactive and harmful radiations are emitted after fission of nuclei occurs. (2) There is a problem of disposal of nuclear waste. (3) Very fatal accidents can happen in nuclear power plant which emits very harmful radiations. Hence, The energy generated from nuclear fuels is not environment friendly.

(2) It is absolutely necessary to control the fission reaction in nuclear power plants. (July '22; March '23)

Ans. (1) Nuclear fission reaction is a type of chain reaction. (2) In nuclear power plants these reactions are closely controlled. (3) If these reactions are not managed properly, there can be more production of neutrons in an uncontrolled way. (4) Each released neutron further causes fission of 3 Uranium (U-235) atoms, such uncontrolled reactions can cause hazardous accidents. Hence it is absolutely necessary to control the fission reaction in nuclear power plants.

(3) The construction of turbine is different for different types of power plants.

Ans. (1) Generators work on the principles of electromagnetic induction. (2) For this the generator must be rotated. (3) For this purpose, there is a turbine for each generator. (4) This rotation needs energy. The turbines are different according to the type of energy source that is used for its rotation. (5) Therefore, the construction of turbine is different for each power plant.

(4) Biofuel is important among the renewable sources of energy. (July '23)

Ans. (1) Using biofuel is eco-friendly. (2) It does not cause pollution. (3) The fuel can be made from discarded biomass. This biomass is plenty and not exhaustible like fossil fuels. (4) Biofuels are renewable and green energy.

(1) Our body gets irritated when it comes in contact with jellyfish. (Sept. '21)

Ans. (1) Jellyfish is a coelenterate that has cnidoblasts bearing tentacles. (2) These cnidoblasts inject toxins to paralyse the prey at the time of feeding. (3) When jellyfish comes in contact with our body, this toxin is released causing reaction to our skin. Therefore, our body gets irritation when we come in contact with jellyfish.

(2) Cockroach belongs to phylum Arthropoda. (March '22)

Ans. (1) Cockroach has jointed appendages. (2) It has chitinous exoskeleton. (3) The body is triploblastic, bilaterally symmetrical and segmented. Therefore, cockroach is included under phylum Arthropoda.

(3) Balanoglossus is connecting link between non-chordates and chordates. (Nov. '20)

Ans. (1) Balanoglossus shows some characters of non-chordates. (2) It also has notochord as in case of chordates. (3) Since it shares the characters of non-chordates and chordates, from the view point of evolution, it is called connecting link between them.

(4) Body temperature of reptiles is not constant.

Ans. (1) Reptiles are cold-blooded animals. (2) The thermoregulatory system is not there in their bodies. (3) Their body temperatures, fluctuate as per the environmental temperatures. Therefore, the body temperature is not maintained at constant level in reptiles.

(1) Enzymes obtained by microbial process are mixed with detergents.

Ans. (1) When detergents are mixed with microbial enzymes, they start working more efficiently. (2) The cleaning process takes place at lesser temperatures. Therefore, for better results, enzymes obtained by microbial process are mixed with detergents.



(2) Microbial enzymes are used instead of chemical catalysts in chemical industry. (March '19; Nov. '20)

OR

Microbial enzymes are said to be eco-friendly.

Ans. (1) Microbial enzymes are active at low temperature, pH and pressure. (2) Due to this property, the energy is saved. The costlier erosion-proof instruments need not be used. (3) In enzymatic reactions, the unnecessary byproducts are not formed as the reactions are highly specific. (4) The expenses on purification of the product are minimised as no unnecessary products are formed. (5) The elimination and decomposition of waste material is avoided and enzymes can be reused again. Hence, microbial enzymes are said to be eco-friendly.

CHAPTER 8 : CELL BIOLOGY AND BIOTECHNOLOGY

• Nowadays, safer vaccines are being produced.

Ans. (1) Before the advent of biotechnology, the vaccines were made from inactive or dead pathogens of that disease. (2) But now the vaccine is made artificially using biotechnological processes. (3) Such vaccines produced some disease symptoms in some cases. (4) The antigen of the disease is researched upon and its genetic code is found out. (5) A similar antigen is made in the laboratories which is used as a vaccine. (6) Such vaccines are more thermostable and remain active for longer duration. Therefore, the vaccines are now safer.

CHAPTER 9 : SOCIAL HEALTH

(1) Girls are facing the problem of stress due to such gender inequality. (Sept. '21)

Ans. (1) In most of the households there are many bindings on girls and excessive freedom for boys. (2) Boys do not participate in the domestic duties whereas girls have compulsion for the same. (3) In society too, girls have to face the problems like teasing and molestation. This creates insecurity in the minds of girls. (4) The social change has made women independent and equal but still the male dominated society and the gender inequality persists causing more stress for young girls.

(2) Tobacco chewing is injurious to health. (July '22)

Ans. (1) Tobacco chewing is an addictive habit. It has toxic substance, nicotine in it. (2) Nicotine has bad impact on nervous system and heart. (3) Cancer of oral cavity and pharynx can be the outcome of tobacco chewing. (4) Fumes of tobacco can also cause lung cancer. Therefore, tobacco chewing is considered to be injurious to health.

(3) Importance of outdoor games is unparalleled.

Ans. (1) Outdoor games give good physical exercise. These games give many physical benefits. (2) It improves personal discipline, interaction with fellow players and creates sense of unity. (3) Through play by driving away the loneliness, mental stress and depression is reduced. (4) The person becomes more social. Therefore, it is said that the importance of outdoor games is unparalleled.

CHAPTER 10 : DISASTER MANAGEMENT

• Mock drill is useful.

Ans. (1) Mock drill is the practice to check whether there is preparedness for dealing with the sudden attack of disaster. (2) For this purpose, virtual or apparent situations that simulate the disaster are created. (3) The reaction time for any type of disaster is checked by such activity. In the presence of trained personnel, the execution of the rescue plans are observed. (4) People also understand their responsibilities at the time of actual disaster. (5) The experts also check execution of plan designed for disaster redressal. (6) By such mock drills, the efficacy of the system can be understood. In future, when actual calamity strikes, there is already preparation for disaster redressal. Therefore, mock drill is useful. (7) e.g. In case of possible fire, the school can have mock drill with the help of fire fighters.



**Question
Set
8**

**RULES, THEORIES, LAWS,
DEFINITIONS AND PRINCIPLES**

[Q. 2 (B) : 2 Marks each]

Remember :

- ☛ This question set is based on theories, rules, definitions and principles.
- ☛ For definitions exact answer is expected.
- ☛ Explanations of theories in 4 points are expected while answering the questions based on theories.

I. Theories :

CHAPTER 1 : HEREDITY AND EVOLUTION

(1) Darwin's theory of natural selection.

Ans. Charles Darwin (1809–1882) proposed the theory of natural selection.

Theory of natural selection : 'The survival of fittest', i.e. organisms which are fit for survival, evolve while those that are not, perish. The natural selection thus acts to produce new species.

(2) Lamarck's principle of 'use or disuse of organs'.

Ans. The theory of use and disuse of organs says that the morphological characters of organism develop because of specific activities that the organisms perform. If some organ is not used it gets degenerated. If excessively used, it develops. Thus, the morphological changes take place due to activities or non-working of a particular body parts in an organism.

II. Give definitions :

CHAPTER 1 : HEREDITY AND EVOLUTION

(1) Heredity : The transfer of biological characters from one generation to another through genes is called heredity.

(2) Transcription : Process of synthesis of m-RNA according to the nucleotide sequence present on DNA is called transcription.

(3) Translocation : The process of movement of the ribosome from one end of m-RNA to other end by the distance of one triplet codon is called translocation.

(4) Mutation : Sudden and drastic change that occurs in the genetic material is called mutation.

CHAPTER 2 : LIFE PROCESSES IN LIVING ORGANISMS : PART - 1

(1) Proteins : Protein is a macromolecule which is formed by many amino acids which are joined by peptide bonds.

(2) Vitamins : Vitamins are a group of heterogeneous compounds essential for proper functioning of the body.

(3) Cellular respiration : Oxidation of glucose and other food components which takes place inside the cell in presence or absence of oxygen, is known as cellular respiration.

(4) Fermentation : Conversion of pyruvic acid produced in the process of glycolysis into other organic acids or alcohol with the help of some enzymes is called fermentation.

CHAPTER 3 : LIFE PROCESSES IN LIVING ORGANISMS : PART - 2

(1) Fragmentation : Fragmentation is the asexual type of reproduction in which the body of parent organism breaks up into many fragments. Each fragment can start living independently.

(2) Regeneration : Regeneration is the asexual reproduction in planaria in which the body is broken up into two parts and resulting each part regenerates remaining part of the body.

(3) Pollination : Transfer of pollen grains from anther to the stigma is called pollination.

(4) Menopause : Stoppage of functioning of female reproductive system due to lack of synthesis of hormones due to advancing age is called menopause.

(5) Placenta : An organ developed in the uterus of the pregnant mother, through which the embryo is given nourishment is called placenta.

(6) Menstrual cycle : The repetitive changes at the interval of every 28–30 days in female reproductive system that take place after puberty, form menstrual cycle.



(7) IVF : In Vitro Fertilization is the technique in which fertilization is brought about outside the female body but in the test-tube and the embryo is implanted in uterus of woman.

(8) Ovulation : Bursting of mature ovarian follicle under the influence of LH hormone to release the oocyte is called ovulation.

CHAPTER 4 : ENVIRONMENTAL MANAGEMENT

(1) Environment : Physical, chemical and biological factors which act upon the living organisms in any possible way is called the environment.

(2) Endangered species : The organisms which are very likely to become extinct in near future are called endangered species.

(3) Rare species : A group of organisms that are very uncommon, in a particular place scarce and not easily spotted are called rare species.

(4) Indeterminate species : Due to behavioural habits organisms appear to be endangered, but their status cannot be determined. Then such organisms are called indeterminate species.

CHAPTER 5 : TOWARDS GREEN ENERGY

(1) Solar photovoltaic effect : Solar photovoltaic effect is the conversion of solar energy directly into electrical energy.

(2) Wind-turbine : The machine which converts the kinetic energy of wind to electrical energy is called wind-turbine.

CHAPTER 6 : ANIMAL CLASSIFICATION

(1) Diploblastic : When in animal body there are two germinal layers, viz. ectoderm and endoderm then such animals are called diploblastic.

(2) Coelom : Coelom is the body cavity present in the animals which is formed during initial embryonic development period from either mesoderm or gut.

CHAPTER 7 : INTRODUCTION TO MICROBIOLOGY

(1) Applied Microbiology : Branch of biology in which study of the enzymes related to some prokaryotes and eukaryotic microbes, proteins, applied genetics, molecular biology, etc. Is performed; is called as applied microbiology.

(2) Antibiotic : An antibiotic is an antibacterial or antimicrobial drug used in the treatment and prevention of bacterial infections by either killing or inhibiting the growth of the bacteria.

CHAPTER 8 : CELL BIOLOGY AND BIOTECHNOLOGY

(1) Stem cell : The special cells having pluripotency and ability to divide and differentiate into new cells are called stem cells.

(2) Biotechnology : Technology that brings about artificial genetic changes and hybridization in organisms for human welfare is called biotechnology.

(3) Genetically modified crops : Crops having desired characters are developed by integrating foreign gene with their genome, such crops have modified genome and are known as genetically modified crops.

(4) Vaccine : The 'antigen' containing material given to a person or animal to acquire either permanent or temporary immunity against a specific pathogen or disease is called a vaccine.

(5) Cloning : Production of replica of any cell or organ or entire organism through biotechnological process is called cloning.

(6) Green revolution : All the methods applied for harvesting maximum yield from minimum land are collectively referred to as green revolution.

CHAPTER 9 : SOCIAL HEALTH

(1) Selficide : Person indulging in selfie is not aware about the world around and the risks. This is called as selficide.

(2) Social health : Ability of a person to establish the positive relationship with other persons and developing a strong personality is called social health.

CHAPTER 10 : DISASTER MANAGEMENT

(1) Disaster : A sudden accident, man-made deliberate activity or a natural catastrophic event that cause great damage or loss of life is called a disaster.

(2) First aid : An immediate help given to a sick or injured person until full medical treatment is available is called a first aid.



**Question
Set
9**

**DISTINGUISH BETWEEN
THE FOLLOWING**
[Q. 2 (B) or 3 : 2 or 3 Marks each]

Remember :

- ☛ In this question set, the points of distinction between two terms are to be written.
- ☛ Since the question carries 2 marks, write minimum 2 or 4 correct points while distinguishing between the two terms that are asked.

Q. Distinguish between :

CHAPTER 2 : LIFE PROCESSES IN LIVING ORGANISMS : PART - 1

(1) Glycolysis and TCA cycle :

Glycolysis	TCA cycle
1. The process of glycolysis occurs in the cytoplasm of the cell.	1. TCA cycle takes place in mitochondria.
2. In glycolysis, one molecule of glucose is oxidised step-by-step to produce two molecules each of pyruvic acid, ATP, NADH ₂ and water.	2. In TCA cycle, molecule of Acetyl Co-A is completely oxidised and in the process CO ₂ , H ₂ O, NADH ₂ , FADH ₂ and ATP is produced.
3. Glycolysis can take place in both aerobic and anaerobic respiration.	3. TCA cycle takes place only during aerobic respiration.
4. The first step in cellular respiration is glycolysis where glucose is converted into pyruvate.	4. The second step in cellular respiration is TCA cycle.
5. Two molecules of pyruvate are obtained in glycolysis.	5. Pyruvate is converted into CO ₂ and H ₂ O during TCA cycle.

Glycolysis	TCA cycle
6. Two molecules of ATP are used up in glycolysis.	6. ATP molecules are not used up in TCA cycle.
7. Four molecules of ATP are produced in glycolysis.	7. Two molecules of ATP are produced in TCA cycle.
8. CO ₂ is not produced during glycolysis.	8. CO ₂ is produced in TCA cycle.

(2) Aerobic and Anaerobic respiration : (Nov. '20) (2 marks)

Aerobic respiration	Anaerobic respiration
1. Oxygen is required for aerobic respiration.	1. Oxygen is not required for anaerobic respiration.
2. Aerobic respiration takes place in nucleus as well as in cytoplasm.	2. Anaerobic respiration occurs only in the cytoplasm.
3. At the end of aerobic respiration, CO ₂ and H ₂ O is formed.	3. At the end of anaerobic respiration, CO ₂ and C ₂ H ₅ OH are formed.
4. Energy is produced in large amount in aerobic respiration.	4. Energy is produced in lesser amount in anaerobic respiration.
5. Glucose is completely oxidised in aerobic respiration.	5. Glucose is incompletely oxidised in anaerobic respiration.
6. 38 molecules of ATP are formed during aerobic respiration.	6. 2 molecules of ATP are formed during anaerobic respiration.

CHAPTER 5 : TOWARDS GREEN ENERGY

• Conventional and Non-conventional sources of energy :

(3 marks)

Conventional sources of energy	Non-conventional sources of energy
1. Conventional sources of energy are largely polluting, they release lot of carbon through its emissions.	1. Non-conventional sources of energy are not polluting. They do not release carbon or other toxic gases.
2. Conventional sources of energy are not eco-friendly.	2. Non-conventional sources of energy are eco-friendly.
3. The fuels produced from the conventional sources of energy are comparatively costlier.	3. The energy obtained from the non-conventional sources of energy are comparatively cheaper.
4. Conventional energy power plants require less area and its management cost is also less.	4. Non-conventional energy power plants require more area and its management cost is also more.
5. Conventional source of energy are non-renewable.	5. Non-conventional source of energy are renewable.
6. Conventional sources of energy are in the form of limited reserves. After few years they will be completely over. e.g. Fossil fuels, coal, crude oil, diesel, petrol, natural gas, etc.	6. Non-conventional energy sources are in abundance on the earth. They are persistent and sustainable. Thus, they will not get over. e.g. Solar energy, wind energy, etc.

CHAPTER 6 : ANIMAL CLASSIFICATION

(1) Class Pisces and Class Reptiles : (March '25) (2 marks)

Class Pisces	Class Reptiles
1. These are aquatic animals living in marine and fresh water.	1. These are mainly terrestrial animals.
2. Respiration occurs with the help of gills.	2. Respiration occurs with the help of lungs.

(2) Butterfly and Bat : (July '22)**(2 marks)**

Butterfly	Bat
1. Butterfly is an insect which is an invertebrate classified under phylum Arthropoda.	1. Bat is a vertebrate belonging to class Mammals.
2. Butterfly is diurnal, poikilothermic and lays eggs.	2. Bat is nocturnal, homeotherm and gives birth to offspring.

(3) Non-chordates and Chordates : (March '23)

Non-chordates	Chordates
1. Non-chordates are less evolved animals and are on the lower levels of evolution.	1. Chordates are more evolved animals and are on the higher levels of evolution.
2. Non-chordates do not have notochord.	2. Chordates have notochord at least in some stage of development.
3. In non-chordates, there are no pharyngeal gill slits.	3. In chordates, there are pharyngeal gill slits.
4. Nerve cord, if present is double and solid.	4. Nerve cord is single and hollow.
5. Nerve cord is located on the ventral side of the body.	5. Nerve cord is located on the dorsal side of the body.
6. Heart if present is on the dorsal side of the body.	6. Heart if present is on the ventral side of the body.



CHAPTER 7 : INTRODUCTION TO MICROBIOLOGY

• Traditional fuels and Biofuels :

(2 marks)

Traditional fuels	Biofuels
1. Traditional fuels are non-renewable fuels.	1. Biofuels are renewable fuels.
2. Traditional fuels are exhaustible.	2. Biofuels are inexhaustible.
3. They are usually used in liquid form and gaseous form.	3. They are usually used in solid, liquid or gaseous form.
4. These were the fuels of yester years.	4. These are fuels of the future.
5. Traditional fuels cause pollution.	5. Biofuels may not cause pollution.

CHAPTER 10 : DISASTER MANAGEMENT

• Natural and Man-made disasters :

(2 marks)

Natural disasters	Man-made disasters
1. Disasters coming suddenly due to natural phenomenon are called natural disasters.	1. Disasters caused either due to accidents or by intentional actions are called man-made disasters.
2. All the people as well as their properties and wealth in affected area are affected due to natural disasters.	2. Specific people are affected who are in the target of man-made disaster.
3. Natural disasters are beyond human control.	3. Man-made disasters can be avoided or controlled.
4. Floods, droughts, tsunami, earthquake are some of the natural disasters.	4. Bomb explosion, terrorism, war, etc. are man-made disasters.

**Question
Set
10**

SHORT NOTES

[Q. 2 (B) or 3 : 2 or 3 Marks each]

Remember :

- ☛ In this question set, short notes are included.
- ☛ The short note should be brief and point-wise. It should have minimum 4 points to obtain 2 marks.

Q. Write short notes :

CHAPTER 1 : HEREDITY AND EVOLUTION

(1) Evolution.

(2 marks)

Ans. (1) The sequential changes in the groups of living organisms that take place very gradually is called evolution.

(2) Evolution is also described as the formation of new species due to natural selection.

(3) The process of evolution takes millions of years for development and speciation of different organisms.

(4) Changes in stars and planets in space and the changes in biosphere occurring on the Earth are all included under study of evolution.

(5) Due to evolution organisms become fit, biodiversity is increased, and new species are created.

(6) Different scientists have put forth theories to explain the process of evolution. Among these Charles Darwin's theory of natural selection and speciation is accepted worldwide.

(2) Connecting link.

OR

Explain the concept of connecting links with two suitable examples. (July '22)

(3 marks)

Ans. For the answer, refer to Q. 3. from Chapter 1 of Q. Set 13.

(3) Embryology.

(2 marks)

Ans. (1) Embryology is the study of developing embryos.



(2) These embryos in their initial stages are very similar to each other.

(3) These similarities decrease later in the development.

(4) This similarity in initial stages indicate that these vertebrates have originated from a common ancestor.

(5) In evolutionary science, comparative study of embryos of various vertebrates provide evidence for evolution.

(4) Lamarckism.

(3 marks)

Ans. (1) Lamarckism consists of two theories which were proposed by Jean Baptiste Lamarck. These are as follows : (a) Use and disuse of the organs (b) Inheritance of acquired characters.

(2) In theory of use and disuse of organs, Lamarck says : The characters of organs develop because specific activities that the organisms perform. If such organ is not used it gets degenerated. Thus, the morphological changes take place due to activities or inactivity of a particular organism.

(3) To emphasise this theory, he quoted following examples. Due to constant extension of neck to eat foliage from the top of the trees, giraffe's neck became long. Similarly blacksmith has strong arms due to constant work.

(4) Flightless ostrich and emu did not fly and hence their wings became useless. Aquatic birds like swan and duck made their feet suitable for swimming by living in water. Snake lost limbs as it tried burrowing mode.

(5) Such acquired characters are passed from one parental generation to the offspring. This is called inheritance of acquired characters.

(6) The theory of inheritance of acquired characters is not accepted as such transmission of acquired character does not take place. Only genetic characters are transmitted.

CHAPTER 2 : LIFE PROCESSES IN LIVING ORGANISMS : PART – 1

• Explain the Krebs cycle with reaction. (2 marks)

Ans. (1) Krebs cycle was proposed by Sir Hans Kreb. This cycle is named after him. It is also called tricarboxylic acid cycle or citric acid cycle.

(2) The acetyl coenzyme-A molecules enter the mitochondria located in the cytoplasm.

(3) They participate in the chemical reactions taking place in Krebs cycle.

(4) In the cyclic chemical reactions, acetyl coenzyme-A is completely oxidised.

(5) It yields molecules of CO_2 , H_2O , NADH_2 , FADH_2 and ATP upon complete oxidation.

CHAPTER 3 : LIFE PROCESSES IN LIVING ORGANISMS : PART – 2

(1) Menstrual cycle.

(3 marks)

Ans. (1) Menstrual cycle is the events of cyclic changes that takes place with the interval of 28 to 30 days in mature woman.

(2) Hormones from pituitary, FSH (Follicle Stimulating Hormone) and LH (Luteinizing Hormone) and hormones from ovary, estrogen and progesterone control the menstrual cycle.

(3) Due to influence of FSH, the ovarian follicle grows along with the oocyte that is present in it.

(4) This growing follicle produces estrogen.

(5) Under the influence of estrogen, the uterine inner layer called endometrium grows or regenerates. In the meantime, the development of follicle is completed.

(6) LH from pituitary stimulates the bursting of ovarian follicle and releases the mature oocyte out of the follicle and the ovarian wall. This process is called ovulation.

(7) The empty ovarian follicle after the ovulation becomes corpus luteum. Corpus luteum produces hormone progesterone.

(8) Under the influence of progesterone, the glands from uterine endometrium start secreting. The oocyte if fertilized is implanted over this endometrium.

(9) If oocyte is not fertilized, the corpus luteum becomes a degenerate body called corpus albicans. The corpus albicans cannot secrete estrogen and progesterone.

(10) Due to lack of these hormones, the endometrial layer of the uterus collapses. The tissue debris, along with unfertilized egg is given out through the vagina as menstrual flow. This results in bleeding for about 5 days.



(11) If woman is not pregnant, then this menstrual cycle keeps on repeating with regularity.

(2) Seed germination.

(2 marks)

Ans. (1) Seed germination is the process in which the seed develops into a new plantlet.

(2) In the plants, after fertilization the ovule develops into seed and ovary turns into fruit.

(3) Seeds fallen on the ground due to bursting of the fruits start germinating. Only under favourable conditions in the soil, this germination takes place.

(4) The zygote present inside the seed uses food stored in endosperm of seed and hence develops further to produce a new plantlet.

CHAPTER 4 : ENVIRONMENTAL MANAGEMENT

(1) Environmental Conservation. (March '23)

(2 marks)

Ans. (1) Due to natural and man-made causes, there are many environmental problems on the earth. These problems affect the existence of various living organisms. In order to save these organisms and maintain the environmental balance, there is need for environmental conservation.

(2) If this is not done, then there will not be any quality of life for the resident humans.

(3) For environmental conservation, the Government has formulated acts and rules.

(4) UN has established UNEP for the conservation programs. The people's participation in the conservation movement is essential.

(5) From school age, the environmental values are inculcated in the young minds. Conservation of environment is the social responsibility of everyone. Judicial use of natural resources is also a way of environmental conservation.

(6) On personal level, we should plant trees and also undertake solid waste management for the conservation of our environment.

(2) Sacred Groves. (Nov. '20)

(2 marks)

Ans. (1) Sacred grove is the green patch of the forest which is conserved by local people in the name of God.



(2) It does not belong to forest department. It is like a sanctuary that is conserved by the common people and tribals in the area.

(3) It is rich in the biodiversity. It is conserved as there is a faith that God or deity reside in the sacred grove. Hence, in local language, they are known as Devrai.

(4) Due to this reason, people do not fell the trees. Also hunting of any wildlife is not done here.

(5) More than 13,000 sacred groves have been reported in India in western Ghats in Maharashtra, Karnataka and Kerala.

CHAPTER 5 : TOWARDS GREEN ENERGY

• Electrical energy generation and Environment. (3 marks)

Ans. The energy obtained through the fossil fuels as well as nuclear energy can cause degradation of the environment. If such energy sources are used, they can cause harm to the environment.

(1) The burning of fossil fuels cause air pollution. The incomplete combustion of fossil fuels cause release of carbon monoxide. Some more toxic gases and soot particles cause various respiratory diseases. The carbon dioxide produced is creating global warming and climate change. The nitrogen dioxide released through burning is responsible for acid rains.

(2) Fossil fuels are limited. They are depleting at a fast rate. It has taken millions of years for the fossil fuels to form. The exploration of such fuels also cause environmental degradation and marine pollution too.

(3) In production of nuclear energy, there is a great risk of accidents. The safe disposal of nuclear waste is also a problem.

(4) Hydroelectric power from water reservoirs, wind power from wind, solar energy from sun and electricity from biofuels are eco-friendly alternatives.

CHAPTER 6 : ANIMAL CLASSIFICATION

• Progressive changes in animal classification. (3 marks)

Ans. There were different methods of classification of animals.

(1) The first classification method was given by the Greek philosopher Aristotle. He took into account the criteria like body



size, habits and habitats of the animals. This method was called artificial method of classification.

(2) The same artificial method was used by other scientists such as Theophrastus, Pliny, John Ray, Linnaeus, etc.

(3) Further due to advances in science, the references were changed and there were some new methods of classification proposed.

(4) The system of classification called 'Natural system of classification' was then proposed. This system of classification was based on criteria such as body organisation, types of cells, chromosomes, bio-chemical properties, etc.

(5) Later, Dobzhansky and Meyer gave the system of classification based on evolution.

(6) In 1977, Carl Woese has also proposed the three domain system of animal classification.

CHAPTER 7 : INTRODUCTION TO MICROBIOLOGY

• Production of cheese.

(3 marks)

Ans. Cheese is made from cow's milk throughout the world. The steps in the process of cheese manufacture are as follows :

(1) Chemical and microbiological testing of milk is done.

(2) Three types of bacteria, viz. *Lactobacillus lactis*, *Lactobacillus cremoris* and *Streptococcus thermophilus* along with some colour is added to the milk.

(3) It imparts sourness to the milk and it is converted into yoghurt like substance.

(4) The water from this yoghurt, i.e. whey is not removed to make the yoghurt denser.

(5) Enzyme, rennet or protease is added to the mixture to make it more denser.

(6) Later cutting the solid yoghurt into pieces, washing, rubbing, salting, and mixing of essential microbes, pigments and flavours is done in suitable steps.

(7) The pressed cheese is then cut into pieces and stored for ripening.

• **DNA fingerprinting. (March '24)**

(2 marks)

Ans. (1) As the fingerprints are unique for every individual, similarly the nucleotide sequence in the DNA molecule is also unique.

(2) By knowing this sequence, one can find out the identity of any person. Such technique to establish the identity of a person by taking into consideration the nucleotide sequence is called DNA fingerprinting.

(3) Its main use is in forensic sciences to confirm the identity of the criminal.

(4) Similarly, identity of parents in case of disputed parentage for any child can be understood by taking DNA fingerprints of both the parents and a child.

CHAPTER 9 : SOCIAL HEALTH

• **Cybercrimes.**

(3 marks)

Ans. (1) No personal information should be shared on the phone, especially the details of bank account, Aadhaar card, PAN card, credit card or debit card number, etc. Cheating persons by using this information is a greatest cybercrime.

(2) If PIN of any debit or credit card is known to a stranger, he or she can make fraudulent transactions.

(3) The PIN number and CVV number should be kept total secret. Otherwise, the bank transactions are done using PIN without the knowledge of consumers.

(4) In on-line purchases, many a times consumers are cheated. In this, the consumers are shown superior items on websites but actually the inferior ones are sold to them.

(5) 'Hacking of information' is done by some programmers in which the confidential information about government, institutes and companies is obtained from internet with the help of computer programs.

(6) Fake Facebook accounts are opened and false information is displayed there. This is for harassing girls or financially exploiting others.



(7) In internet piracy, written literature, software, photos, videos, music, etc. of other persons are misused or illegally used.

(8) Misuse of electronic media sending derogatory messages, spreading vulgar pictures and provocative statements is also a cybercrime. Very rapid exchange of information through media like email, Facebook and WhatsApp takes place these days. But we have to take care about leaking of our own important information. However, when our personal information and phone numbers are automatically spread and reached to fraudulent people, then they commit malpractices which can hinder the function or shut of the cell phones or computers.

All these are cybercrimes which are also indicative of mental health.

CHAPTER 10 : DISASTER MANAGEMENT

(1) Nature of disaster management.

(2 marks)

Ans. Disaster management involves either prevention of disasters (Pre-disaster management) or creating preparedness to face them (Post-disaster management). The action plans are prepared for managing disasters. This is done after studying the different aspects such as preventive measures, rehabilitation and reconstruction plans. The disasters are tackled by executing action plans in the following steps : Preparation, redemption, preparedness, action during actual disaster, response, resurgence and restoration. At every level there are other voluntary organisations and Government meteorological institutions for their help.

(2) Mock drill. (Sept. '21)

(2 marks)

Ans. For the answer, refer to question from Chapter 10 of Q. Set 7.

**Question
Set
11**

EXPLANATIONS AND CONCEPTS

[Q. 3 : 3 Marks each]

Remember :

- ☛ In this question set, questions based on explanations and concepts are included.
- ☛ You are expected to understand the underlying concept for writing the explanations.
- ☛ Minimum 6 explanatory statements are to be written while answering such questions.

Q. Explain the given statements :

CHAPTER 1 : HEREDITY AND EVOLUTION

(1) Study of fossils is an important aspect of study of evolution.

Ans. (1) Fossils offer palaeontological evidence for the evolutionary process.

(2) Due to some natural calamities the organisms got buried during ancient times.

(3) The impressions and remnants of such organisms remain preserved underground. The hot lava also traps some organisms or their impressions. All such formations form fossils.

(4) Study of fossils help the researcher to understand the characteristics of the organisms that existed in the past.

(5) Carbon dating method helps in finding out exact age of the fossil. According to the structure of earth's crust, the fossils are obtained at specific depths.

(6) The oldest ones are obtained at the depth while the relatively recent ones occupy the upper surface. Thus, fossils of invertebrates were seen in very old Palaeozoic era. Later were seen fossils of Pisces, Amphibia and Reptilia. The Mesozoic era was dominated by reptiles while Coenozoic era showed presence of mammals.

(7) In this way, study of fossils unfold the evolutionary secrets.



(2) Human evolution began approximately 7 crore years ago. Explain this statement.

Ans. (1) Approximately around 7 crore years ago the ice age began on the earth. In such conditions, dinosaurs became extinct. The evolution and diversity of mammals started during this time. Due to change in climate the forest cover also declined rapidly.

(2) Ancestors of monkey-like animals were Lemur like animals which evolved during this time period.

(3) The tails of these monkey-like creatures started vanishing very gradually around 4 crore years ago.

(4) The body and brain both increased in volume forming first ape-like animals. The monkey-like ancestors gave rise to two evolutionary links to apes and human-like animals.

(5) Later, the human evolution took place by changes in the brain volume, the ability to walk upright, excessive use of hand for manipulations.

(6) This journey of human evolution began 7 crore years ago. But the true wise and intelligent man arose around 50,000 years ago.

CHAPTER 2 : LIFE PROCESSES IN LIVING ORGANISMS : PART - 1

(1) Oral rehydration solution (Salt-sugar-water) is frequently given to persons experiencing loose motions.

Ans. (1) Loose motions cause lot of loss of water from the body.

(2) This may result in dehydration. This can be lethal if ignored.

(3) Especially in case of young children this is a very serious fatal problem.

(4) Thus, to bring back the normal proportion of water and electrolytes, oral rehydration solution or ORS is given to the patient who suffers from loose motions.

(2) After complete oxidation of a glucose molecules, 38 number of ATP molecules are formed.

Ans. I. **Glycolysis** : No. of ATP molecules formed = 4

No. of ATP molecules used = 2

II. **Krebs cycle** : No. of ATP molecules formed = 2

III. **ETC Reaction** :

$\text{NADH}_2 : 10 \text{ NAD}_2 \times 3 \text{ ATP} = 30 \text{ ATP}$

$\text{FADH}_2 : 2 \text{ FADH}_2 \times 2 \text{ ATP} = 4 \text{ ATP}$

$$\begin{aligned}\text{ATP Total ATP molecules produced} &= (4 + 2 + 34) \\ &= 40 \text{ ATP}\end{aligned}$$

$$\text{ATP molecules used} = 2 \text{ ATP}$$

Therefore, total ATP molecules = 38 ATP

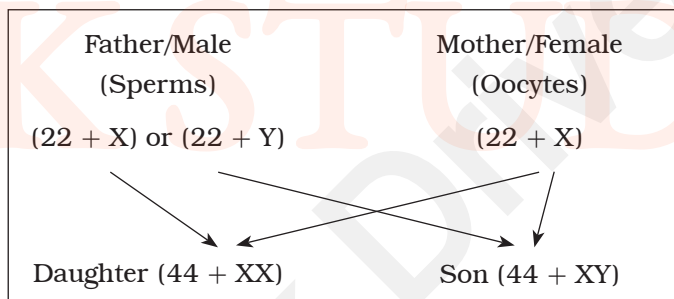
(3) Excess of carbohydrates are stored in liver and muscles in the form of glycogen.

Ans. The carbohydrates which are not used to produce energy cannot be stored in the body in the form of glucose. This glucose is therefore converted into complex compound called glycogen. Glycogen is stored in muscles and liver.

CHAPTER 3 : LIFE PROCESSES IN LIVING ORGANISMS : PART - 2

(1) Gender of child is determined by the male partner of couple. Explain with reasons whether this statement is true or false.

Ans.



Sex determination in Human being

(1) The statement Gender of child is determined by the male partner of couple is true.

(2) It is clearly seen from the diagram that there are two types of sperms produced by males. One sperm has a X-chromosome while the other has a Y-chromosome, apart from autosomes. The mother on the other hand has all X-bearing oocytes. Thus, the sperm that fertilizes the oocyte decides the sex of the child.

(3) If X-bearing sperm fertilizes the oocyte, daughter is born and when Y-bearing sperm fertilizes the oocyte, son is born.

(4) Thus, father or male partner is responsible for the determination of the sex.



(2) Explain the term : Modern technology of reproduction IVF. (Nov. '20)

Ans. (1) IVF means In Vitro Fertilization (IVF).

(2) This is the technique in the modern medical field where childless couples can be blessed by their own child.

(3) IVF technique is used for childless couples who are faced with problems such as less sperm count, obstacles in oviduct, etc.

(4) The IVF technique is done by removing the oocyte from the mother and artificially fertilizing by the sperms collected from father. This fertilization is done in a test-tube. Thus, it is also called test tube baby. The embryo formed is implanted in uterus of real mother or a surrogate mother at appropriate time.

CHAPTER 4 : ENVIRONMENTAL MANAGEMENT

(1) Explain the statement – ‘We have got this Earth planet on lease from our future generations and not as an ancestral property from our ancestors.’

Ans. (1) The earth was inhabited by older generations before us. We have replaced them.

(2) But during their life time, they have created hazardous impact on the earth. The industrialisation, the quest for more and more natural resources, wars fought, the construction activities such as dams, roads and bridges, extensive deforestation, etc. were their thoughtless activities.

(3) All these activities were for development of mankind. But most of them have destroyed the delicate balance between the producers and different levels of consumers.

(4) Due to ever increasing population of human beings there is shortage of food, clothing and shelter. To procure these basic needs, we have exploited many natural resources causing destruction of the earth's natural ecosystem.

(5) Now it is our turn to protect the earth as on the same planet the next generations have to survive. We have to hand over the ecosystems of the earth which are in perfect balance to the new generations.

(6) The future generations need a good quality of air, water and land along with all other living organisms.

(7) Due to problems like climate change, global warming, pollution, droughts, etc. the environment is impacted, thus in order to keep sustainability of earth, we have to remember that the earth has not been obtained only as ancestral property but we have to save it for future generations.

(2) Attempts at various levels are performed for conserving environment. Which role would you like to perform. Give two actions each :

(a) Prevention (b) Control (c) Production (d) Awareness
(e) Conservation. **(March '20)**

Ans. I like the role of control for environmental conservation. By control we can change the attitude of population. We can carry out significant project to explain about harmful effects of plastic.

(a) Prevention : (i) I will not allow anybody to cut down the trees. (ii) I will take care of birds and animals.

(b) Control : (i) I will control the use of water. (ii) I will control the excessive use of plastic.

(c) Production : (i) I will help in creating gardens. (ii) I will help in growing medicinal plants.

(d) Awareness : I will work for important issues such as climate change by carrying out awareness campaigns.

(e) Conservation : (i) I will not waste food, clothing. (ii) I will help in foresting.

CHAPTER 5 : TOWARDS GREEN ENERGY

(1) Explain the statement : Energy obtained from fossil fuels is not green energy.

Ans. Fossil fuels like petrol, diesel or natural gas when burnt, emit toxic gases and soot particles. Thus, fossil fuels cause air pollution. Burning of fossil fuels cause increased levels of carbon dioxide, carbon monoxide and nitrogen dioxide. The increased carbon dioxide emission results in global warming. Nitrogen oxide results later in acid-rain. Soot particles generated through burning of fuels cause respiratory problems like asthma.



Moreover, the fossil fuels are non-renewable and exhaustible fuels. They have to be explored from the deeper layers of the earth causing lots of environmental problems. Green energy is sustainable, renewable and abundant. It never creates any environmental problems and is non-polluting.

Thus, energy obtained from fossil fuels is not at all a green energy.

(2) Explain the statement : Saving energy is the need of the hour.

Ans. In modern civilisation, continuous energy supply is needed for the technology and development. The energy has become a basic need for man. Most of the energy used in India is obtained from thermal power plant. For this energy generation, various fuels are used. The coal and fossil fuels are limited. Due to over-exploitation, these reserves are getting fast depleted. Use of fossil fuels is also resulting in pollution and climate change. Nuclear energy can be very hazardous. Lot of research is being done in the field of green energy, but the tremendous human population always is in need of more energy. Therefore, each and every person should save the energy, as saving energy is the need of the hour.

CHAPTER 6 : ANIMAL CLASSIFICATION

Q. Animals like ghariyal and crocodile live in water as well as on land. Are they amphibians or reptiles?

Ans. Ghariyal and crocodile are reptiles. They can swim in water and crawl on land. But they can respire only with the help of lungs. Their breathing is through nostrils. Even when in water, they have to inhale and exhale by coming up to the surface of water for air. Amphibians can breathe through the skin when in water and by lungs when on land. They also have hard exoskeleton which amphibians do not have. Hence, ghariyal and crocodile are not amphibians, but they are reptiles.

Q. 'It is necessary to ban the use of plastic bags', explain this statement.

Ans. (1) Plastic is a non-biodegradable substance. It cannot be degraded back into its original constituents. It remains just like that for many hundreds of years.

(2) It causes solid waste pollution in any environment wherever it is thrown indiscriminately.

(3) If burnt, it releases very toxic gases. If dumped in landfills, it obstructs the other decomposition processes.

(4) If thrown in water bodies, it causes harm to aquatic life.

(5) Cattle graze on plastic unknowingly and are killed by it as it clogs inside their alimentary canal.

(6) The gutters and rain water drains get clogged due to plastic bags and this causes cities to submerge in water during heavy rains.

(7) Nowadays, the fishermen get more than half of plastic if they cast their net in the sea. People use the plastic bags indiscriminately without any thought towards their environmental impact.

(8) There are better alternatives for plastic bags such as cloth bags which can be reused again and again.

Therefore, it is absolutely necessary to ban the use of plastic bag.

CHAPTER 10 : DISASTER MANAGEMENT

Q. Explain the role of district disaster control unit after occurrence of any disaster.

Ans. (1) District control unit looks after the disaster management of the district.

(2) It is immediately formed either after the impact of disaster or if warning is given about some upcoming disaster.

(3) District-wise Disaster Control Unit performs following role :

(i) The review of various aspects of disasters is done.

(ii) Through the disaster control unit, there is continuous contact established with various agencies like army, air force, navy, telecommunication department, paramilitary forces, etc. for obtaining help.

(iii) The unit also coordinates with various voluntary organisations for their help in disaster management.



**Question
Set
12**

**REMEDIES/MEASURES
ON PROBLEMS
[Q. 3 : 3 Marks each]**

Remember :

- ☛ In this question set, the sub-questions based on effects, problems and their remedial measures are included.
- ☛ Some problems encountered in daily life will be given and the measures to solve these have to be suggested.

Q. Answer the following questions :

CHAPTER 3 : LIFE PROCESSES IN LIVING ORGANISMS : PART – 2

(1) Which precautions will you follow to maintain the reproductive health? (July '22) (2 marks)

Ans. About reproductive health one should have scientific and authentic information. The cleanliness of body is very essential but keeping the mind clean is also important to maintain good reproductive health. One should be careful about sexual relationships. These things should not be experimented in young age. Mistakes committed like these can change the sexual health forever. The cleanliness and hygiene during menstruation, the cleanliness of genitals and other private parts are the aspects of personal hygiene. When living in a society, one should always be away from cross-infections of venereal type.

(2) Explain the following concepts in short :

(a) Surrogacy (b) In Vitro Fertilization (IVF) (c) Sperm Bank.
(March '19-'20)

Ans. (a) Surrogacy : In woman, if there is problem regarding the implantation of embryo in uterus, then help of another woman is taken. This woman is called surrogate mother.

Oocyte from real mother is taken out and fertilized with sperms collected from her husband. These gametes are fertilized outside in a test-tube and then the fertilized zygote is implanted in the surrogate mother.



(b) In Vitro Fertilization (IVF) : *For the answer, refer to Q. 2. in Chapter 3, Q. Set 11.*

(c) Sperm bank : If man has problems with the sperm production, then the sperms are collected from the sperm bank. Sperm bank is the place where the donors donate the sperms and such sperms are kept stored. The donors identity is kept secret and he should also be physically and medically fit person.

CHAPTER 4 : ENVIRONMENTAL MANAGEMENT

Q. Which projects will you run in relation to environmental conservation? How? OR

Write six strategies implemented by you for conservation of the environment.

Ans. Initially, assessment of the environmental problems will be done. The nature and severity of these problems will be understood by detailed study of the same. Then the projects can be undertaken to combat these problems.

(1) Tree plantation is one such easier project that can be undertaken to conserve environment. The further nurturing of the tree will also be our responsibility. While selecting the tree, the local and sturdy varieties will be selected. Such trees can survive in polluted environment too and even under the pressure of urbanization.

(2) Solid waste management is another very important project that should be undertaken by every society, colony or school. Segregation of waste into dry and wet types and then its proper disposal will be taught to all the people in the neighbouring area.

(3) To ban the plastic and make people aware about harmful effects of plastic is another very significant project.

(4) Fossil fuels are non-renewable and polluting. Therefore, their use should be reduced as far as possible. Therefore, using bicycle, or walking down for shorter distances or using public transport systems are the better alternatives. The awareness drive about these facts will be taken up as a project.

(5) To take care of stray animals, provide shelter, feeding endangered birds like sparrows and allowing them to survive with our support is also one of the essential acts to conserve other species.



(6) Attempts will be made for bringing awareness among minds of everyone.

Such small acts can bring about major shift in the attitude of the people. This will certainly help in the environmental conservation.

CHAPTER 7 : INTRODUCTION TO MICROBIOLOGY

(1) Which precautions are necessary for proper decomposition of domestic waste? (March '20; July '22)

Ans. The domestic waste should be properly segregated into biodegradable (wet waste) and non-biodegradable (dry waste). After segregation, these wastes should be stored separately into two different containers. The non-biodegradable substances should be either reused or sent for recycling. The biodegradable substances are decomposed naturally.

The decomposition process can be done at household level too in a pot or a tank. This decomposition will yield a rich manure. The pot should be covered by a thin layer of soil and it should be kept in a dark but airy place.

The non-biodegradable things such as plastic articles, glass pieces, metal objects, unused medicines, e-waste should never be thrown in wet wastes. The toxic substances and the insecticides if added to wet waste, will never allow the natural decomposition process. Therefore, only after taking proper precautions we can aim at proper decomposition of domestic wastes.

(2) There is an oil layer on the water surface of river in your area. What will you do? (March '19) (1 mark)

Ans. If there is an oil layer on the water surface, we shall use hydrocarbonoclastic bacteria like *Pseudomonas* to clean up the oil spill.

(3) How can the soil polluted by acid rain be made fertile again?

Ans. (1) The soil polluted by the acid rain is made fertile again by using bacteria.

(2) *Acidophillium* spp. and *Acidobacillus ferrooxidans* are the bacteria which have the capacity to use sulphuric acid as their energy source.

(3) Since this sulphuric acid present in the acid rain, can be controlled by these bacteria.

(4) In this way, bacteria can control the soil pollution occurring due to acid rain, making the soil fertile again.

CHAPTER 9 : SOCIAL HEALTH

Q. 1. What will you do?

(1) Child of your neighbour is addicted to tobacco chewing.

(July '19; March '20 – '22; Nov. '20)

Ans. The hazards of tobacco chewing will be explained to this child. Different photographs and videos showing the conditions of oral cancer will be shown to this child to persuade him, so that he can stay away from tobacco. This addiction has to be removed, so help of his parents will be taken. They will be told about the child's habit and asked to help him free from his addiction.

(2) Your sister has become incommunicative. She prefers to remain alone. (July '19; March '20–'22)

Ans. The individual who prefers to be incommunicative has lots of thoughts in his/her mind.

If this is the case with sister, she will be taken into confidence and the reason behind this lack of communication will be found out. Most often such persons have depression. So she will not be left alone. Her friends will be invited at home, so that she can converse with them. She should be motivated to mix with her favourite people. She should be encouraged to pursue her hobbies. She should be helped in selecting such work. If nothing changes her, then the help of counsellor should be taken.

(3) You have to use free space around your home for good purpose. (Nov. '20; March '22; July '22)

Ans. The free space around our home can be used to make a small garden. The garden-soil can be bought and spread in this free space. Small saplings can be planted here and nurtured for further growth. Nursery of saplings can also be started in this free space.

The space can also be used for outdoor games. The net for Badminton can be fixed and evening times can be spent in playing the game. Also care will be taken to keep the space clean and without any garbage.



(4) Your friend has developed the hobby of snapping selfies.

(July '19; March '20; Nov. '20)

Ans. The habit of continuously taking selfie is bad. It shows that the friend is constantly thinking of himself only. His self-centredness is to be removed by counselling him. The reason behind this behaviour should also be understood. He should be diverted and motivated to take some other tasks so that his habit can be lessened. Taking selfies is not a hobby. It is a bad habit if someone is repeatedly engaged in it.

(5) Your maid's husband has started drinking and beating her. (Sept. '21)

Ans. We shall try to tell maid's husband about the ill-effects of alcoholism and will help him for de-addiction. We also ask maid to be self-reliant and able to protect herself. In order to stop social, mental and societal problems, help of police can also be taken.

(6) Your younger brother plays games on cellphone while eating. (Sept. '21)

Ans. We will explain him ill-effects of over use of cellphone. We will also tell him that paying attention towards proper nutrition is important. If he does not obey, we will take help of parents.

(7) Your brother studying in Class XII has developed the stress. (March '22; July '22)

Ans. The syllabus for Class XII is vast. If the studies are not taken seriously from the beginning of the academic year, then the stress develops due to the fear of examination and result. Therefore, instead of being stressed, he should practise time management and study schedule. He should think of only one subject at a time. The atmosphere in the house should be maintained happy and tension-free. Everybody in the house should interact with him so that he gets a feeling that he is not alone. He should be convinced, "study is for you and you are not for study".

Q. 2. What are the various ways to minimise mental stress?

OR

Write four methods to reduce stress. (March '25) (2 marks)

Ans. Methods to reduce stress are as follows :

- (1) Photography (2) Reading (3) Cooking (4) Drawing
(5) Sculpturing (6) Rangoli (7) Laughter club (8) Outdoor games, etc.

CHAPTER 10 : DISASTER MANAGEMENT

(1) The building in which you are residing has caught fire on the ground floor. What necessary rescue steps will you take?

(July '19)

Ans. (1) We shall call out for help.

(2) We shall immediately call fire brigade.

(3) We shall try to extinguish fire with the help of other people.

(4) We will give first aid to people who are injured, till the medical help arrives.

(2) Write down the reasons, effects and remedial measures taken for any one disaster seen or heard by you. (March '25)

Ans. Disaster : Flood

(1) **Reasons for floods :** Floods are often caused by heavy rainfall, rapid snow melt or tsunami in coastal areas.

(2) **Effects of flood :**

(a) Loss of human life and property occurs.

(b) The fields get waterlogged.

(c) Spreads water-borne diseases.

(3) **Remedial measures :**

(a) In preventive measures, planting trees to prevent soil erosion, construction of dams to control, river water flow can be done.

(b) In post-Flood measures, providing relief by giving food, medical help to victims of flood, repairing roads, bridges, preventing the spread of diseases through clean drinking water and hygiene awareness can be done.

Note : Students are expected to write the answer based on their own experience.



**Question
Set
13**

DIAGRAM BASED QUESTIONS

[Q. 1 (B), 2 (B), 3, 4 : 1/2/3/5 Marks each]

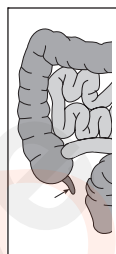
Remember : Diagram based questions are clubbed together in this question set.

CHAPTER 1 : HEREDITY AND EVOLUTION

(1) Identify and define the evidence of evolution shown by the arrow in the given figure. (2 marks)

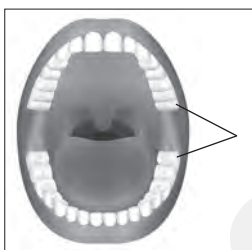
Ans. The evidence of evolution shown in the diagram is appendix which is vestigial organ.

Definition of vestigial organ : The degenerated or underdeveloped organ which cannot perform any function is known as vestigial organ.



(2) Write the name of indicated part in the diagram :

(March '23) (1 mark)



Ans. Wisdom teeth

Human jaw

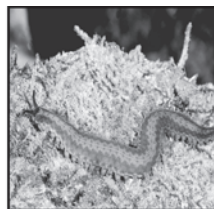
(3) Observe the picture given and discuss the characters observed. (5 marks)



Duck-billed



Lung fish



Peripatus

Ans. The pictures given show connecting links. These are as follows :

(1) **Duck-billed platypus** : This is a connecting link between reptiles and mammals. Like reptiles it lays eggs but like mammals it has mammary glands and hairy skin.

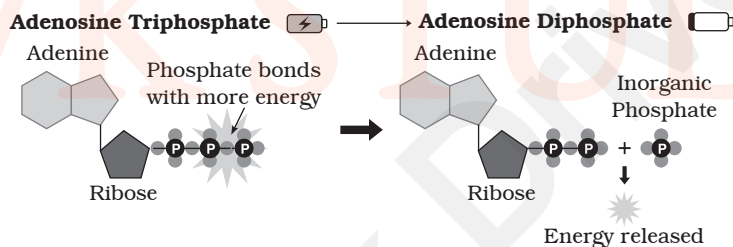
(2) **Lung fish** : Lung fish is a connecting link between fishes and amphibians. Though a fish, it shows lungs for respiration as in amphibian animals.

(3) **Peripatus** : *Peripatus* is the connecting link between Annelida and Arthropoda. It shows characters of both animal phyla. Like annelid worm, it shows segmented body, thin cuticle and parapodia. Like an arthropod, it shows open circulatory system and tracheal system for respiration.

CHAPTER 2 : LIFE PROCESSES IN LIVING ORGANISMS : PART - 1

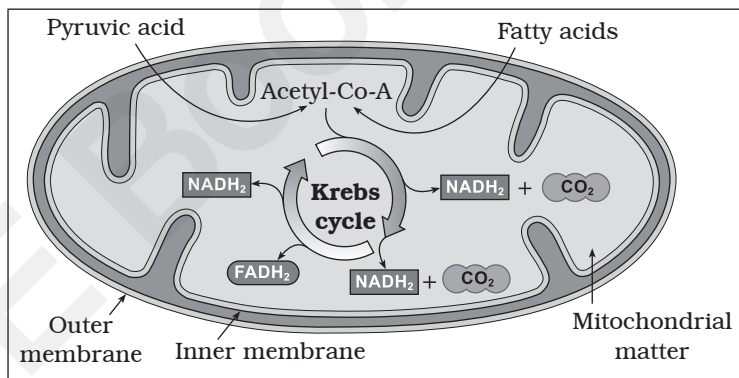
(1) **Sketch and label the diagram to show ATP—the energy currency of the cell.** (2 marks)

Ans.



ATP – the Energy Currency

(2) **With the help of given diagram, answer the following questions :** (2 marks)



Mitochondria and Tri-carboxylic acid cycle

(a) Which co-enzymes are shown in the diagram?

(b) Which chemical reaction takes place in the mitochondria?

Which molecules are produced in this reaction?

OR

Sketch and label mitochondria showing TCA cycle.

Ans. (a) The co-enzymes NADH_2 and FADH_2 are shown in the above diagram.

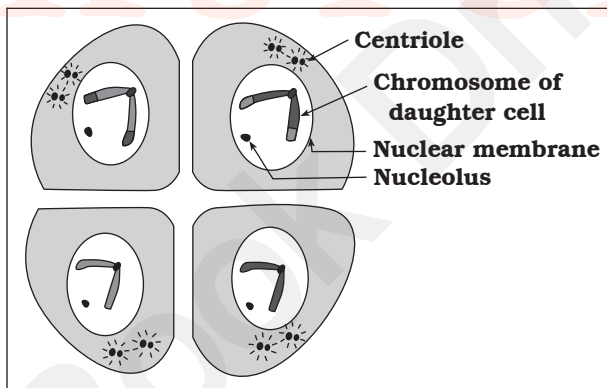
(b) The chemical reaction that takes place in the mitochondria is called Electronic Transport Chain reaction. The molecules of H_2O , carbon dioxide and energy in the form of ATP are produced in this reaction.

(3) Draw a neat and labelled diagram of mitochondria and tri-carboxylic acid cycle. (July '23) (2 marks)

Ans. For this, refer to Q. Set 13, Chapter 2, Q. (2).

(4) Label the diagram below. Which phase of cell division is seen in the diagram given below? (3 marks)

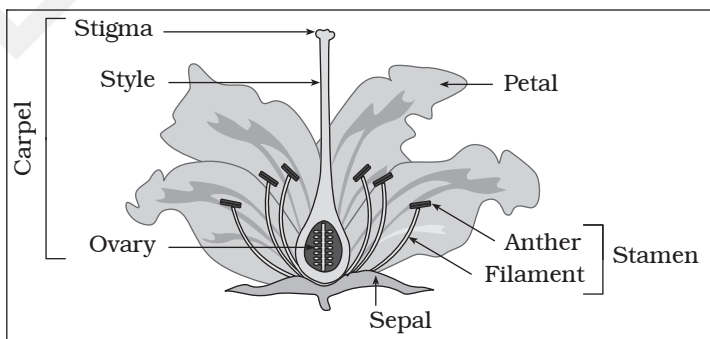
Ans. (Answers are given directly in bold.)



The diagram given above shows Telophase-II of Meiosis-II.

(1) Draw a neat and labelled diagram of 'Parts of Flower'. OR Flower with its sexual reproductive organs. (2 marks)

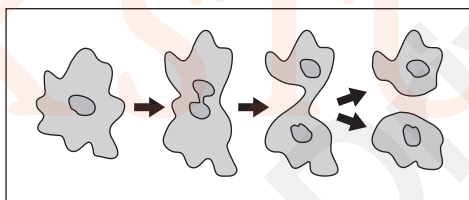
Ans.



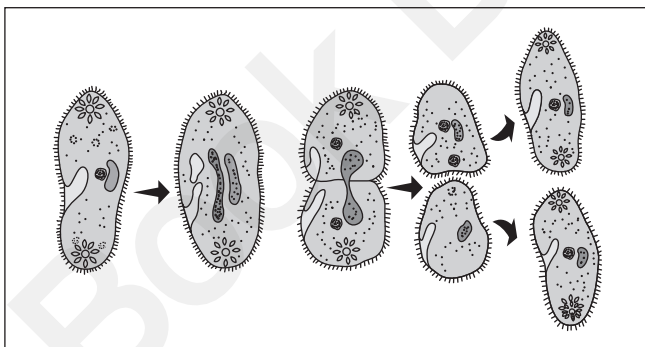
L.S. of a flower showing different parts

(2) Identify the type of asexual reproduction. In which organism do you find it? (5 marks)

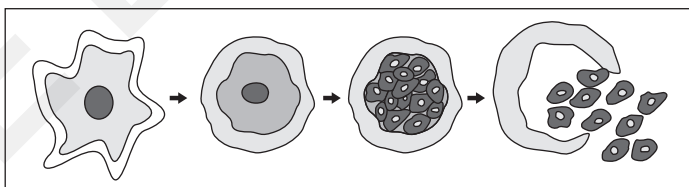
(a)



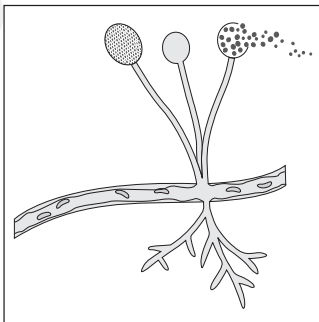
(b)



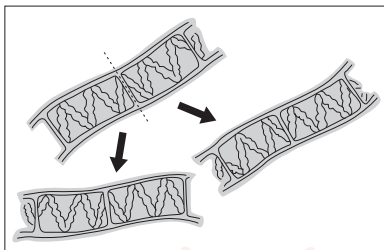
(c)



(d)



(e)

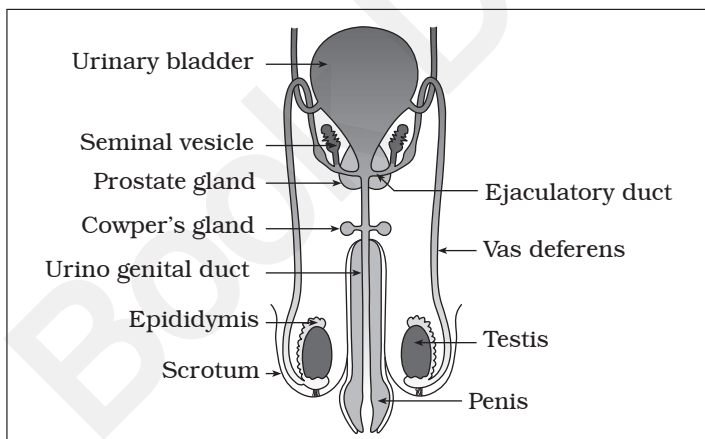


Ans. (a) Simple binary fission – Amoeba **(b)** Transverse binary fission – Paramecium **(c)** Multiple fission – Amoeba **(d)** Spore formation – Mucor **(e)** Fragmentation – Spirogyra.

(3) Sketch and label the given diagram : (2 marks)

Human male reproductive system.

Ans.

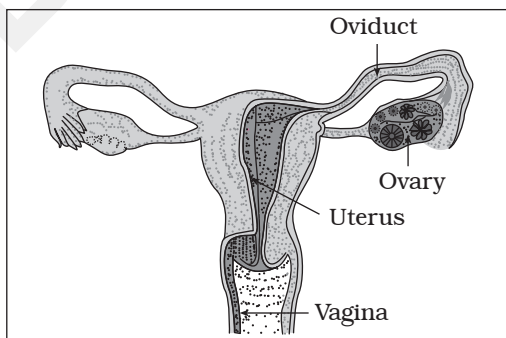


Male reproductive system of human

(4) Draw a neat and labelled diagram of female reproductive system and give the functions of :

(a) Oviduct (b) Ovary (c) Uterus. (July '22) (5 marks)

Ans. For the functions, refer to Q. Set. 6, Chapter 3, Q. 5, 3 and 4.



Human female reproductive system

(5) Answer the following questions in complete sentence based on female reproductive system : (July '23) (5 marks)

- (a) How many immature oocytes are present at the time of birth in the ovary of female foetus?
- (b) What is ovulation?
- (c) What is menopause?
- (d) Draw scientifically correct diagram of female reproductive system.
- (e) Label the following parts :
 - (i) Ovary
 - (ii) Uterus.

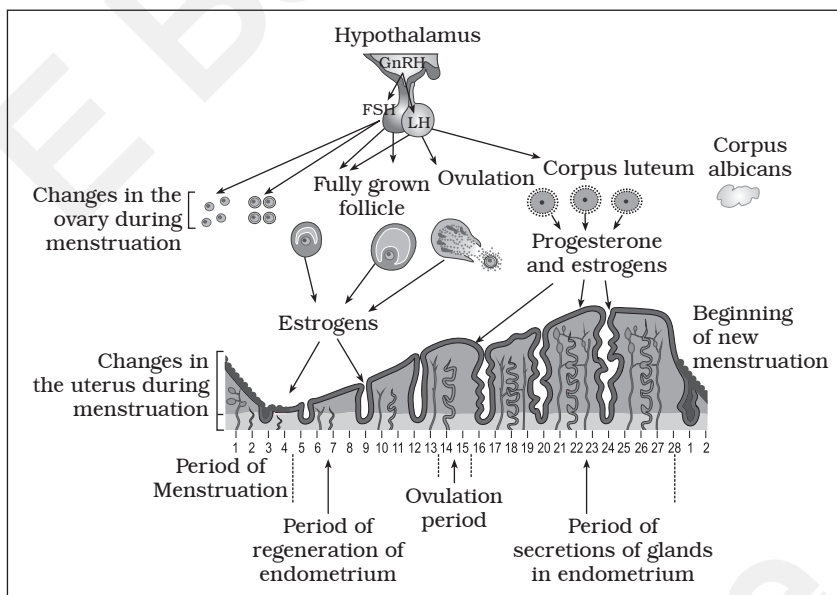
Ans. (a) At the time of birth, there are about 2 to 4 million of immature oocytes in the ovary of female foetus.

(b) and (c) For this, refer to Q. Set 8, Chapter 3, Q. (8), (4).

(d) and (e) For this, refer to Q. Set 13, Chapter 3, Q. 4.

(6) Answer the following questions by observing the diagram of menstrual cycle. (March '23) (5 marks)

- (a) What is the period of menstruation?
- (b) On which day does ovulation occur during menstrual cycle?
- (c) In menstrual cycle which reproductive organs undergo changes?
- (d) Which period is said to be period of regeneration of endometrium?
- (e) Which period is said to be period of secretions of glands in endometrium?



Ans. (a) 1 to 5 days is the period of menstruation.

(b) Ovulation occurs on 14th or 15th day.

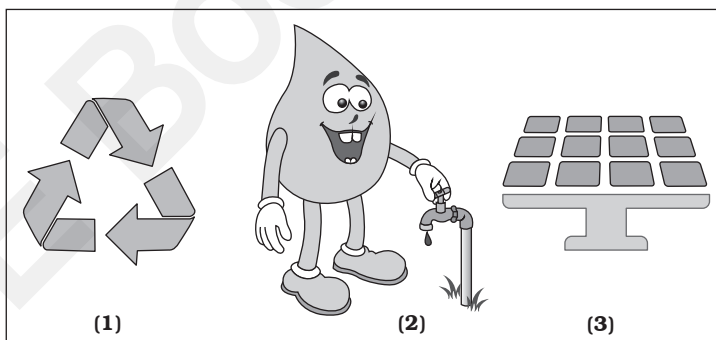
(c) Ovary and uterus undergo changes during menstrual cycle.

(d) In menstrual cycle, days 5 to 14 are period of regeneration of endometrium.

(e) Period of secretions of glands in endometrium is 15 to 28 days.

CHAPTER 4 : ENVIRONMENTAL MANAGEMENT

(1) What are the meanings of the following symbols? Write your role accordingly. (July '19-'22) (1 mark each)



Ans. (1) The first symbol is for giving the message, “Reduce, reuse and recycle”. This is important mantra for the utilization of natural resources.

The second gives the message about ‘Save water’.

The third advocates the use of solar energy.

(2) These symbols inculcate the importance of being eco-friendly. The first symbol is essential to maintain the natural resources by reusing and recycling them. One should reduce the excessive use of resources.

(3) The message about saving water or to make judicious use of water should be spread far and wide because the water problems are rising everywhere.

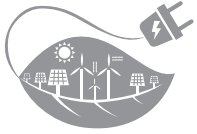
(4) The solar energy is the renewable energy option. By using solar energy, we can replace the polluting and exhaustible fuels.

Such symbols, teach us environment conservation.

(2) Explain the meaning of following symbols A and B.

(1 mark each)

Ans.

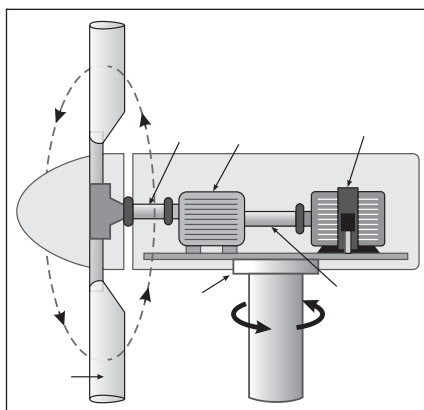
(A) 	The symbol shows types of green energy such as solar energy and wind energy. This symbol promotes use of eco-friendly energy.
(B)  (March '19-'22)	This symbol is giving the message ‘Save water’. Sustainable use of water is necessary for our future.

(1) Observe the given picture and answer the questions :

Questions :

- Identify the given diagram.
- What is the source of energy in this project?
- Why this source is supposed to be eco-friendly?

(March '24) (3 marks)



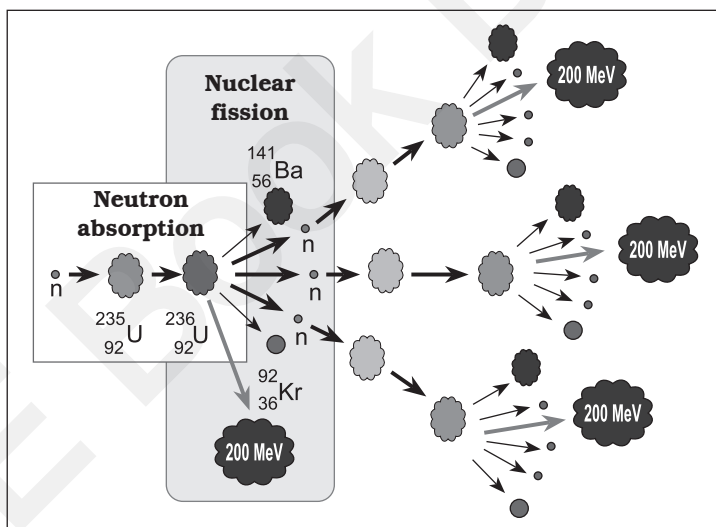
Ans. (a) Schematic of wind mill is shown in the given diagram.

(b) The kinetic energy of wind is the source of energy in this project.

(c) This source is eco-friendly because it does not cause pollution. Wind energy is green energy which is non-exhaustible, perpetual and sustainable.

(2) Observe the figure and answer the questions given below.

(Sept. '21) (3 marks)



Nuclear fission

(a) Which reaction is in the figure?

(b) Where is this process used?

(c) Which element is used in this reaction?

OR

Identify the process shown in figure and name it.

(March '19) (1 mark)

Ans. (a) The reaction shown is nuclear fission is used in nuclear power plants where electricity is generated.

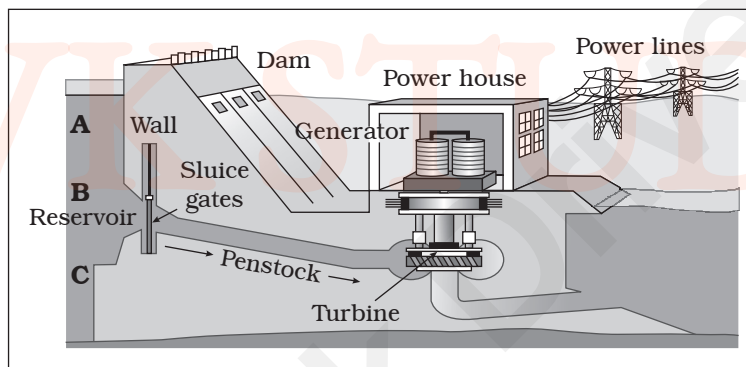
(c) Uranium or chain reaction.

(b) This re-235 is used in the nuclear fission reactions. OR

The above figure shows nuclear fission chain reaction of Uranium-236.

(3) Answer the following questions from figure.

(March '23) (3 marks)



(i) With reference to point B, potential energy of how much water reservoir in the dam will be converted into kinetic energy?

(ii) What will be the effect on electricity generation, if the channel taking water to turbine starts at point A?

(iii) What will be the effect on electricity generation, if the channel taking water to turbine starts at point C?

Ans. (i) When the sluice gate at point B is opened, the water from reservoir will start flowing. The potential energy of the stored water will become kinetic energy of the quantity of water that is let out through the sluice gates.

(ii) If the channel taking water to turbine starts at point A, then the water will flow with a greater speed. Since point A is at height,

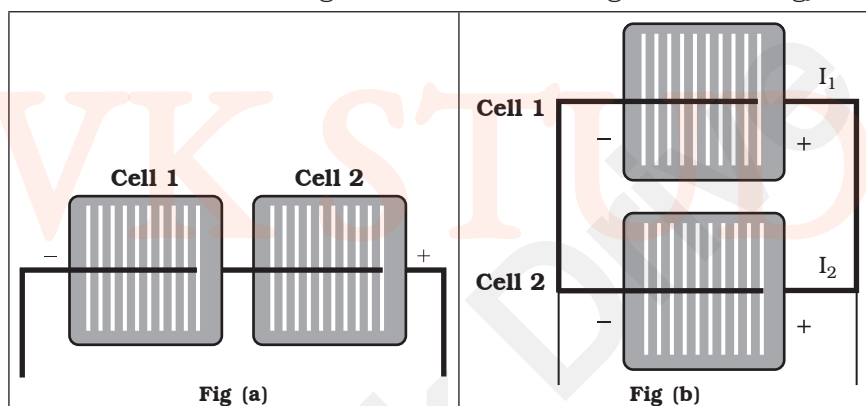


water will acquire speed. This will result into more efficient rotation of the blades of turbine. The electricity generation will thus become more efficient.

(iii) If the channel taking water to turbine starts at point C, it will affect the electricity generation adversely. Point C is on the lower height as compared to the channel that carries water to the turbine. The flow of the water thus will be affected resulting into improper rotation of blades of turbine. This will certainly affect the electricity generation.

(4) Observe the connections of cells shown in the following images : **(3 marks)**

- (i) Which connection will give maximum potential difference?
- (ii) Give one advantage and one disadvantage of this energy.



Ans. (i) The solar cells shown in the diagram are connected in series. This gives maximum potential difference.

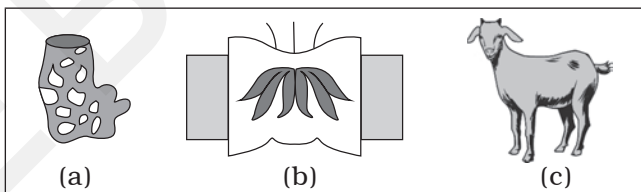
(ii) Advantage of solar energy : Solar energy is eco-friendly which does not create pollution. It is boundless source.

Disadvantage of solar energy : Solar energy is available only when sun is in the sky. Therefore, it has to be stored in batteries.

CHAPTER 6 : ANIMAL CLASSIFICATION

(1) Identify the types of body symmetry and give one example of each part from the given figure. (July '23) (3 marks) OR

Identify and explain from the given figure different types of symmetry. (Nov. '20) (3 marks)



Ans. (a) Asymmetrical body : In asymmetrical body, there is no imaginary axis that can pass through body and divide it into two equal halves, e.g. Amoeba, Sponge, Paramoecium.

(b) Radial symmetry : In this type of body, if imaginary cut passes through central axis, in any plane of body, then it can give two equal halves. e.g. starfish, sea anemone, sea cucumber.

(c) Bilateral symmetry : In such type of symmetry, there is only one imaginary axis that can pass through the body dividing it into two equal halves. e.g. fishes, frog, birds, mammals.

(2) (a) Identify the class of the animals shown in the figure :



(b) Write two characteristics of that class.

(July '22) (3 marks)

Ans. (a) Animals shown in the figure are frog and toad. They belong to class Amphibia.

(b) (i) Amphibians live in water as well as on land by breathing in both environments. (ii) The exoskeleton is absent and the skin is soft, slimy and moist. (iii) Body is divided into head and trunk. Neck is absent. (iv) Digits are without claws. Two pairs of limbs present.

(3) Name the locomotory organ of animal shown in the given figure : (Nov. '20) **(1 mark)**



Ans. Tube feet.

(4) Observe the following diagram. Write the answer of the following questions : (March '20) (3 marks)

(a) To which phylum does the animal included in the diagram belong?

(b) What is the exoskeleton made up of?

(c) What is the symmetry?



Ans. (a) This animal is cockroach. It belongs to phylum Arthropoda and class Insecta.

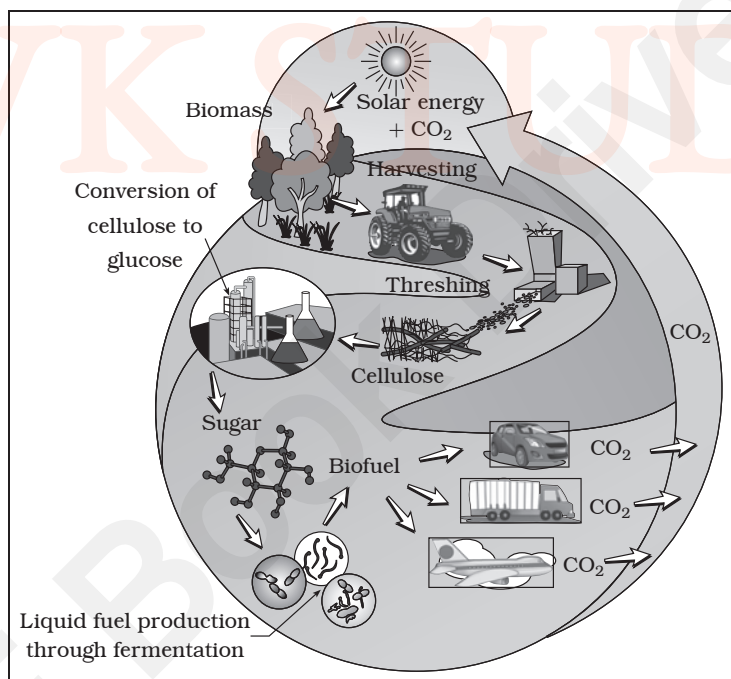
(b) Its exoskeleton is made up of chitin.

(c) It has bilateral symmetry.

CHAPTER 7 : INTRODUCTION TO MICROBIOLOGY

(1) Observe the diagram and answer the questions :

(March '22) (3 marks)



Biofuel production

(a) Which process is shown in the above given picture?

(b) Give two examples for liquid fuels and solid fuels obtained by this process.

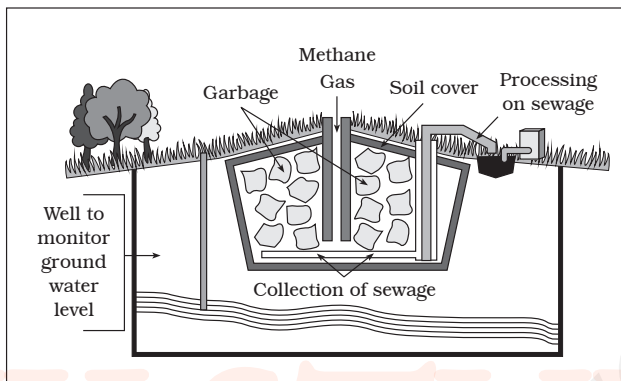
Ans. (a) The diagram shows process of production of biofuels.

(b) (i) Liquid fuels : Vegetable oil, alcohol.

(ii) Solid fuels : Coal, dung, crop, residue.

(2) Observe the diagram and answer the following questions :

(March '19) (3 marks)



Modern landfill site

(a) Identify the process shown in the figure.

(b) Explain the process in short.

Ans. (a) The figure shows modern landfill site where microbial biodegradation process is carried out.

(b) (1) In the landfilling sites, the degradable wastes are transferred.

(2) Large pits are dug in open spaces which are lined with plastic sheets. Therefore, the leaching of toxic and harmful materials is avoided to reduce the chance of soil pollution due to leachates.

(3) Compressed waste is put in the pit and is covered with layers of soil, saw dust, leafy waste.

(4) Specific biochemical substances are added for speedy decomposition. Bioreactors which are mixtures of bacteria are mixed at some places. Soil microbes and other top layers decompose the waste.

(5) Soil slurry is used to seal the pits completely.

(6) After a certain period, best quality compost is formed. Such landfilling sites can be reused after removal of compost.

(1) Rearrange and write proper alphabetical sequence only and complete the given blanks : (July '23) (3 marks)

Inject virus in the plant



Entire plant is developed from the leaf pieces containing genes of human pathogen.

(A)

(B)

Consumption of raw potatoes helps to develop immunity against that pathogen.

(C)



Virus infects the pieces of potato leaves.

(D)

(E)

Ans. (1) Correct arrangement of the given diagrams is :

B → E → D → A → C

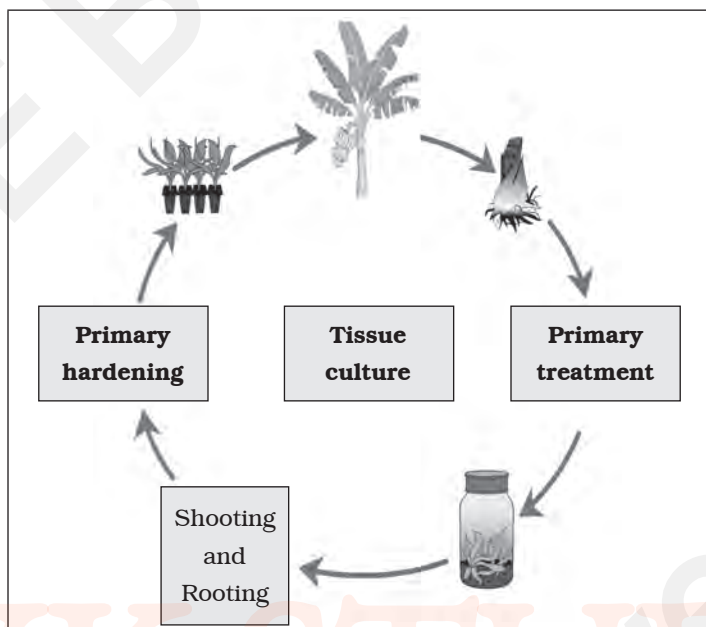
(2) Blank places :

(B) : Isolation of desired gene from human beings

(E) : Transgenic plant virus

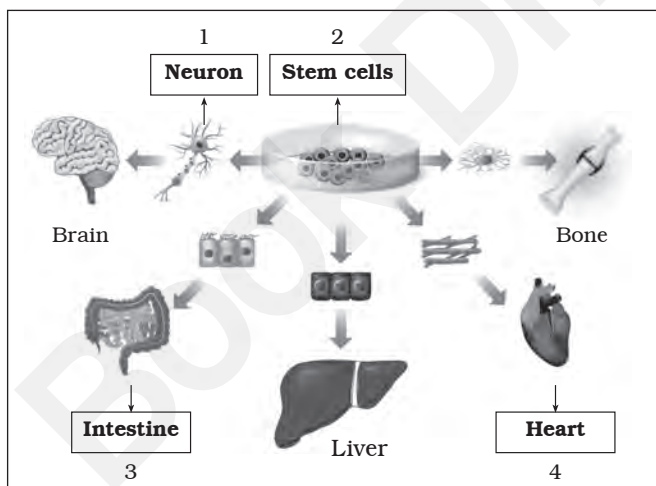
(2) Complete the diagram showing various processes in tissue culture : (March '23) (3 marks)

Ans. (Answers are given directly in bold.)



(3) Observe the given diagram and write proper names in the blocks. (July '22) (2 marks)

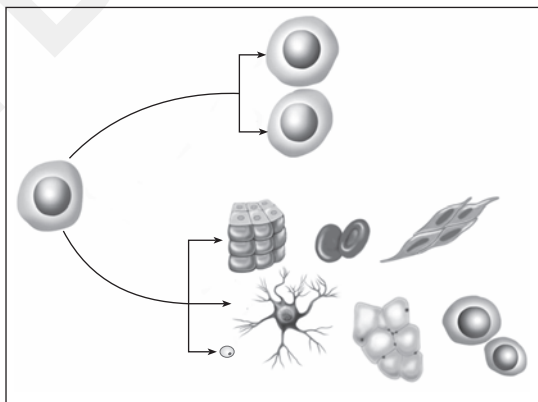
Ans. (Answers are given directly in bold.)



(4) Answer the following questions after observing the diagram : (March '22) (3 marks)

- Which process is shown in the given diagram?
- What is the importance of this process?

(c) Which organs can be transplanted in this process?



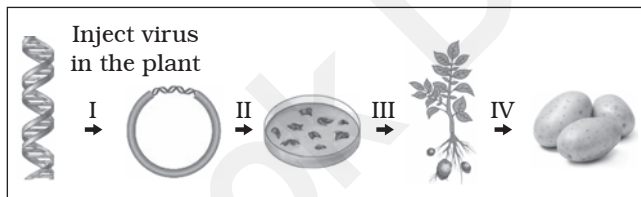
Ans. (a) The process shown in the diagram is stem cell therapy.

(b) Stem cells are used in treatment of diseases such as diabetes, myocardial infarction, Parkinson's disease, etc. to replace the damaged cells with new functioning cells. The stem cell therapy can be used in organ transplantation also, to save a life of a patient.

(c) Kidney and liver can be replaced by using stem cell therapy technique.

(5) Observe the figure and answer the following questions :

(Nov. '20) (5 marks)



(a) Identify the process given in the above figure.

(b) Write the importance of above process.

(c) Write any two benefits of Biotechnology.

Ans. (a) The process of making transgenic potatoes is shown in the figure.

(b) (1) These potatoes act against bacteria causing cholera e.g. vibrio cholerae. Also they are effective against ***Escherichia coli***.

(2) Consumption of raw potatoes produces immunity against cholera and colitis – disease caused by *Escherichia coli*.

(c) (1) Per hectare yield is increased though there are limitation in crop agriculture.

(2) Expenses of disease control are minimised due to development of resistant varieties.

CHAPTER 9 : SOCIAL HEALTH

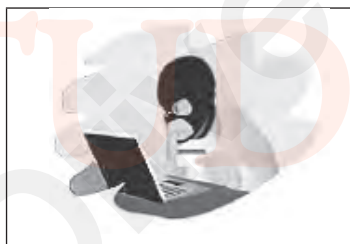
(1) Observe the following pictures and answer the following :

(July '23) (2 marks)

(a) Whether the incidence shown in the picture is rational or irrational? Justify your answer.



(b) Identify the picture. What is the punishment given to the person shown in the picture?



Ans. (a) In one way, the picture looks proper as the beggar may be causing inconvenience to the people in that house. He is unhygienic and may spread the infection.

But from the humanitarian point of view, he may be needing help. He may be starving. He may be sick. In such a case, he should be given food and help.

(b) The picture shown is of cybercrime of some kind, such as hacking of information or cheating the consumers.

The criminal should be punished according to the law against cybercrime. There is provision of imprisonment for three years and a fine of up to 5 lakh rupees.

(2) Observe the pictures and answer the questions.

(March '19) (3 marks)

(a) Playing games on mobile while eating is right or wrong. Justify.

Ans. The boy taking his lunch is shown in the adjoining picture. He is busy with his mobile while having his food. His nutrition may affect due to such behaviour.



(b) What do you conclude from the adjacent picture?

Ans. Cigarette contains carcinogenic nicotine. It should never be smoked. Similarly, always stay away from addictions such as drugs, alcohol, gutkha, etc. The pictures give message for control of addictions.



(c) Observe the adjacent picture and state what can be the outcome.

Ans. In the picture, a young man is taking selfie while standing in the busy road. He is not aware of the approaching car too. This may cause an accident.



(3) Identify the picture and name it : (March '22) (1 mark)



Ans. Laughter club

(1) Observe the following picture and write any two effects of this disaster :
(March '25) (2 marks)



Ans. The disaster shown in pictures is overflowing due to heavy rain. Effects of this disaster are as follows :

- (1) Due to overflowing, bridges and roads are affected.
- (2) It causes loss of human life and property.

(2) The following are some pictures of disasters. Which preliminary actions you will take during those disasters?

(Sept. '21) (3 marks)



(a)



(b)



(c)

Ans. The given pictures show three disasters. The first and second are man-made disasters while the third is a natural disaster.

(a) The picture shows leakage of gas cylinder. The preliminary action to be taken is to put a damp cloth over a cylinder and move out of the vicinity of such cylinder.

(b) The picture is showing flight among the school children. We should immediately call teachers or seniors to intervene and stop fight.

(c) The picture is showing excessive rain and flooding. We have to move immediately out of such low-lying areas and go to safer place which is on heights, where flood water will not reach.

(3) Following are some pictures of disasters. Which precautions would you take during these disasters?

(July '22) (3 marks)



(a)



(b)



(c)

Various disasters

Ans. The pictures shown above are showing (a) earthquake, (b) fire and (c) snake bite respectively.

In the above disasters, the initial precautions to be taken are as follows :

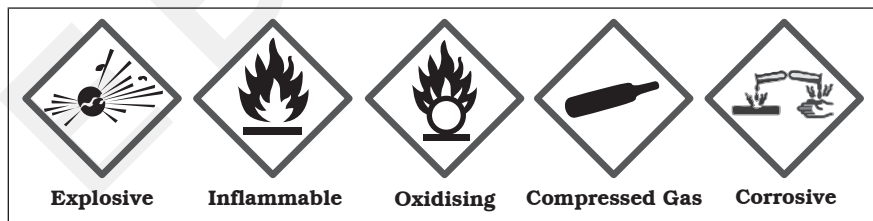
(a) Earthquake : In case of earthquake, one should immediately come out of house and stand in the open ground. If this is not possible, one has to go below table or any other cover to protect from head injury. Switch off the power supply. If in journey, stay inside the vehicle.

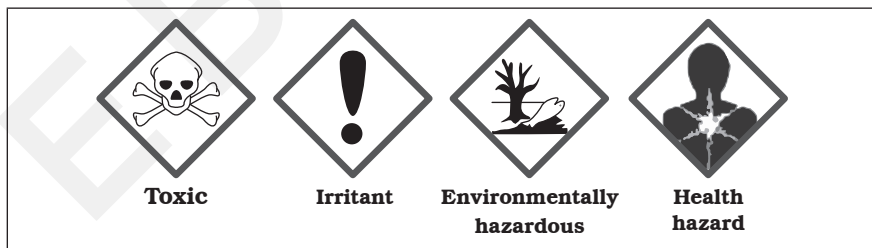
(b) Fire : First and foremost is to save ourselves from fire. Then one can help others in rescue operations. Help others to extinguish fire. Call the fire department for immediate action.

(c) Snake bite : The tourniquet should be tied in the region above the snake bite, so that the venom, should not rise and reach vital organs. The wound should be made near the bite-wound so that the blood will ooze out and some venom can automatically flow out. Though these are first-aid measures, the victim should be rushed to a qualified doctor for an injection of antivenin. Victim should be given psychological support.

(4) Some symbols are given below. Explain those symbols. Which disasters may occur if those symbols are ignored?

(1 mark each)





Ans. The above signs are warning symbols which should never be ignored.

The meaning of each is given below. They are giving warnings about explosive, inflammable, oxidising, compressed gas, corrosive, toxic, irritant, environmentally hazardous and health hazard.

(1) **Explosive** : Some materials are explosive. If explosion occurs, there would be a major disaster causing great loss of life and property.

(2) **Inflammable** : The inflammable materials can also catch fire easily.

(3) **Oxidising** : Some chemical substances are oxidising. They carry out chemical reactions with a rapid speed. e.g. If potassium permanganate falls on the cloth, it starts the reaction on its C-C bonds. Due to such property of carrying out reactions, the cloths may catch fire.

(4) **Compressed** : Compressed substances are filled under pressure in some containers. If mishandled, they can come out of the container by bursting it open.

(5) **Corrosive** : The corrosive substances are very reactive. The mere touch or smell of corrosive substances can cause destruction of skin, eyes, respiratory passages, digestive organs, etc. rapidly.

(6) **Toxic** : Tasting or smelling a toxic substance can lead to death.

(7) **Irritant** : When skin, eyes, nasal mucosa and skin comes in contact with the irritant substance, it can cause harmful reaction.

(8) **Environmentally hazardous** : Many substances cause harm to the environment due to their toxicity. By them air, water or soil can be polluted. Pollution can cause harm to human health.

(9) **Health hazard** : Materials marked with health hazard can cause severe toxicity; therefore they should be distanced from us.

**Question
Set
14**

**CONCEPT DIAGRAMS, TABLES AND
CHART-BASED QUESTIONS**

[Q. 2 (B), 3 or 4 : 2, 3 or 5 Marks each]

Remember :

☛ In the questions based on charts and tables, following subtypes can be asked :

(1) Draw a chart/table (2) Complete the chart/table (3) Fill in the blanks in the given chart/table (4) Correct the incorrect chart given. (5) Making a concept diagram. While making concept diagram take into account the correctness of figure and the interrelationships between the different components in the concept.

Such questions can be asked under Q. 2 (B) or Q. 3 or Q. 4.

Q. Complete the following chart :

CHAPTER 1 : HEREDITY AND EVOLUTION

(1) Complete the following diagram : (Sept. '21) (2 marks)

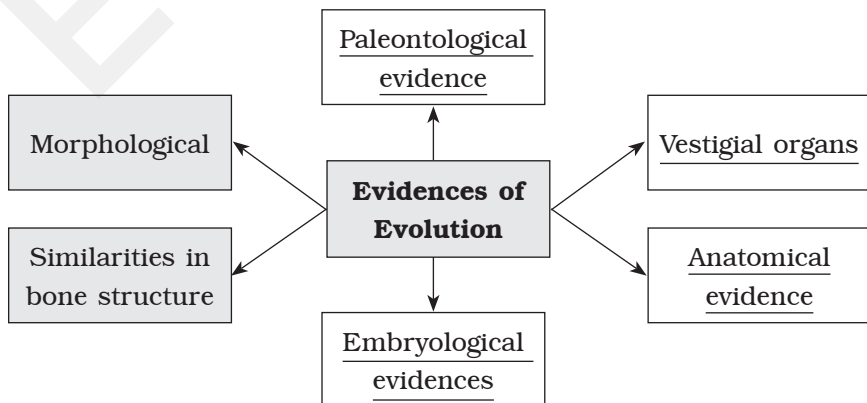
Ans. (Answers are directly given and underlined.)

Human hand		Cat's/Bull's <u>foreleg</u>
	Anatomical evidences	
<u>Flipper of Whale</u>		Patagium of Bat

(2) Complete the following diagram :

(March '22; July '22) (2 marks)

Ans. (Answers are directly given and underlined.)



CHAPTER 2 : LIFE PROCESSES IN LIVING ORGANISMS : PART - 1

(1) Complete the given chart : (July '23)

(3 marks)

Ans. (Answers are directly given and underlined.)

Sr. No.	Body Constituents	Proteins/Amino acids
1.	Blood	<u>Haemoglobin/Antibodies</u>
2.	<u>Bones</u>	Ossein
3.	Skin	<u>Melanin/Keratin</u>
4.	<u>Pancreas</u>	Insulin
5.	Muscle	<u>Actin/Myosin</u>
6.	<u>Cell</u>	Various proteins of cell membrane

(2) Complete the chart : (March '23)

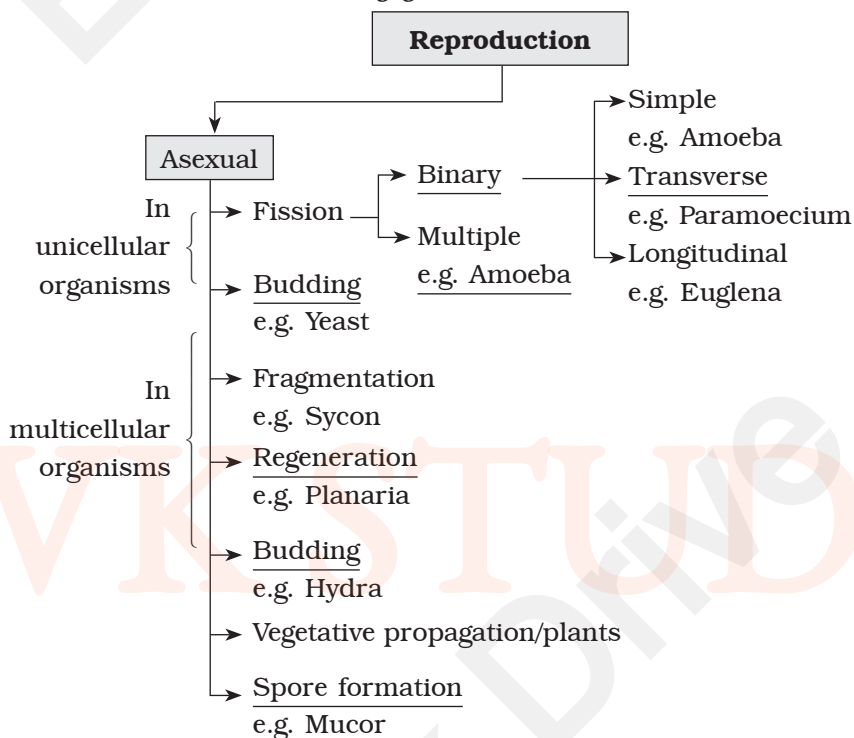
(3 marks)

Ans. (Answers are directly given and underlined.)

Sr. No.	Proteins	Name of the organs
1.	Melanin, Keratin	<u>Skin</u>
2.	<u>Insulin, Trypsin</u>	Pancreas
3.	Haemoglobin, antibodies	<u>Blood</u>

Q Complete a concept map to show different methods of asexual reproduction : (3 marks)

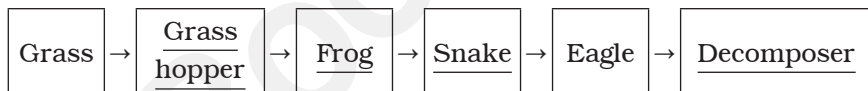
Ans. (Answers are directly given and underlined.)



CHAPTER 4 : ENVIRONMENTAL MANAGEMENT

(1) Complete the following food chain : (March '25) (2 marks)

Ans. (Answers are directly given and underlined.)



(2) Complete the flow chart. Colour of the pages in Red List of Endangered species : (March '24) (1 mark)

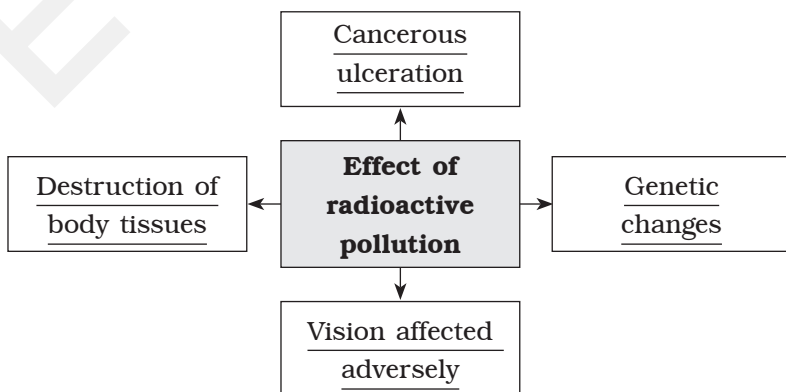
Ans. (Answers are directly given and underlined.)



(3) Complete the following chart :

(March '20; July '23) (2 marks)

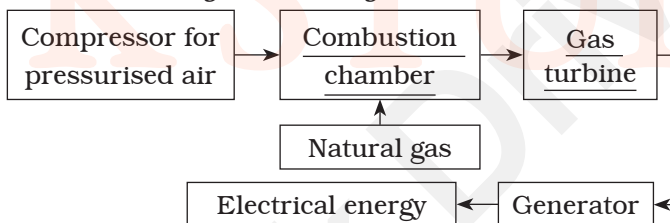
Ans. (Answers are directly given and underlined.)



CHAPTER 5 : TOWARDS GREEN ENERGY

(1) Complete the given flow chart for electrical energy production based on energy of natural gas. (2 marks)

Ans. (Answers are given directly and underlined.)



(2) Make a table based on forms of energy and corresponding devices. (March '20) (3 marks)

Ans. (Answers are directly given and underlined.)

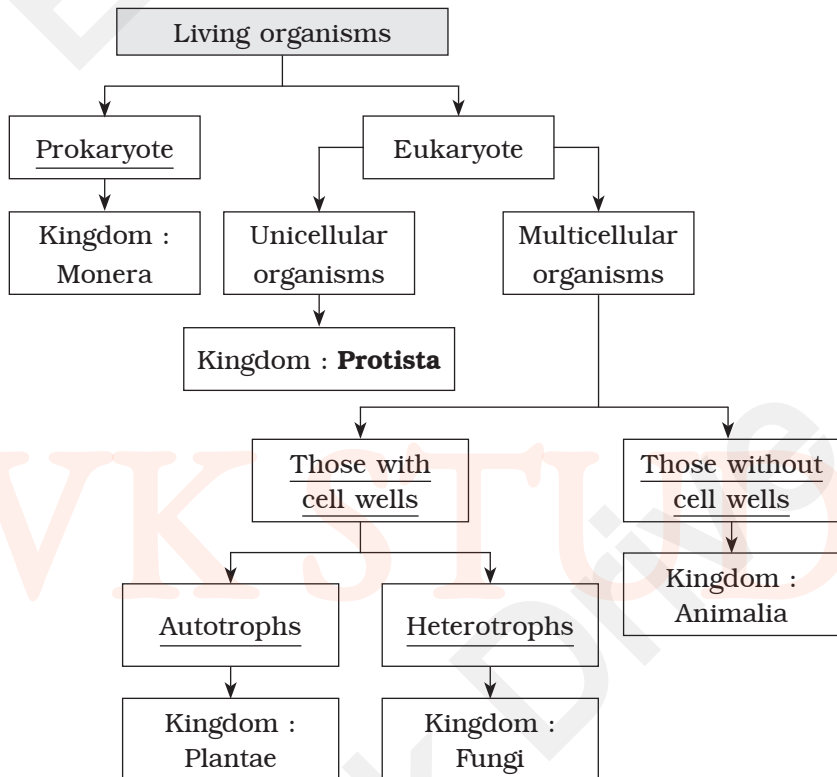
Forms of energy	Devices based on this type of energy
(1) Electric	Electric iron, Geyser, Heater, Oven, Refrigerator, Fans, Lights, Elevator.
(2) Mechanical	Sewing machine, Car, Bicycle, Different machines.
(3) Thermal	Chulha, Furnace, Steam engine.
(4) Solar	Solar cooker, Solar heater.



CHAPTER 6 : ANIMAL CLASSIFICATION

Q. Complete the chart by taking into consideration the criteria for classification : (3 marks)

Ans. (Answers are directly given and underlined.)



CHAPTER 7 : INTRODUCTION TO MICROBIOLOGY

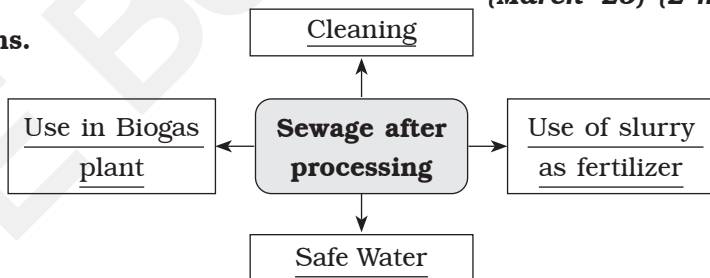
(1) Complete the following conceptual picture :

(Answers are directly given and underlined.)

(i) Complete the following conceptual picture :

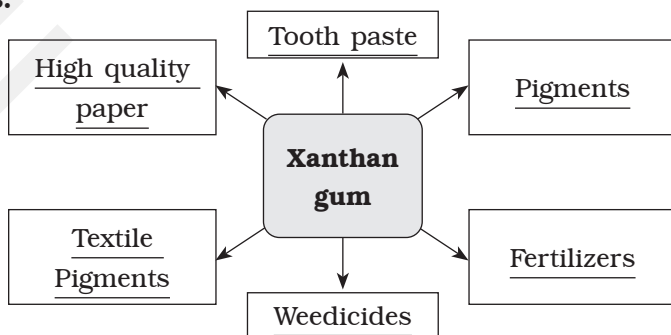
(March '25) (2 marks)

Ans.



(ii) Complete the following conceptual picture with respect to its uses : (3 marks)

Ans.



(2) Complete the following table : (July '19) (2 marks)

Ans. (Answers are directly given and underlined.)

Source	Microbe	Amino acid
(1) Sugar molasses and salt	<u>Aspergillus niger</u>	Citric acid
(2) <u>Molasses, corn steep liquor</u>	<u>Lactobacillus delbrueckii</u>	<u>Lactic acid</u>
(3) Corn steep liquor	<u>Aspergillus itaconius</u>	<u>Itaconic acid</u>

(3) Complete the chart : (Nov. '20) (3 marks)

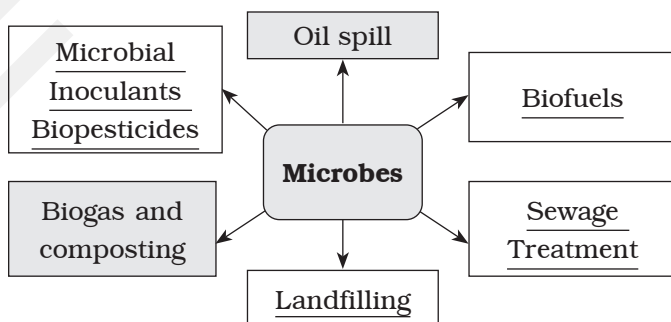
Ans. (Answers are directly given and underlined.)

Substance obtained by microbial processing	Roles
(1) Citric acid	<u>To impart acidity</u>
(2) Ascorbic acid	<u>Antioxidants, vitamins</u>
(3) Beta carotene	<u>Edible colours</u>
(4) Glycolipid	<u>Emulsifiers</u>
(5) Vanillin	<u>Essence</u>
(6) Xylitol	<u>Artificial sweetener (low caloric)</u>

(4) Complete the following conceptual picture :

(March '20) (2 marks)

Ans. (Answers are directly given and underlined.)

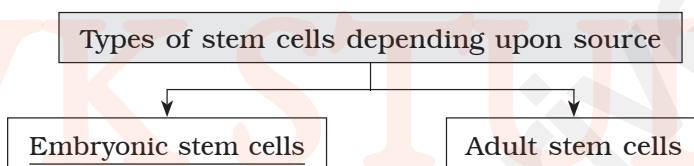


CHAPTER 8 : CELL BIOLOGY AND BIOTECHNOLOGY

(1) Complete the given flow chart : (July '23)

(1 mark)

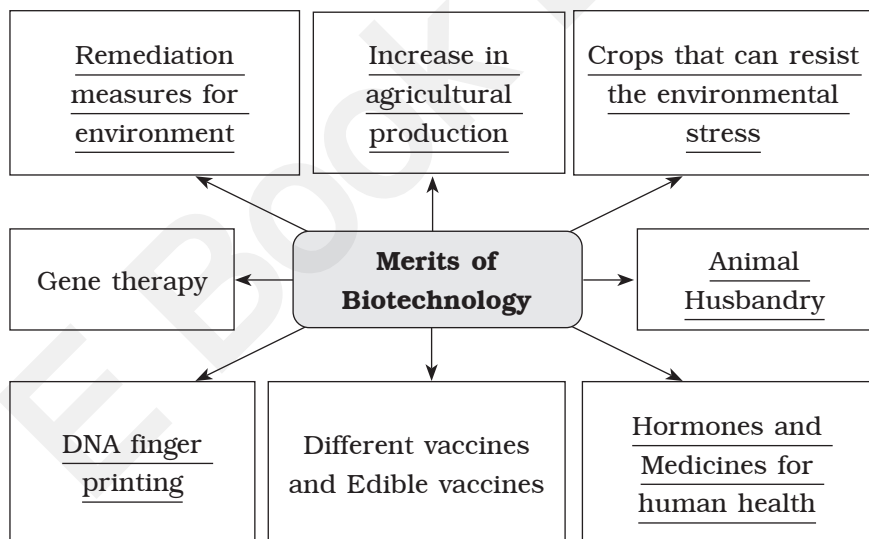
Ans. (Answers are directly given and underlined.)



(2) Complete the following chart :

(3 marks)

Ans. (Answers are directly given and underlined.)



(3) Complete the following table : (July '19) (2 marks)

Ans. (Answers are directly given and underlined.)

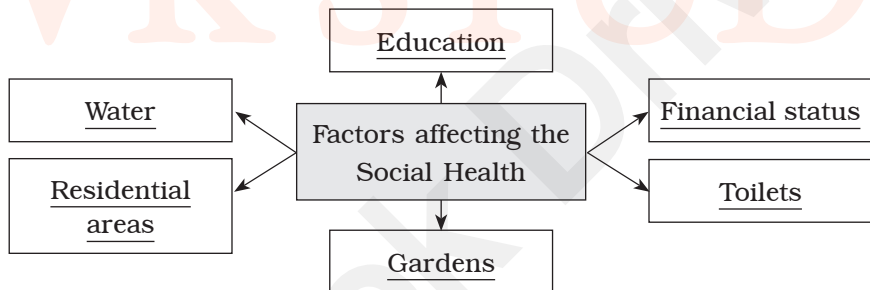
Plant/Microbes	Functions
(1) Pteris vitata	<u>Absorbs arsenic from soil</u>
(2) Pseudomonas	<u>Separates hydrocarbon and oil from water and soil</u>
(3) <u>Sunflower</u>	Absorption of uranium and arsenic
(4) <u>Deinococcus radiodurans</u>	Absorption of radiations of nuclear waste

CHAPTER 9 : SOCIAL HEALTH

(1) Complete the following chart :

(July '23) (3 marks); (March '24) (2 marks)

Ans. (Answers are directly given and underlined.)



(2) Solve the following cross word : (March '23) (5 marks)

1. Continuous consumption of alcoholic and tobacco materials.

(March '25)

2. This app may cause the cybercrimes. **(March '25)**

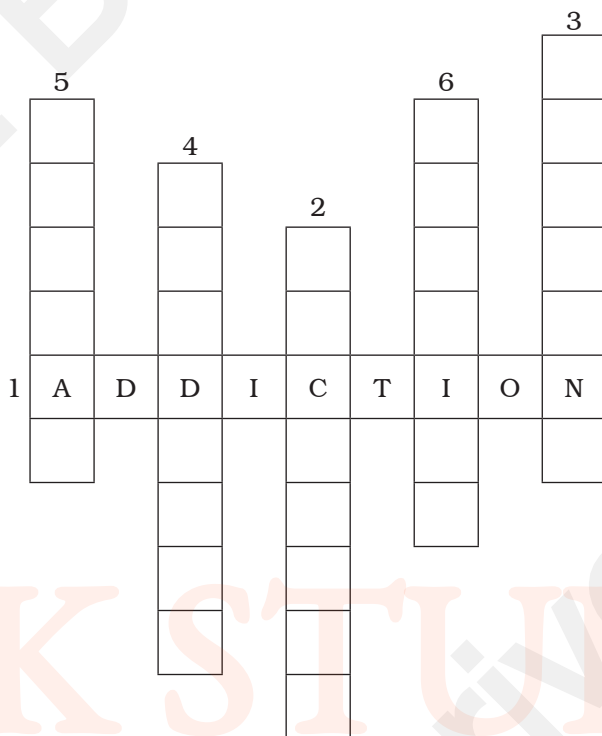
3. A remedy to resolve stress. **(March '25)**

4. Requirement for stress free life. **(March '25)**

5. Various factors affect health.

6. Art of preparing food items.

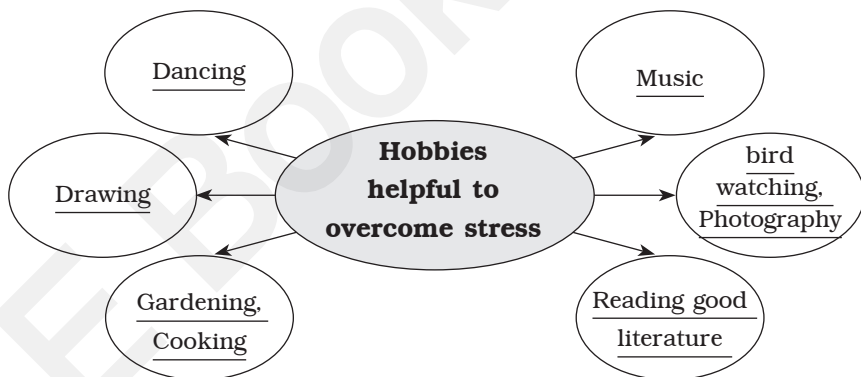




Ans. (1) Addiction (2) Facebook (3) Singing (4) Good food
(5) Social (6) Cooking.

(3) Complete the following chart. (March '24) (3 marks)

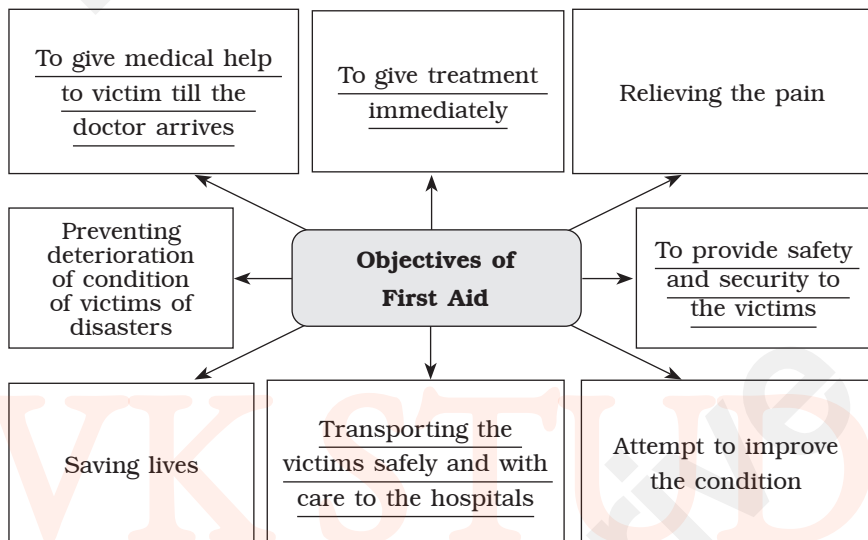
Ans. (Answers are directly given and underlined.)



(1) Complete the chart :

Complete the chart as per the objectives of the first aid :

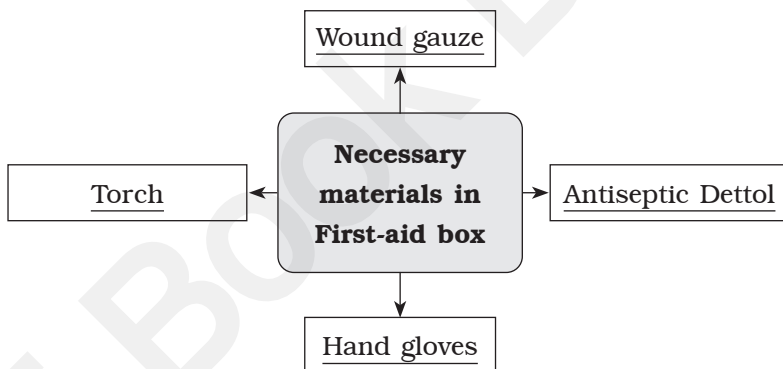
Ans. (Answers are directly given and underlined.)



(2) Complete the diagram :

(Nov. '20) (2 marks)

Ans. (Answers are directly given and underlined.)



**Question
Set
15**

COMPLETE THE PARAGRAPH

[Q. 3 : 3 Marks each]

Remember :

- ☛ In this question set, some paragraphs are given which have blanks. The suitable words have to be inserted in the blanks and the paragraph has to be completed.
- ☛ The words will be provided in the bracket. While answering, write the paragraph completely and underline the answers.
Any six blank spaces should be correctly filled to procure 3 marks.

Q. Complete the paragraph using appropriate words given in the bracket :

CHAPTER 1 : HEREDITY AND EVOLUTION

Q. (*m-RNA, codon, Har Govind Khorana, 50, Anand Chakravarti, Nobel, 20, nucleotides*)

The code for each amino acid consists of three It is called triplet Dr., a scientist of Indian origin has made an important contribution in discovery of triplet codons for amino acids. He was awarded prize for this discovery in 1968. Each is made up of such thousands triplet codons.

Ans. The code for each amino acid consists of three nucleotides. It is called triplet codon. Dr. Har Govind Khorana, a scientist of Indian origin has made an important contribution in discovery of triplet codons for 20 amino acids. He was awarded Noble prize for his discovery in 1968. Each m-RNA is made up of such thousands triplet codons.

CHAPTER 2 : LIFE PROCESSES IN LIVING ORGANISMS : PART - 1

Q. (*gamete, crossing over, haploid, Meiosis-II, meiosis-I, diploid*)
..... is just like mitosis. In this stage, the two haploid daughter cells formed in undergo division by separation of recombined

sister chromatids and four daughter cells are formed. Process of production and spore formation occurs by meiosis. In this type of cell division, four haploid (n) daughter cells are formed from one cell. During this cell division, occurs between the homologous chromosomes.

Ans. Meiosis-II is just like mitosis. In this stage, the two haploid daughter cells formed in meiosis-I undergo division by separation of recombined sister chromatids and four haploid daughter cells are formed. Process of gamete production and spore formation occurs by meiosis. In this type of cell division, four haploid (n) daughter cells are formed from one diploid cell. During this cell division, crossing over occurs between the homologous chromosomes.

CHAPTER 3 : LIFE PROCESSES IN LIVING ORGANISMS : PART - 2

9. (*Luteinizing hormone, endometrium of uterus, follicle stimulating hormone, estrogen, progesterone, corpus luteum*)

Growth of follicles present in the ovary occurs under the effect of This follicle secretes estrogen. Ovarian follicle along with oocyte grows/regenerates under the effect of estrogen. Under the effect of, fully grown up follicle bursts, ovulation occurs and is formed from remaining part of follicle. It secretes and Under the effect of these hormones, glands of are activated and it becomes ready for implantation.

Ans. Growth of follicles present in the ovary occurs under the effect of follicle stimulating hormone. This follicle secretes estrogen. Ovarian follicle along with oocyte grows/regenerates under the effect of estrogen. Under the effect of Luteinizing hormone, fully grown up follicle bursts, ovulation occurs and corpus luteum is formed from remaining part of follicle. It secretes estrogen and progesterone. Under the effect of these hormones, glands of endometrium of uterus are activated and it becomes ready for implantation.



CHAPTER 4 : ENVIRONMENTAL MANAGEMENT

9. (conserving, pledge, biodiversity, judicious, wildlife, diversity, destruction)

I am aware that the on the Earth is for the existence of me, my family and the entire mankind. I am aware about the responsibility of and protecting the rich diversity. I am aware about the fast-declining number of, plants and animals. I am accepting the responsibility of use of natural resources and management of I for adopting the following principles for happy and satisfactory life of all organisms on the Earth.

Ans. I am aware that the diversity on the Earth is for the existence of me, my family and the entire mankind. I am aware about the responsibility of conserving and protecting the rich diversity. I am aware about the fast declining number of wildlife, plants and animals. I am accepting the responsibility of judicious use of natural resources and management of biodiversity. I pledge for adopting the following principles for happy and satisfactory life of all organisms on the Earth.

CHAPTER 5 : TOWARDS GREEN ENERGY

9. (marginal, array, string, current, power station, potential difference)

Many solar panels are connected in series and in parallel to generate required and Solar cell is the basic unit in solar electric plant. Many solar cells come together to form a solar panel. Many solar panels connected in series form a solar, and, many solar strings connected in parallel form a solar As we can obtain as much electrical power as needed, they are used in applications which need power (e.g. calculators that run on solar energy) to of MW capacity.

Ans. Many solar panels are connected in series and in parallel to generate required current and potential difference. Solar cell is the basic unit in solar electric plant. Many solar cells come together to form a solar panel. Many solar panels connected in

series form a solar string, and, many solar strings connected in parallel form a solar array. As we can obtain as much electrical power as needed, they are used in applications which need marginal power (e.g. calculators that run on solar energy) to power station of MW capacity.

CHAPTER 6 : ANIMAL CLASSIFICATION

Q. (*Dobzhansky, Carl Woese, Theophrastus, Artificial method, Aristotle, Natural system, Traditional system*)

Time to time, different scientists have tried to classify the animals. Greek philosopher was the first to perform the animal classification. Aristotle classified the animals, according to the criteria like body size, habits and habitats. Classification proposed by Aristotle is known as Besides Aristotle, artificial method of classification was followed by, Pliny, John Ray and Linnaeus. Later on, '..... of classification' was followed. Natural system of classification was based on various other criteria. By the time, system of classification based on evolution was also brought into practice. It was used by and Meyer. Recently, has also proposed the animal classification.

Ans. Time to time, different scientists have tried to classify the animals. Greek philosopher Aristotle was the first to perform the animal classification. Aristotle classified the animals, according to the criteria like body size, habits and habitats. Classification proposed by Aristotle is known as 'Artificial method'. Besides Aristotle, artificial method of classification was followed by Theophrastus, Pliny, John Ray and Linnaeus. Later on, 'Natural system of classification' was followed. Natural system of classification was based on various other criteria. By the time, system of classification based on evolution was also brought into practice. It was used by Dobzhansky and Meyer. Recently, Carl Woese has also proposed the animal classification.

CHAPTER 7 : INTRODUCTION TO MICROBIOLOGY

(1) (*nutritive, ethanol, salt, actinomycetes, fermentation, carbon dioxide, lactic acid, Saccharomyces cerevisiae*) (**July '23**)

Different types of breads are produced from flour of cereals. Dough is formed by mixing of baker's yeast –, water, and other necessary materials with flour. Due to of carbohydrates by

yeast, sugar is converted into and Dough rises up due to CO_2 and the bread becomes spongy. Bread and other products produced with the help of yeast become

Ans. Different types of breads are produced from flour of cereals. Dough is formed by mixing of baker's yeast – *Saccharomyces cerevisiae*, water, salt and other necessary materials with flour. Due to fermentation of carbohydrates by yeast, sugar is converted into carbon dioxide and ethanol. Dough rises up due to CO_2 and the bread becomes spongy. Bread and other products produced with the help of yeast become nutritive.

(2) Complete the given paragraph by filling the blank spaces from the options given in the bracket :

(oxygen, pyridines, mechanical, CO_2 , petroleum, fatal, polyester, Nocardia) **(March '20 – '24)**

Spilling of oil occurs in ocean due to various reasons. This oil may prove and toxic to aquatic organisms. It is not easy to remove the oil layer from surface of water by method. However, bacteria like *pseudomonas* spp. and *Alcanovorax borkumensis* have the ability to destroy the and other chemicals. Hence, these bacteria are used to clear the oil spills. These are called as hydrocarbonoclastic bacteria (HCB). HCB decompose the hydrocarbons and bring about the reaction of carbon with and water is formed in this process.

Ans. Spilling of petroleum oil occurs in ocean due to various reasons. This oil may prove fatal and toxic to aquatic organisms. It is not easy to remove the oil layer from surface of water by mechanical method. However, bacteria like *Pseudomonas* spp. and *Alcanovorax borkumensis* have the ability to destroy the pyridines and other chemicals. Hence, these bacteria are used to clear the oil spills. These are called as hydrocarbonoclastic bacteria (HCB). HCB decompose the hydrocarbons and bring about the reaction of carbon with oxygen. CO_2 and water is formed in this process.

Note : Similar question was asked in March '20 with few changes in blanks.

CHAPTER 8 : CELL BIOLOGY AND BIOTECHNOLOGY

Q. (*degenerated, red bone marrow, adipose connective tissue, blastocyst, umbilical cord, Differentiation*)

..... of stem cells can form various tissues in the body. Stem cells are present in the by which the foetus is joined to the uterus of the mother. Stem cells are also present in the stage of embryonic development. Stem cells are present in and of adult human beings. It has become possible to produce different types of tissues and the part of any organ with the help of these stem cells.

Ans. Differentiation of stem cells can form various tissues in the body. Stem cells are present in the umbilical cord by which the foetus is joined to the uterus of the mother. Stem cells are also present in the blastocyst stage of embryonic development. Stem cells are present in red bone marrow and adipose connective tissue of adult human beings. It has become possible to produce different types of tissues and the degenerated part of any organ with the help of these stem cells.

CHAPTER 9 : SOCIAL HEALTH

Q. (*personality, thoughts, virtual, health, converse, relative*)

When we interact with the, it becomes easier to mix with other strangers later. The moulds when we talk and interact with different people. We can exchange the We learn to in a rightful way. When we go to playground and take part in outdoor games, we get benefits. Sitting at home and spending time in just mobile or computer games, does not benefit in any way. Most of the serials on the television are of no use for any kind of personality development, instead they push us in a world.

Ans. When we interact with the relative, it becomes easier to mix with other strangers later. The personality moulds when we talk and interact with different people. We can exchange the thoughts. We learn to converse in a rightful way. When we go to playground and take part



in outdoor games, we get health benefits. Sitting at home and spending time in just mobile or computer games, does not benefit in any way. Most of the serials on the television are of no use for any kind of personality development, instead they push us in a virtual world.

CHAPTER 10 : DISASTER MANAGEMENT

9 (Delhi, 12, 2005, army, National Disaster Response Force, rescue work, State Reserve Police, 2015) (**March '23**)

National Disaster Response Force has been established as per the Disaster Management Act, Divisions of this force are working in army. Overall, divisions are working in the country. Its headquarter is in and it is in action all over the country with the help of In Maharashtra, Response Force is in action through State Reserve Police Force. Personnel of this force have substantial contribution in in disasters like cyclones, cliff-sliding, building collapse, etc.

Ans. National Disaster Response Force has been established as per the Disaster Management Act, 2005. Divisions of this force are working in army. Overall, 12 divisions are working in the country. Its headquarter is in Delhi and it is in action all over the country with the help of army. In Maharashtra, National Disaster Response Force is in action through State Reserve Police Force. Personnel of this force have substantial contribution in rescue work in disasters like cyclones, cliff-sliding, building collapse, etc.

**Question
Set
16**

PARAGRAPH-BASED QUESTIONS

[Q. 4 : 2/3/5 Marks each]

(2/3/5 questions based on a given paragraph. 1 mark each)

Remember :

- ☛ In this question set, paragraph-based questions are asked.
- ☛ In activity sheet a paragraph will be given. This paragraph may be textual or from allied matter but out of the textbook.
- ☛ Five questions based on this paragraph will be asked. Each one will bear 1 mark.
- ☛ Answer each question correctly in one sentence only.

Q. Read the given paragraph or information and answer the questions below.

CHAPTER 1 : HEREDITY AND EVOLUTION

With the help of RNA, the genes present in the form of DNA participate in the functioning of cell and thereby control the structure and functioning of the body. Information about protein synthesis is stored in the DNA and synthesis of appropriate proteins as per requirement is necessary for body. These proteins are synthesized by DNA through the RNA. This is called 'Central Dogma'. m-RNA is produced as per the sequence of nucleotides on DNA. Only one of the two strands of DNA is used in this process. The sequence of nucleotides in m-RNA being produced is always complementary to the DNA strand used for synthesis. Besides, there is uracil in RNA instead of thymine of DNA. This process of RNA synthesis is called 'transcription'.

Questions and Answers :

(1) Which part of the cell control the structure and functioning of the body?

Ans. Genes present in the form of DNA along with RNA control the structure and functioning of the body.

(2) What is the similarity between m-RNA and DNA?

Ans. The sequence of nucleotides on DNA is copied on m-RNA. The nucleotide sequence on m-RNA is thus complementary to DNA.



(3) Give one difference between RNA and DNA.

Ans. RNA has uracil instead of thymine which is present in DNA.

CHAPTER 2 : LIFE PROCESSES IN LIVING ORGANISMS : PART – 1

Dietary fibre — found mainly in fruits, vegetables, whole grains and legumes — is probably best known for its ability to prevent or relieve constipation. But foods containing fibre can provide other health benefits as well, such as helping to maintain a healthy weight and lowering your risk of diabetes, heart disease and some types of cancer. Dietary fibre, also known as roughage or bulk, includes the parts of plant foods your body can't digest or absorb. Unlike other food components, such as fats, proteins or carbohydrates — which your body breaks down and absorbs — fibre isn't digested by your body. Instead, it passes relatively intact through your stomach, small intestine and colon and out of your body.

Questions and Answers :

(1) Which food items provide rich fibre content?

Ans. Fruits, vegetables, whole grains and legumes give rich amount of dietary fibre.

(2) Enlist the advantages of fibres in diet.

Ans. Fibres help to relieve constipation and help in maintaining a healthy weight and lowering risk of diabetes, heart disease and some types of cancer.

(3) What is a roughage?

Ans. Roughage is the fibre content of the food which consists of plant matter which cannot be digested by the human enzymes, hence form undigested bulk matter in the faeces.

CHAPTER 3 : LIFE PROCESSES IN LIVING ORGANISMS : PART – 2

Reproduction is the process by which the living species continues its existence. Lower organisms carry out asexual reproduction while higher plants and animals always show sexual reproduction. Plants reproduce asexually by methods such as fragmentation, vegetative propagation, budding, spore formation. For sexual reproduction they form gametes. In animal kingdom, budding, fission of different types and parthenogenesis are some of the methods that do not require

both the sexes. Though regeneration also forms new individual, it is not considered to be a reproductive process because, basically it is a repair process. The ability to regenerate is lost in higher phyla. In human beings it is restricted only to wound healing. Sexual reproduction is also undergoing lots of experimentation such as cloning which may make females capable of producing their own baby without intervention of any male.

Questions and Answers :

(1) How do living species continue their existence?

Ans. Through the process of reproduction, living species continue their existence.

(2) Which are asexual methods of reproduction in kingdom Animalia?

Ans. Fission, budding and parthenogenesis are the asexual methods of reproduction in kingdom Animalia.

(3) Why is regeneration not true method of reproduction?

Ans. Regeneration is the repair process than a reproductive process. It is not done with the intention of producing offspring, but is for healing or repairing the lost part.

(4) What are methods of reproduction in plants?

Ans. Plants reproduce by asexual as well as sexual methods. Asexual reproduction is by fragmentation, vegetative propagation, budding, spore formation, while by formation of gametes, sexual reproduction is done.

(5) What is the modern method of reproduction aimed at in higher organisms?

Ans. Cloning is the modern method of reproduction by which production of young one can be aimed at.

CHAPTER 5 : TOWARDS GREEN ENERGY

Renewable energy is energy produced from sources that do not deplete or can be replenished within a human's life time. The most common examples include wind, solar, geothermal, biomass and



hydroelectric power. This is in contrast to non-renewable sources such as fossil fuels. Most renewable energy is derived directly or indirectly from the sun. Sunlight can be captured directly using solar technologies. The sun's heat drives winds, whose energy is captured with turbines. Plants also rely on the sun to grow and their stored energy can be utilised for bioenergy. Not all renewable energy sources rely on the sun. For example, geothermal energy utilises the Earth's internal heat, tidal energy relies on the gravitational pull of the moon, and hydroelectric power relies on the flow of water.

Questions and Answers :

(1) What is renewable energy?

Ans. Renewable energy is energy that is produced from sources which will not get exhausted within a human's life time.

(2) Give the examples of renewable energy.

Ans. Wind, solar, geothermal, biomass and hydroelectric power are some examples of renewable energy.

(3) Why will energy from fossil fuel be over soon?

Ans. Fossil fuels are exhaustible in their amount. We have been using these extensively in the past 100 years and hence it may get over soon. It is a non-renewable resource.

CHAPTER 7 : INTRODUCTION TO MICROBIOLOGY

Remediation is the process of removing dangerous or poisonous substances from the environment, or limiting the effect that they have on it. When any biological organism is used for remediation, it is called bioremediation. When plant species are used for the purpose of remediation, it is called phytoremediation. When any microbes are used then it is named as microbial remediation. The methods of such remediation have helped to clean the environment from toxic effluents, especially sewage and crude oil. Dr. Anand Chakraborty, a scientist of Indian origin, has worked on *Pseudomonas aeruginosa* which have reduced the crude oil films into carbon dioxide and water.

Questions and Answers :

(1) What is the meaning of remediation?

Ans. Remediation is the process by which the dangerous or toxic substances are removed from the environment.

(2) What is the difference between phytoremediation and microbial remediation?

Ans. When any plant species are used for remediation process, then it is called phytoremediation, whereas when any microbe species used for remediation then it is called microbial remediation.

(3) Which environmental pollutant is mainly removed through bioremediation processes?

Ans. Toxicants released through sewage and crude oil are removed by bioremediation processes.

CHAPTER 8 : CELL BIOLOGY AND BIOTECHNOLOGY

A liberal view behind the concept of organ and body donation is that after death our body should be useful to other needful persons so that their miserable life would become comfortable. Awareness about these concepts is increasing in our country and people are voluntarily donating their bodies.

Life of many people can be saved by organ and body donation. Blinds can regain their vision. Life of many people can be rendered comfortable by donation of organs like liver, kidneys, heart, heart valves, skin, etc. Similarly, body can be made available for research in medical studies. Many government and social organisations are working towards increasing the awareness about body donation.

(March '19) (2 marks)

Questions and Answers :

(1) What is the liberal view behind the organ and body donation?

Ans. By body donation, research in medical studies is possible. The needy persons can get vital organs which can save their lives.

(2) Name any four organs that can be donated.

Ans. Liver, kidneys, heart, eyes, skin, etc. can be donated.



**Question
Set
17**

CLASSIFICATION-BASED QUESTIONS

[Q. 1(B), 3, 4 : 1/3/5 Marks each]

Remember :

- ☛ The questions based on classification are included in this question set.
- ☛ Most likely the questions based on the animal classification are included here.

Q. Classification-based questions :

CHAPTER 6 : ANIMAL CLASSIFICATION

(1) Read the statements and answer accordingly :

(March '24) (3 marks)

- (a) Calcareous spines are present on the body. Tube feet are used for locomotion and capturing prey also.
— Identify the phylum and give *one* example.
- (b) I live in your small intestine. Pseudocoelom is present in my thread like body.
— Identify the phylum and give *one* example.
- (c) My body is spindle-shaped. I respire with the help of gills.
— Identify my class and give *one* example.

Ans. (a) Phylum : Echinodermata

Examples : Star fish, Sea cucumber, Brittle star, Sea urchin.

(b) Phylum : Aschelminthes

Examples : *Ascaris*, *Loa loa*, Filarial worm.

(c) Class : Pisces

Examples : Pomfret, Shark, Sting ray, Sea-horse.

(2) Identify the class of given animals and write one characteristic of each animal :

- (a) Kangaroo (b) Penguin (c) Crocodile (d) Frog (e) Sea-horse.

(July '19) (5 marks)

Ans. (a) Kangaroo : Class Mammalia. It is a marsupial animal with pouch for development of offspring. Long hind limbs for jumping.

(b) Penguin : Class Aves. It is flightless bird. Body covered with thick feathery coat. Oviparous mode.

(c) Crocodile : Class Reptilia. It is a large animal seen near waterbodies. Can swim in water but cannot respire in water. Body covered with exoskeleton of scaly plates. Limbs very weak in comparison with huge bodies.

(d) Frog : Class Amphibia. Shows aquatic as well as terrestrial mode. Can breathe with lungs and skin. No exoskeleton and skin is slimy.

(e) Sea-horse : Class Pisces. Bony fish. Highly modified body structure showing brood pouch for development of offspring, gills for respiration, fins for swimming.

(3) Identify my class/phylum and give one example of it :

(March '19) (3 marks)

(a) I have mammary glands and exoskeleton in the form of hair.

(b) We form the highest number of animals on the planet. We have bilateral symmetry and our exoskeleton is in the form of chitin.

(c) I live in your small intestine, my body is long and thread like and pseudocoelomate.

Ans. (a) Class : Mammalia, Example : Cat, Dog, Man.

(b) Phylum : Arthropoda, Example : Prawn, Crab.

(c) Phylum : Aschelminthes, Example : Ascaris or round worm, Filarial worm.

(4) Identify me :

I am metamerically segmented, blood sucking, ectoparasite. I have suckers. Who am I and to what phylum do I belong to?

OR

(i) Who am I? **(July '19)**

(1 mark)

I have suckers. I am blood sucking.

Ans. Leech, Phylum Annelida.



(ii) I have chitinous exoskeleton, I have three pairs of walking appendages, what phylum do I belong to? **(Sept. '21) (1 mark)**

Ans. Phylum : Arthropoda; Class : Insecta.

(iii) I have calcareous spines and I move with tube feet, which phylum do I belong to? **(July '22) (1 mark)**

Ans. Echinodermata.

(5) Tell me who am I? What is my class/phylum?

(1 mark each)

(i) I am sedentary marine animal drinking water all the time through numerous pores on the body.

(ii) I am male as well as female. I am endoparasite having a coelomate and bilaterally symmetrical and flattened body.

(iii) I stay inside two shells. My body is divided into head, foot and visceral mass.

(iv) My body is divided into proboscis, collar and trunk. I am marine animal.

Ans. (i) Sponge; Phylum : Porifera.

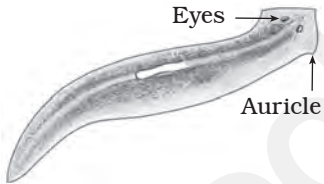
(ii) Liver fluke or tape worm; Phylum : Platyhelminthes.

(iii) Bivalve or Oyster; Phylum : Mollusca.

(iv) Balanoglossus; Phylum : Hemichordata.

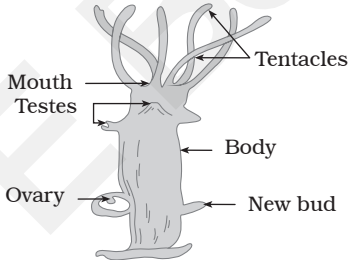
(6) Sketch, label and classify the following organisms :

(1) Planaria :

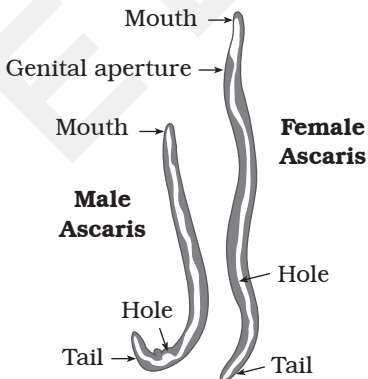
 The diagram shows a lateral view of a Planaria. At the anterior end, there are two small eyes labeled 'Eyes'. On the side of the head region, there is a small, ear-like structure labeled 'Auricle'.	Classification : Kingdom : Animalia Division : Non-chordata Phylum : Platyhelminthes Example : Planaria
---	---

(2) Hydra :

(2 marks each)

 The diagram shows a Hydra with various labeled parts. At the top, there are 'Tentacles'. Below them is the 'Mouth'. On the sides of the upper body, there are 'Testes'. The main part is the 'Body'. At the bottom, there is an 'Ovary' and a 'New bud' growing from the side.	Classification : Kingdom : Animalia Division : Non-chordata Phylum : Coelenterata Example : Hydra
---	---

(3) Round worm :

 Male Ascaris Female Ascaris	Classification : Kingdom : Animalia Division : Non-chordata Phylum : Aschelminthes Example : Ascaris (Round worm)
---	---

(7) Observe the given animal and answer the following questions : (July '23) (2 marks)

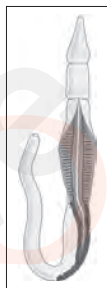
(a) Identify the given animal and its phylum.

(b) Write any two characteristics of that phylum.

Ans. (a) The given animal is *Balanoglossus*. It belongs to phylum Hemichordata.

(b) (1) Body is divisible into proboscis, collar and trunk. Pharyngeal gill slits present.

(2) Notochord present in the region of proboscis.



(8) Write the characters of each of the following animals with the help of classification chart : (3 marks each)

(1) Jellyfish :

Ans. Classification :

Kingdom : Animalia

Subkingdom : Non-chordata

Phylum : Cnidaria or Coelenterata

Characters :

- Multicellular organisms without cell wall
- Tissue grade organisation
- Radially symmetrical
- Diploplastic and Acoelomate

Jellyfish or Aurelia is a coelenterate. Its body is medusa. It appears as a transparent balloon seen floating in the marine waters. Since it has appearance like a jelly, it is known commonly as jellyfish. There are tentacles provided with cnidoblasts or stinging cells. Tentacles are used for catching the prey. Cnidoblasts are used to secrete a toxin which paralyses the prey.

(2) Bath sponge :

Ans. Classification :

Kingdom : Animalia

Subkingdom : Non-chordata

Phylum : Porifera

Characters :

- Multicellular organisms without cell wall
- Cellular grade organisation.
- Asymmetrical body
- Acoelomate

Bath sponge is a marine animal. Blackish in colour and round in shape having porous body. It has spongin fibres and spicules which serve as skeleton. Bath sponges have good water-holding capacity. It is sedentary animal which is fixed to some substratum in the aquatic environment. Reproduction is by budding. It also has a good regeneration capacity.

**Question
Set
18**

ANSWER THE QUESTIONS – 1

[Q. 2 [B] : 2 Marks each]

Remember :

- ☛ This question set consists of 'Answer the following' type of questions.
- ☛ Write minimum 2 points for each question. In some questions 4 points are expected.
- ☛ Q. 2 [B] consists of questions carrying two marks. Moreover, in Q. 3 or Q. 4 sometimes 2 marks questions may be included, if more than two questions are clubbed together.

CHAPTER 1 : HEREDITY AND EVOLUTION

(1) What is meant by carbon dating method?

Ans. (1) Carbon dating method is technique used for determining the age of fossils.

(2) After the death of the organisms, their consumption of carbon stops. But right from that moment the decaying process of C-14 occurs continuously.

(3) This results in change in the ratio between C-14 and C-12. C-12 is not radioactive as C-14.

(4) Thus, the time passed since the death of a plant or animal is calculated by measuring the radioactivity of C-14 and ratio of C-14 to C-12 present in their body.

(5) The points noted during carbon dating are :

- (i) The period after the organism has been dead.
- (ii) The activity of C-14 in the dead organism.
- (iii) Ratio between C-14 and C-12.

(2) Describe briefly the Darwin's theory of Natural Selection.

(Nov. '20)

Ans. For the answer, refer to Q. 1 (1) of Chapter 1 in Q. Set 8 in the same book.



(1) Many players are seen consuming some food stuffs during breaks of the game. Why may be the players consuming these food stuffs?

Ans. (1) Players require energy in greater amount.

(2) They perspire heavily at the time of game or sport which results in the loss of water and electrolytes from their body.

(3) This may affect their performance in sport. To prevent such unfavourable effect, they are given juices or drinks.

(4) This helps them to restore the balance of water and electrolytes in their body. It also gives enhanced energy required for the performance.

(2) Write the forms to which the following food materials are converted after digestion :

(a) Milk (b) Potato (c) Oil (d) Chapati.

Ans. (a) Milk : Proteins (caesin) are converted into amino acids. Lactose sugar is converted into glucose. Lipids are converted into fatty acids and glycerol.

(b) Potato : Carbohydrates (starch) are converted into glucose.

(c) Oil : Lipids are converted into fatty acids and glycerol.

(d) Chapati : Carbohydrates (starch) are converted into glucose.

(3) Which molecules are formed after whole oxidation of acetyl-co-enzyme A?

Ans. By the process of Krebs cycle, acetyl-Co-A is oxidised forming CO_2 , H_2O , NADH_2 , FADH_2 and molecules of ATP.

(4) Write uses of fatty acids.

Ans. (1) Fatty acids are very essential for the health.

(2) After digestion, fatty acids are absorbed into the blood and transported to the cells.

(3) Different types of cells produce their own substances from these fatty acids. e.g. (a) Plasma membrane is produced from phospholipids. (b) Hormones like testosterone, progesterone, estrogen, aldosterone are produced from fatty acids. (c) The axonal coverings around the neurons are also made from fatty acids.

(1) Why has the Government of India enacted the law to fix the minimum age of marriage as 18 in girls and 21 in boys?

Ans. The full growth of female body is not completed till the age of 18. Till 18 years of age the physical and emotional maturity is not attained. Therefore, she is not suitable for marriage, sexual relationship and pregnancy. Similarly, boy attains complete growth only by the age of 21. Therefore, to keep individuals and their progeny safe and healthy, the Government of India enacted the law to fix the minimum age of marriage as 18 in girls and 21 in boys.

(2) Explain the process of reproduction in fungus.

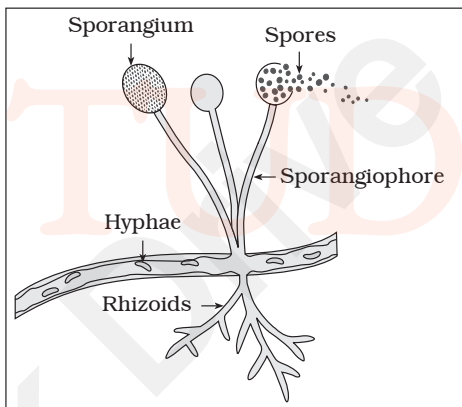
Ans.

(1) Mucor reproduces asexually by spore formation.

(2) It has filamentous body that possess sporangia.

(3) When the spores are formed, the sporangia burst. The spores are released which settle down at suitable places.

(4) They germinate in moist and warm place forming a new fungal colony.



Spore formation

(3) What precautions should be followed to maintain the reproductive health? (July '22)

Ans. (1) Genitals should be kept clean. There should not be lack of awareness about reproductive health.

(2) Social customs, traditions, illiteracy, shyness, etc. should not result in negative attitude towards reproductive health.

CHAPTER 4 : ENVIRONMENTAL MANAGEMENT

(1) What is environment? Write two types of environment.

(July '22)

Ans. (1) **Environment** : Physical, chemical and biological factors

affecting the living organisms in any possible way is collectively called as environment.

(2) **Two types of environment** : Natural environment and Artificial environment.

(2) Write four adverse effects of radiations on human body.

(July '23; March '24)

Ans. (1) Cancerous ulcerations due to X-rays.

(2) Destruction of body tissues.

(3) Change in the genes.

(4) Adverse effect on vision.

(3) What do we learn from the story of Jadav Molai Payeng?

(March '22)

Ans. Jadav Molai Payeng is a common man who was just a simple forest worker. But he has conscience about plants and tree plantations. He single-handedly planted thousands of trees. He converted a barren patch of land into forest which is spread over 1360 acres. For these plantations he continuously worked. He has shown that a single determined person, can establish a new forest! We understand the values of hard work, sincerity and devotion to the nature through the story of Jadav Molai Payeng. Even a common man can contribute a lot for the conservation and protection of the environment by learning the story of Payeng.

(4) Explain any two types of threatened species. (Sept. '21)

Ans. (1) Threatened species are of four main types i.e. endangered species, rare species, vulnerable species and indeterminate species.

(2) **Endangered species** : Number of organisms belonging to such type of species is declining and habitat is shrinking at great speed. If conservative measures are not implemented, such species will be extinct in near future. e.g. Lesser florican.

(3) **Rare species** : Number of organisms belonging to this type is considerably declined. Their organisms are endemic and also on the verge of extinction. e.g. Red Panda.

CHAPTER 5 : TOWARDS GREEN ENERGY

(1) Write advantages of solar energy.

Ans. (1) While generating the power through solar radiations, no fuel is burnt.

(2) Solar energy generation thus does not create any type of pollution. The technology can be completely utilised in regions with abundant sunlight.

(3) Solar energy is eco-friendly, green energy.

(2) Write the limitations of solar energy.

Ans. (1) Sunlight is available only during day time. Thus, solar cells can generate power only during day.

(2) In rainy season and in cloudy conditions, solar power generation suffers.

(3) The power present in the solar cells is DC while most of the domestic equipments work on AC.

(3) Why steam is used to rotate the turbine?

Ans. When water is heated, it converts into steam. This steam is under high pressure and temperature. This force of steam rotates the turbines. The same steam is again condensed and converted back into water. The excess heat in turbine again makes this water turn into steam. In this way turbine rotates very effectively due to steam.

(4) Write the advantages of hydroelectric power generation.

(March '19) (2 marks) (Nov. '20) (3 marks)

Ans. (1) Hydroelectric energy does not cause pollution.

(2) Generation of hydroelectric energy does not involve burning of fossil fuels.

(3) If sufficient water storage is available, then electricity generation can be done as per requirement.

(4) Rainwater can replenish the water storage and power generation can thus be done uninterrupted.

CHAPTER 6 : ANIMAL CLASSIFICATION

(1) Write four distinguishing characters of phylum – Echinodermata.

Ans. Distinguishing characters of Echinodermata :

(1) Marine organisms with skeleton made up of calcareous spines. Calcareous material on the body hence the name is Echinodermata. Some are sedentary while some are free swimming.



(2) Body is triploblastic, eucoelomate and radially symmetrical when adult. The larvae are bilateral symmetrical.

(3) Locomotion with the help of tube-feet which are also used for capturing the prey.

(4) Echinoderms have regeneration capability. Hence they can restore their lost parts.

(5) Most of them are unisexual.

Examples : Starfish, sea-urchin, brittle star, sea-cucumber, etc.

(2) In what way the leech is used in ayurvedic system of treatment?

Ans. Leeches are blood sucking ectoparasite. In Ayurveda, leech is used to remove impure blood and blood clots. Such blood is sucked up by leeches and then the patient gets some relief. In the leech body there is a substance called hirudine which prevent blood clotting as it sucks up the blood. This hirudine is also used for medicinal purpose.

CHAPTER 7 : INTRODUCTION TO MICROBIOLOGY

(1) How can the soil polluted by acid rain be made fertile again?

Ans. (1) The soil polluted by the acid rain is made fertile again by using bacteria.

(2) *Acidophilium* spp. and *Acidobacillus ferrooxidans* are the bacteria which have the capacity to use sulphuric acid as their energy source.

(3) Since this sulphuric acid present in the acid rain, can be controlled by these bacteria.

(4) In this way, bacteria can control the soil pollution occurring due to acid rain, making the soil fertile again.

(2) What are the benefits of mixing ethanol with petrol and diesel? (March '23)

Ans. (1) When only diesel or petrol is used as fuel, there is increased air pollution. Moreover, since these are non-renewable and exhaustible fuels, they will be finished in next some years. (2) When petrol and diesel is mixed with ethanol, the proportion of CO₂, CO and hydrocarbons which are emitted in the atmosphere becomes lesser. The particulate pollutants which otherwise are emitted

through combustion of petrol and diesel are not formed when fuels are mixed with ethanol. (3) By adding ethanol to the fuels, the cost of expensive petrol or diesel also becomes less. (4) The ethanol burns more efficiently, hence ethanol is mixed with petrol and diesel.

(3) Name two fuels obtained by microbial processes. Why is it necessary to increase the use of such fuels? (March '24)

Ans. (1) Fuels obtained by microbial processes, which are :
(i) Methane, (ii) Ethanol (iii) Hydrogen gas.

(2) The conventional fuels are exhaustible. After few hundred years, they will be over completely. Moreover, these fossil fuels cause lot of air pollution due to emission of carbon dioxide. The fuels obtained by the microbial processes are not polluting. Therefore, it is necessary to increase the use of eco-friendly fuels.

(4) Which plants are cultivated to obtain fuel?

Ans. (1) The ethanol is obtained from wheat, maize, beet, sugarcane and molasses of sugarcane.

(2) For biodiesel, the soyabean, rapeseed, jatropha, mahua, flaxseed, mustard, sunflower, palm, jute and some types of algae are cultivated.

CHAPTER 8 : CELL BIOLOGY AND BIOTECHNOLOGY

(1) What are edible vaccines?

Ans. (1) Edible vaccines are those which are given as a food by incorporating them into the foodstuff.

(2) Such edible vaccines are produced through biotechnology.

(3) Transgenic potatoes are produced with the help of biotechnology which contain vaccine that act against bacteria like *Vibrio cholerae*, *Escherichia coli*.

(4) If raw potatoes are consumed, then the immunity is generated in the body of a person. However, eating only raw potatoes generates the immunity against cholera and the disease caused due to *E. coli*.

(2) Define genetically modified crops and give any two examples. (March '22)

Ans. (1) Definition of genetically modified crops :

Crops having desired characters are developed by integrating foreign gene with their genome, such crops have modified genome and are known as genetically modified crops.



(2) Examples of genetically modified crops : BT cotton, BT Brinjal and Golden rice are the GMOs.

(3) What is DNA Fingerprinting? Where is it mainly useful?

(March '24)

Ans. (1) **DNA fingerprint** : The nucleotic sequence present on the DNA of each person is unique just like the fingerprint, thus for establishing the identity of any person DNA can be analysed, this technique is known as DNA fingerprinting.

(2) **Uses of DNA fingerprinting** : (i) Investigations used in forensic sciences. (ii) To establish identity of the criminal. (iii) To identify parents of any child in case of disputed parentage.

CHAPTER 9 : SOCIAL HEALTH

(1) Write down impact of alcoholism on human health.

(July '19)

Ans. (1) Consuming liquor becomes an addiction for a long-term. Due to alcohol, the efficiency of nervous system and especially the brain is affected.

(2) Other vital organs such as kidneys and liver are adversely affected.

(3) Lifespan of an alcoholic decreases due to constant drinking and malnourishment.

(4) Especially in adolescent age, if alcohol is consumed the brain functioning does not take place properly. The mental ability of memorisation and learning becomes slow. There is lack of concentration in studies.

(5) The alcoholic person lacks the rational thinking and hence faces social, mental and familial problems along with physical illness.

(2) How is the excessive use of social media and technology proving harmful?

Ans. Excessive use of social media and modern technology is disturbing the social health. It is also affecting physical and mental health. Increase in cybercrimes take place. People waste their time by watching useless and obscene material. Violence develops by watching few weird cartoon serials. Dependency on machine rises and persons lose self-reliance.

(1) State any four objectives of disaster management.

(March '19) OR

State any two objectives of disaster management.

(March '23; July '23)

Ans. For the answer, refer to Answer of Q. 3 (viii) from Activity Sheet of July '24.

(2) Write any four important facts which must be thought over while considering the scope of disaster. (March '24)

Ans. Important facts for consideration of scope of disaster are as follows :

(1) Pre-disaster phase (2) Warning phase (3) Emergency phase (4) Rehabilitation phase (5) Recovery phase (6) Reconstruction phase.

(Write any four of the above facts.)

(3) Why is it essential to get the training of first aid?

(July '19)

Ans. When there is a disaster, we need to immediately help the victim. Till the medical help arrives, one should be in position to treat the injured and save his or her life. In such cases, knowing first-aid is essential. Such kind of a need may arise in case of our parents, our siblings at home or with friends in school. Those who are injured should be treated at once. If we know about techniques of first aid, we can save such person before the medical help arrives. Therefore, it is essential to get the training of the first aid.

**Question
Set
19**

ANSWER THE QUESTIONS – 2

[Q. 3/Q. 4 : 3/5 Marks each]

Remember :

- ☛ This question set consists of 'Answer the following' type questions.
- ☛ Q. 3 will consist of questions carrying 3 marks each, write 6 points in the answer.
- ☛ Q. 4 will consist of questions carrying 5 marks each.
- ☛ If the question is asked under Q. 4, then suitable, well-labelled diagram is necessary along with important points in the answer.
- ☛ In some of the previous Board Activity Sheets, 5 sub-questions carrying 1 mark each have been asked.

CHAPTER 1 : HEREDITY AND EVOLUTION

(1) Define heredity. Explain the mechanism of hereditary changes. (March '25)

Ans. (1) Heredity : Heredity is the process by which the biological characters from parental generation are transmitted to the next generation through genes.

(2) The mechanism of hereditary changes :

(i) Mutation : Sudden change in the parental DNA can cause mutations. This results into changes in the hereditary characters.

(ii) At the time of meiosis, the crossing over takes place. This creates new recombination of the genetic information. Therefore, the haploid gametes produced carry changed hereditary characters.

(2) Answer the following questions : (March '19) (3 marks)

- (a) What do you mean by central dogma?
- (b) What is transcription?
- (c) What is meant by triplet codon?

Ans. (a) Information about protein synthesis is present in DNA. As per this information, proteins are produced by DNA through RNA molecules. This is called central dogma.

(b) The process of synthesis of m-RNA as per the nucleotide sequence present in DNA is called transcription. The nucleotide sequence on m-RNA is complimentary to that of the single DNA strand used in synthesis. Instead of thymine, m-RNA possesses uracil.

(c) The code for each amino acid always consists of three nucleotides which is known as triplet codon.

(3) Explain with suitable examples importance of anatomical evidences in evolution. (July '19; March '20) (3 marks)

Ans. (1) There are similarities in the structure and anatomy of different animal groups. e.g. Human hand, forelimb of bull, patagium of bat and flipper of whale are all similar in their internal anatomy. There is similarity in the bones and joints of all these specimens.

(2) External morphology does not show any similarity. Use of each of the organ is also different in different animals. Structurally, they may not be related.

(3) However, the similarities in the anatomy is an evidence that they may have a common ancestor.

In this way, the anatomical evidence throws light on the process of evolution.

(4) Write about Darwin's theory of Natural Selection.

(March '23) (3 marks)

Ans. For the answer, refer to Q. 1 (1) of Chapter 1 from Q. Set 8.

(5) Write the objections raised against Darwin's theory of Natural Selection. (March '22) (3 marks)

Ans. Objections to Darwin's theory of Natural Selection :

(1) There are other factors too for evolution and just not the Natural Selection.

(2) Arrival of useful and useless modifications were not explained by Darwin, though he said about the survival of the fittest.

(3) He has not given any explanation about slow changes and abrupt changes occurring during evolution.

(4) Darwin did not give any explanation for arrival of the fittest.



(1) Write any four stages of mitosis and explain any two of them. (March '25) (3 marks)

Ans. Four stages of mitosis are as follows :

(1) Prophase (2) Metaphase (3) Anaphase (4) Telophase

Explanation :

(1) **Prophase** : During prophase, condensation of chromosomes starts. The thin and thread like chromosomes start thickening. They are seen with their pair of sister chromatids. In animal cells, the centrioles are seen to duplicate and move to opposite poles of the cell. Nuclear membrane and nucleolus disappear.

(2) **Metaphase** : Chromosomes complete their condensation and each one is seen with its sister chromatids. The chromosomes are seen in equatorial plane of the cell. The spindle fibres are formed from polar region, where centrioles are present, and they attach themselves to the centromere of each chromosome. Nuclear membrane now disappears completely.

(3) **Anaphase** : The centromeres of the chromosomes now divide forming two daughter chromosomes. The spindle fibres pull apart the chromosomes from equatorial region to the opposite poles. Chromosomes moving to the poles appear like bunch of bananas. One set of chromosomes reach each pole by the end of the anaphase.

(4) **Telophase** : Telophase is reverse of events that occurred in prophase. The thickened chromosomes decondense. They again assume the thin and thread like appearance. Nuclear membrane and nucleolus appear again. The spindle fibres are completely lost. The cell looks as if it has two nuclei in one cytoplasm.

(2) Explain the glycolysis in detail. (5 marks)

Ans. (1) Carbohydrates are converted to glucose after the process of digestion is completed. The oxidation of glucose for releasing energy is called glycolysis which takes place in cytoplasm.

(2) Glycolysis can occur in presence of oxygen or without oxygen too. The first type of glycolysis takes place in aerobic respiration and the second type is in anaerobic respiration.

(3) In aerobic respiration, there is step-wise oxidation of glucose molecule forming two molecules each of pyruvic acid, ATP, NADH_2 and water.

(4) Later the pyruvic acid formed in this process is converted into molecules of Acetyl-Coenzyme-A along with two molecules of NADH_2 and two molecules of CO_2 .

(5) During anaerobic respiration along with glycolysis there is fermentation too. This is incomplete oxidation of glucose and thus it results in formation of lesser energy.

(6) The process of glycolysis was discovered by Gustav Embden, Otto Meyerhof and Jacob Parnas. Therefore, in their honour, glycolysis is also called as Embden-Meyerhof-Parnas pathway (EMP pathway). For the discovery they had performed experiments on muscles.

(3) Answer the following questions : (July '22) (3 marks)

- (a) Write two sources of lipids.
- (b) How much energy is obtained from lipids?
- (c) What is produced after digestion of lipids?

Ans. (a) Sources of lipids : Oil, butter, ghee.

(b) 9 kcal/gm of lipid.

(c) After digestion of lipid, fatty acid and glycerol are formed.

(4) (a) What are vitamins? (March '24) (3 marks)

- (b) Classify them according to their solubility.
- (c) Give one example of each type as per above classification.

Ans. (a) Vitamins are a group of heterogeneous compounds essential for proper functioning of the body.

(b) Fat soluble vitamins and water soluble vitamins.

(c) Fat soluble vitamins : A, D, E, K. Water soluble Vitamin : B and C.

CHAPTER 3 : LIFE PROCESSES IN LIVING ORGANISMS : PART - 2

(1) Answer the following questions : (March '19) (5 marks)

- (a) "Gender of child is determined by the male partner of couple."

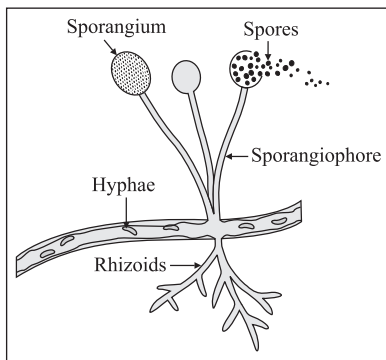
Draw a diagram explaining the above statement.



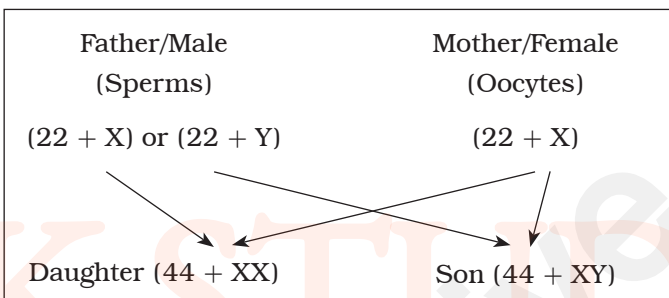
(b) Prepare a slogan for campaign against female foeticide.

(c) In the following figure, explain how new fungal colonies of mucor are formed.

(d) Identify and state the type of reproduction represented in the above figure.



Ans. (a)



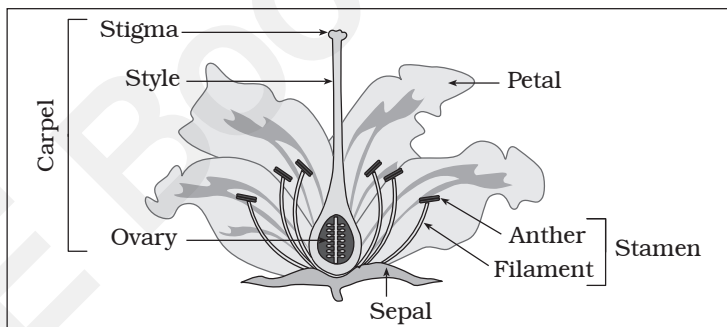
Sex determination in Human being

(b) (1) Save the girl child. (2) Daughters give lot of joy, it is not only the boy.

(c) For the answer, refer to Q. 2 of Chapter 3 from Q. Set. 18.

(d) The spore formation is asexual type of reproduction seen in Mucor.

(2) With the help of neat labelled diagram, explain any three functions of parts of a flower. (Sept. '21) (5 marks)



L.S. of a flower showing different parts

Ans. (1) The structural unit of sexual reproduction in plants is flower. There are total four floral whorls. Of these two are accessory floral whorls while two are essential floral whorls.

(2) Calyx and corolla are accessory whorls. They are protective in nature.

(3) Members of calyx are known as sepals. They are usually green in colour. They protect the inner whorls.

(4) The members of corolla are called petals. They can be of different colours.

(5) Androecium and gynoecium are essential whorls as they participate in sexual reproduction.

(6) The male whorl androecium is made up of stamens. Each stamen has a filament with anther located at the upper end. In the anther there are four locules. Inside the locules the meiosis takes place forming pollen grains. During suitable time, the pollen grains are released from anther lobes.

(7) Gynoecium is made up of carpels, either in separate form or are united. Each carpel is formed of ovary at the basal end, hollow 'style' and the stigma at the tip of style. There are one or many ovules inside the ovary.

(8) In bisexual flowers both androecium and gynoecium are located in the same flower. e.g. Hibiscus.

(9) In unisexual flowers, androecium is present in male flowers and gynoecium is present in the female flowers. e.g. Papaya.

(3) Answer the following questions : (March '22) (5 marks)

(a) Sketch and label the essential and accessory whorls of flower.

(b) What is pollination?

(c) Give any two examples of agents of pollination.

Ans. (a) *For the answer, refer to the previous question.*

Essential whorls : Androecium and gynoecium

Accessory whorls : Calyx and corolla

(b) Pollination : The process of transfer of pollen from anthers to the stigma is called pollination.

(c) Agents of pollination :

Biotic : Insects, birds, few animals.

Abiotic : Water and wind.



(4) Answer the following questions based on human male reproductive system : (March '24) (5 marks)

- (a) In which organ sperms are formed?
- (b) What is the length of a sperm?
- (c) What is the work of vas deferens?
- (d) By which type of cell division, sperms are formed?
- (e) Name any *two* unpaired organs of male reproductive system.

Ans. (a) Sperms are formed in testes.

(b) Length of sperms is about 60 micrometres.

(c) Carrying the sperms from testes to penis by helping their forward movement, is the function of vas deferens.

(d) Sperms are formed by meiosis.

(e) Penis, Scrotum, Prostate gland are the unpaired organs of male reproductive system.

CHAPTER 4 : ENVIRONMENTAL MANAGEMENT

(1) Write the factors affecting environment.

(March '23) (3 marks)

Ans. (1) The biotic and abiotic factors affect the environment. Among abiotic factors, the physical and chemical factors can alter the conditions of the environment.

(2) Abiotic factors are either natural or man-made.

(3) The various interrelationships between different living organisms can also affect environment.

(4) The natural disasters such as earthquake, forest fires, cyclones, cloud bursting, drought, etc. change the environment.

(5) The human activities such as deforestation, urbanisation, constructions, etc. cause permanent damage to the ecosystems. Due to man-made impact, there can be large scale changes in the environment.

(2) Explain the three different levels of biodiversity.

(Nov. '20) (3 marks)

Ans. Biodiversity is documented on the following three levels, viz. genetic diversity, species diversity and ecosystem diversity.

(1) **Genetic Diversity** : Diversity seen among the organisms of same species due to genetic differences is called genetic diversity. e.g. The individual human beings are different from each other. No two animals or plants are exactly alike.

(2) **Species Diversity** : The difference between the different species is the species diversity. e.g. All the species of plants, animals and microbes which are seen in any natural environment.

(3) **Ecosystem Diversity** : In one region there may be different ecosystems, such diversity in the ecosystems is called ecosystem diversity. Ecosystems are natural or artificial. Every region shows different types of ecosystems such as aquatic, terrestrial, desert or forest ecosystems. Each ecosystem has its own habitats with resident flora and fauna.

(3) Which are three endangered heritage places of India? Explain with reasons. Suggest any two measures for their protection. (Sept. '21) (5 marks)

Ans. (1) Three endangered heritage places of India are as follows : (a) Western Ghats (b) Manas sanctuary – Assam (c) Sunderban sanctuary – West Bengal.

(2) In Western Ghats, mining industry and exploration for natural gas is taking place. Due to many development projects this place has become endangered. Asiatic lion and wild bison are under threat due to disturbed habitats in this region.

(3) In Manas sanctuary, there is construction of dams and indiscriminate use of water. Rhinoceros and tigers are endangered in this area.

(4) In Sunderban, the environment is threatened due to dams, deforestation, excessive fishing, trenches dug for fishing, etc. In addition to this, there is sea level rise in this area due to global warming causing threat for tigers.

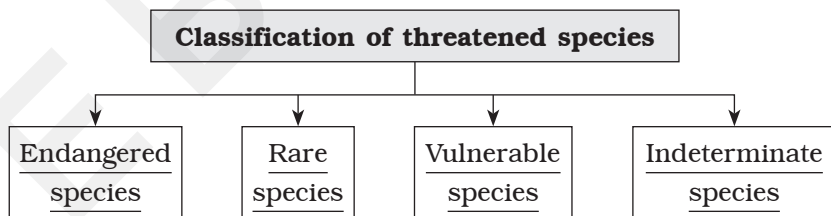
(5) **Measures** : (i) To curtail the developmental projects, environmental impact assessment should be done before taking up such projects. (ii) Afforestation and stopping destruction of environment.

(4) Answer the following questions based on environmental conservation and biodiversity : (March '24) (5 marks)

(a) Complete the flow chart :

Ans. (Answers are directly given and underlined.)





(b) How can biodiversity be conserved? Give any four points.

Ans. Biodiversity can be conserved by the following ways :

- (1) Protection of the rare species of plants and animals.
- (2) Creating habitats for the animals and plants by establishing national parks and sanctuaries.
- (3) Declaration of bioreserves, the areas which are protected through conservation.
- (4) Projects for conservation of special species.
- (5) Conserving of all flora and fauna.
- (6) Strict observance of the acts and rules.
- (7) Use of traditional knowledge and maintaining record of traditional knowledge.

(Write any four points.)

(c) Complete the flow chart :

Ans. Colour of the pages in Red List of Endangered Species.



CHAPTER 5 : TOWARDS GREEN ENERGY

(1) How is nuclear fission reaction carried out in nuclear power plants?

Ans. (1) In nuclear power plants, neutrons are bombarded on atom of Uranium-235.

(2) This causes conversion of Uranium-235 into its isotope U-236.

(3) U-236 is very unstable and thus forms atoms of Barium and Krypton by nuclear fission. This forms 3 neutrons and 200 MeV energy.

(4) In a similar way three more Uranium-235 atoms are subjected to nuclear fission which then releases energy.

(5) The neutrons released are again used for further nuclear fission reactions. In this way, nuclear fission reactions are carried out in controlled manner in nuclear power plants.

(2) Draw schematic of power plant based on natural gas and answer the following questions : (July '19) (5 marks)

(a) At which place natural gas power plant is situated in Maharashtra?

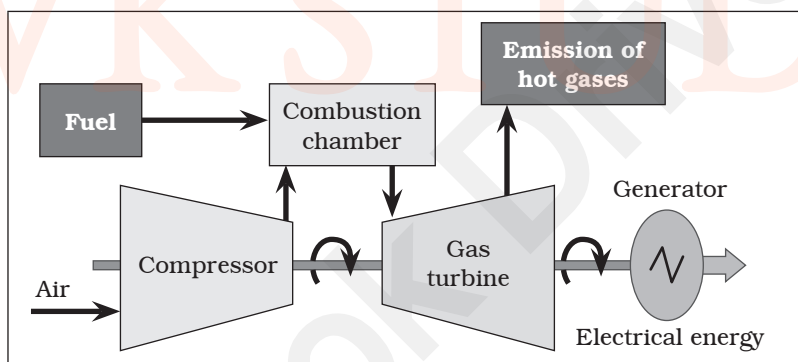
(b) How is pollution reduced in natural gas based power plant?

(c) Give two examples of eco-friendly electricity process.

Ans. (a) Natural gas power plant is situated at Anjanvel in Maharashtra.

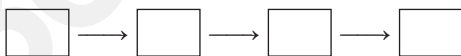
(b) Natural gas does not contain sulphur. Burning of such natural gas does not produce pollution.

(c) Solar energy and wind energy are two examples of eco-friendly energy.



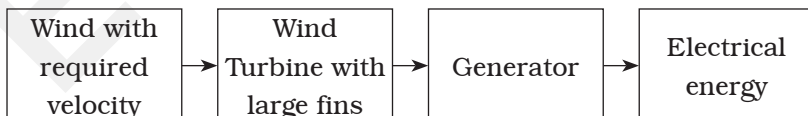
(2) Answer the following questions : (March '22) (3 marks)

(a) Complete the stages in electric generator using wind energy.



(b) Write limitations in an electric generator using wind energy.

Ans. (a) Stages in electric generator using wind :



(b) The installation of windmills to produce electricity from wind energy depends upon following factors :

- (1) The wind velocity available at the site of installation.
- (2) Geographical factors.

If these factors such as wind velocity are not suitable, use of wind energy is limited.

CHAPTER 6 : ANIMAL CLASSIFICATION

(1) Write three characteristics of phylum-Mollusca.

(March '25)

Ans. Characteristics of phylum-Mollusca :

- (1) Body of these animals is soft and slimy.
- (2) These animals are aquatic or terrestrial.
- (3) These animals are unisexual.
- (4) This is second largest phylum in animal kingdom.

(Any three characteristics, each 1 mark)

(2) What is the exact difference between grades of organisation and symmetry? Explain with examples. (5 marks)

Ans. (I) Grades of organisation :

(1) The grades of organisation mean the way an organism has different body formation.

(2) Unicellular organisms like amoeba have a single cell in the body and hence the organisation in its body is called protoplasmic grade of organisation.

(3) Some organisms have only cells in their body which is called cellular grade of organisation. e.g. Poriferans.

(4) Some have tissues. e.g. Coelenterates. They are said to have tissue grade organisation. Some have organs, they are said to have organisation-organ grade. e.g. Platyhelminthes. All other higher animals have organ-system grade organisation.

(II) Symmetry :

(1) Symmetry on the other hand shows the base of the body formation.

(2) The symmetry can be understood by taking an imaginary cut through the animal body.

(3) Based on the symmetry there can be three types.

(4) In asymmetric animals, there is no symmetry in any plane.
e.g. Amoeba.

(5) The bilateral symmetry is the one in which an imaginary axis can pass through only one median plane to divide the body into two equal halves. Most of the animals have bilateral symmetry and hence their organs are arranged in symmetric way on both the sides.

(6) The imaginary cut passing through the central axis but any plane of body can give more than one equal half. The organs of such animals are arranged in a radius of an imaginary circle.
e.g. Cnidarians and some echinoderms.

Both grades of organisation and symmetry are the bases for classifying animals into different phyla.

(3) Name the largest phylum in animal kingdom. Write any two characters of that phylum. (Sept. '21) (3 marks)

Ans. (1) Arthropoda is the largest phylum in animal kingdom.

(2) **Characteristics of Arthropoda :**

(1) Eucoelomate, triploblastic, bilaterally symmetrical and segmented body.

(2) Body covered over by chitinous exoskeleton. Jointed appendages present.

(4) Write any three characteristics of class Reptilia.

(March '22) (3 marks)

Ans. Characteristics of class Reptilia :

(1) Reptilia consists of weak limbed terrestrial animals that crawl on the ground. The limbs cannot support their body weight. Hence the name Reptilia or creeping animals. Few are seen in water such as crocodile or turtle, but they cannot breathe in water.

(2) The body is divided into head, neck and trunk and is covered with dry and scaly skin.

(3) Four limbs which are provided by clawed digits. (Except in snakes.)

(4) External ear is absent. Respiration is by lungs.



(5) Cockroach, frog, sparrow, starfish from these animals which can regenerate its broken body part?

(March '23) (3 marks)

Ans. (1) Out of cockroach, frog, sparrow and starfish only starfish can show regeneration.

(2) Starfish belongs to phylum Echinodermata. Only in this phylum, organism can show capacity to regenerate the broken body part.

(3) Remaining animals cannot show regeneration, so only starfish is able to regenerate.

CHAPTER 7 : INTRODUCTION TO MICROBIOLOGY

(1) What are the constituents of acid rain? What are its two effects on earth's surface? (March '25)

Ans. Constituents of acid rain are sulfur dioxide (SO_2) and nitrogen dioxide (NO_2). Effects of acid rain :

(1) Erosion of metal statues and bridges.

(2) Erosion of historical monuments occurs due to acid rain.

(3) It also causes soil pollution.

(2) Answer the following questions : (March '22) (3 marks)

(a) Which substance is used to produce cheese?

(b) Which enzyme was used earlier for cheese production?

(c) Which enzyme is used to produce vegetarian cheese?

Ans. (a) Mostly cow's milk is used to produce cheese throughout the world.

(b) An enzyme obtained from alimentary canal of cattle, called rennet was traditionally used for making cheese.

(c) Now enzyme obtained from fungi is used to produce vegetarian cheese.

(3) Which fuels are obtained from biomass?

(March '23) (3 marks)

Ans. From biomass, the biogas and biodiesel are mainly obtained. The biogas is obtained from dung of cattle. The fermentation of cattle dung gives rise to methane. From methane, methanol is obtained. Ethanol is obtained from molasses of sugarcane and some other crops. In some countries, special crops are cultivated for the biodiesel.

(1) What is stem cell? Write any four uses of stem cell.

(March '25)

Ans. Stem cells : The special cells having pluripotency and ability to divide and differentiate into new cells are called Stem cells. They are present in multicellular living beings.

Uses of stem cells :

- (1) Stem cells are used for regenerative therapy.
- (2) In case of diseased conditions like diabetes, myocardial infarction, Alzheimer's disease, Parkinson's disease, etc. Stem cells can be used to replace the damaged or functionless, cells.
- (3) In conditions such as anaemia, thalassaemia, leukaemia, etc., there is always the need of newer blood cells. Here, stem cells can be used to restore the number of blood cells.
- (4) In techniques of organ transplantation, stem cells can be used and they can help in the transplantation of new organs such as kidney and liver. The defective organs can be replaced by those that defective organs are produced with the help of stem cells and transplanted.

(2) Answer the following questions :

- (a) What is Biotechnology?
- (b) Give one use of Biotechnology.
- (c) Give one commercial use of Biotechnology.
- (d) Write two bacterial examples of biofertilizer.
- (e) Write two names of crops genetically developed.

(March '20) (5 marks)

Ans. (a) Biotechnology : Technology that brings about artificial genetic changes and hybridisation in organisms for human welfare is called biotechnology.

(b) Biotechnology is very useful in improving the yield and variety of agricultural products.

(c) Commercial use of Biotechnology : Production of hormones, interferons, antibiotics and different vaccines.

(d) *Rhizobium* and *Azotobacter* are used to prepare natural fertilizers.

(e) Genetically modified crops : Golden Rice and Bt. Cotton.



(3) Explain the following terms :

(a) Stem cells

(b) Cloning

(c) Genetically modified crops (**March '24**)

(3 marks)

Ans. For the respective answers, of refer to Q. 1, Q. 5 and Q. 3 from Chapter 8 of Q. Set 8.

(4) Which precautions will you take during spraying of pesticides? (Sept. '21) **(3 marks)**

Ans. (1) Pesticides are toxic chemicals. By using them indiscriminately, they contaminate the water, soil and also crops.

(2) The D.D.T., chloropyriphos and malathion are very dangerous. They spread through the food chain causing biomagnification.

(3) Therefore, we shall not use such insecticides and pesticides. We shall use organic pesticides.

(4) At the time of spraying, nose, eyes and skin will be covered and protected.

(5) Care will be taken not to allow children or domestic animals to come in contact with a pesticide.

(6) Excessive use will not be done.

(5) Answer the following questions :

(3 marks)

(a) What is meant by green revolution?

(b) Who is a father of green revolution in India?

(c) What is AMUL and who was the founder director of AMUL?

Ans. (a) Green revolution is collective method used in agriculture practices where different methods are used to harvest maximum yield from minimum land.

(b) Dr. M.S. Swaminathan is a father of green revolution in India.

(c) AMUL means Anand Milk Union Limited whose founder director was Dr. Verghese Kurien.

CHAPTER 9 : SOCIAL HEALTH

Q. Write any six factors affecting social health.

(Nov. '20) (3 marks)

Ans. (1) In order to maintain the social health of any community there should be good amenities for the people. e.g. Food, water, shelter, clothing, medicines and medical help.

(2) There should be equal opportunities for education, cleanliness of the surroundings, transport facilities, etc. should be properly provided.

(3) The social and political conditions of the surrounding should be such that there should not be any connections with world of criminals. The presence of such criminal ties can affect the social health to a great extent.

(4) The gardens, playgrounds, the empty plots for outdoor games, sports clubs, etc. are important criteria for overall development of the society. This results into personality development and make people happy and strong.

(5) Addictions, criminal tendencies, pervert behaviour and perverse thinking affects other people in the society and this reflects negatively on the social health.

(6) Having large number of friends and relatives, proper use of time when alone and when along the peer group, trust in others, respect and acceptance for others build stronger social health.

CHAPTER 10 : DISASTER MANAGEMENT

(1) What is meant by disaster? Write any two examples of each natural and man-made disaster. (March '22) (3 marks)

Ans. (1) Disaster is the incidence that occurs suddenly causing heavy damage to life and property. The disaster can be man-made or due to natural reason.

(2) Natural disasters : Flood, earthquake, drought, landslide, volcano.

(3) Man-made disasters : War, bomb explosion, gas leakage, terrorism.

(2) Answer the following. (Nov. '20) (5 marks)

(i) What is meant by disaster management?

Ans. Disaster management is either prevention of disasters or making arrangements to face it or at least achieve the abilities to face it.

(ii) Classify the following activities into pre-disaster management.

(a) Identifying the pre-disaster areas.



- (b) Quick establishment of help centre.
- (c) Participation of preferably local peoples saved from the disaster in arranging the help to victims.
- (d) Getting special training for disaster management.

Ans. (a) Pre-disaster management.

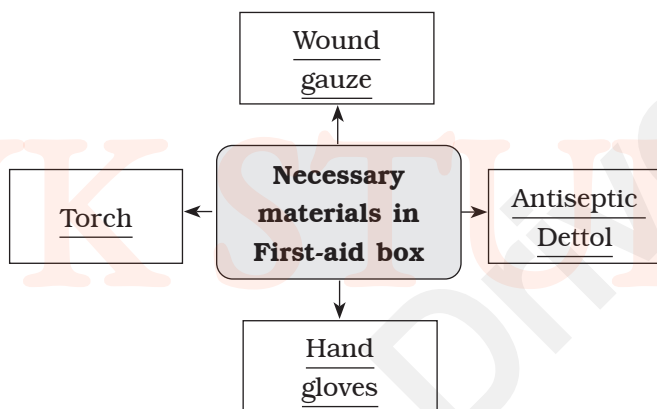
(b) Post-disaster management.

(c) Post-disaster management.

(d) Pre-disaster management.

(iii) Complete the diagram :

Ans. (Answers are directly given and underlined.)



(3) Write names of any six materials which are to be in the first aid box. (March '24) (3 marks)

Ans. (1) For wrapping the wounds : Bandage strips of different size, Wound gauze, Triangular and circular bandage, Medicated cotton, Band aid.

(2) For cleanliness : Hand gloves, Clean and dry cloth pieces, Soap

(3) Medicines : Antiseptic (Dettol/Savlon), Petroleum jelly

(Any six materials to be written in answer.)

Q. 1 (B) : 1 Mark each**Q. State True/False :**

- (1) Geographic and Reproductive isolation does not lead to speciation. **(Ch. 1)**
- (2) Glucose is oxidised step by step in the cells during the process of respiration at the body level. **(Ch. 2)**
- (3) Glycolysis is also called Embden-Meyerhof-Parnas pathway. **(Ch. 2)**
- (4) Out of 2 – 4 million ova, approximately only 400 oocytes are released up to the age of menopause. **(Ch. 3)**
- (5) Energy obtained from nuclear fission is eco-friendly. **(Ch. 5)**
- (6) Mammals breathe with the help of lungs. **(Ch. 6)**
- (7) The Pseudomonas bacteria are useful for cleaning the hydrocarbon and oil pollutants from soil and water. **(Ch. 8)**

Q. 1 (B) : 1 Mark each and Q. 2 (B) : 2 Marks each**Q. Name the following :**

- (1) Fish can breathe through lungs. **(Ch. 1)**
- (2) Three scientists who proved that living organisms have DNA as genetic material. **(Ch. 1)**
- (3) Place where electron transfer chain reaction takes place. **(Ch. 2)**
- (4) Agents of pollination. **(Ch. 3)**
- (5) Components of semen. **(Ch. 3)**
- (6) The modern techniques in reproduction. **(Ch. 3)**

Note : Students can find the answers to all the questions of Section 3 in **Navneet's Science and Technology (Part 2) Digest : Std. X.**

- (7) Any two environmental laws. (Ch. 4)
- (8) Forms of energy. (Ch. 5)
- (9) Antibiotics. (Ch. 7)
- (10) Microbe used in preparation of wine and cider. (Ch. 7)
- (11) Microbes used for production of beverages. (Ch. 7)
- (12) Two important aspects of human health management. (Ch. 8)
- (13) Mental/Psychological disorders. (Ch. 9)
- (14) Name the disasters from the statements given below : (Ch. 10)
- (i) On 26th July 2005, entire suburban Mumbai was waterlogged.
 - (ii) Elephants in the Bandipur forest started running helter and skelter due to smoke.
 - (iii) Many innocent people died in the bomb blast that occurred on 11th July 2006 in local trains.
 - (iv) In Kutch, suddenly many school children were buried under the rubble.
 - (v) Because of lack of grains, people from Vidarbha are migrating to other regions.
 - (vi) The huge waves in Chennai engulfed many human lives in December 2004.
-

9. Write functions :

- (1) D.N.A. (Ch. 1)
- (2) Phospholipids (Ch. 2)
- (3) Sporangium (Ch. 3)
- (4) Calyx (Ch. 3)
- (5) Androecium (Ch. 3)
- (6) Gynoecium (Ch. 3)
- (7) Corpus luteum (Ch. 3)
- (8) Vagina (Ch. 3)
- (9) Turbine (Ch. 5)

Q. 2 (A) : 2 Marks each

Q. Give scientific reasons :

- (1) Vertebrates have been slowly originated from invertebrate.
(Ch. 1)
- (2) During human evolution, the hands became available for use.
(Ch. 1)
- (3) Peripatus is said to be a connecting link between Annelida and Arthropoda. (Ch. 1)
- (4) Water is an essential nutrient. (Ch. 2) (July '22)
- (5) Many times we experience dryness in mouth. (Ch. 2)
- (6) By the age of 45–50, women get menopause. (Ch. 3)
- (7) Older mothers have greater chance of conceiving abnormal children. (Ch. 3)
- (8) Destroying trees is to destroy everything. (Ch. 4)
- (9) Human beings have important place in environment. (Ch. 4)
- (10) Energy in the coal is called chemical energy. (Ch. 5)
- (11) Wind energy is called eco-friendly. (Ch. 5)
- (12) All vertebrates are chordates but all chordates are not vertebrates. (Ch. 6)
- (13) Use of mutant strains has been increased in industrial microbiology. (Ch. 7)
- (14) Awareness about organ donation after death is increasing.
(Ch. 8)
- (15) We need to keep the PIN number of the debit card secret.
(Ch. 9)
- (16) Effective disaster management makes us well prepared for future. (Ch. 10)



Q. 2 (B) : 2 Marks each

Q. Write down the following theories :

- (1) Theory of evolution. (Ch. 1)
- (2) Theory of speciation. (Ch. 1)

Q. Give definitions :

- (1) Translation. (Ch. 1)
- (2) Aerobic respiration. (Ch. 2)
- (3) Embryo sac. (Ch. 3)
- (4) Corpus luteum. (Ch. 3)
- (5) Food web. (Ch. 4)
- (6) Electromagnetic induction. (Ch. 5)
- (7) Nuclear fission. (Ch. 5)
- (8) Triploblastic. (Ch. 6)
- (9) Bioremediation. (Ch. 7)
- (10) DNA fingerprinting. (Ch. 8)

Q. 2 (B) or 3 : 2 or 3 Marks each

Q. Distinguish between :

- (1) Transcription and Translation. (Ch. 1)
- (2) Mitosis and Meiosis. (Ch. 2) (Sept. '21)
- (3) Human male and female reproductive systems. (Ch. 3)
- (4) Asexual and Sexual reproduction. (Ch. 3)
(March '22; July '24)
- (5) Aves and Mammalia. (Ch. 6) (March '20)
- (6) Platyhelminthes and Aschelminthes. (Ch. 6)
- (7) Amphibia and Reptilia. (Ch. 6)
- (8) Green revolution and White revolution. (Ch. 8)

Q. 2 (B) or 3 : 2 or 3 Marks each

Q. Write short notes :

- (1) Vestigial organs. (Ch. 1)
- (2) Carbon dating. (Ch. 1)
- (3) Anaerobic respiration. (Ch. 2)
- (4) Importance of water and fibre in diet. (Ch. 2)
- (5) Multiple fission. (Ch. 3)
- (6) IVF. (Ch. 3)
- (7) Radioactive pollution. (Ch. 4)
- (8) Biodiversity. (Ch. 4)
- (9) Chipko movement. (Ch. 4)
- (10) Lake tapping. (Ch. 5)
- (11) Benefits of classification. (Ch. 6) (March '19)
- (12) Landfilling sites. (Ch. 7)
- (13) Uses of stem cells. (Ch. 8)
- (14) Addiction. (Ch. 9)

Q. 3 : 3 Marks each

Q. Explain the given statements :

- (1) Geographical and reproductive isolation of organisms gradually leads to speciation. (Ch. 1)
- (2) "Solving the problem of pollution is an effective way of environmental management." Justify the statement. (Ch. 4)
- (3) Explain the step by step energy conversion that occurs in
(i) Thermal power plant (ii) Hydroelectric project (iii) Solar thermal power plant. (Ch. 5)

Q. 3 : 3 Marks each

Q. Answer the following questions :

- (1) Despite various diagnostic tests, a couple could not have a child.
In this situation, which remedies will you suggest?

(Ch. 3) (July '19)

- (2) Which precautions will you follow to maintain the reproductive health? **(Ch. 3)**

- (3) What will you do? **(Ch. 9)**

(a) You are spending more time in internet/mobile games, phone, etc. **(March '22; July '22)**

(b) Your friend's younger brother has started smoking.

(Sept. '21)

Q. 3 : 3 Marks each

Q. Complete the paragraphs using proper words :

- (1) (*translation, anticodon, t-RNA, m-RNA, amino acids, triplet codon, transcription, DNA*) **(Ch. 1)**

The formed in nucleus comes in cytoplasm. It brings in the coded message from DNA. The message contains the codes for amino acids. The code for each amino acid consists of three nucleotides. It is called a '.....'. Each m-RNA is made up of thousands of triplet codons. As per the message on m-RNA, are supplied by the For this purpose, t-RNA has '.....' having complementary sequence to the codon on m-RNA. This is called '.....'.

- (2) (*adipose, axons, progesterone, fatty, phospholipids, plasma*)

(Ch. 2)

Molecules called as are essential for producing membrane are formed from acids. They are also used for production of hormones like estrogen, testosterone, aldosterone, etc. Covering around of nerve cells is also produced from them. Excess of lipids are stored in connective tissue in the body.

(3) (*species, biodiversity, genetic, ecosystems, sacred, society*)

(Ch. 4)

..... is the richness of living organisms in nature due to presence of varieties of organisms, and variations within a species. Possibility of wiping out arise when there is decrease in diversity. groove is the area of rich biodiversity which is conserved by

(4) (*Nocardia, Geobacter, Ideonella sakaiensis, Alcanovorax borkumensis, hydrocarbonoclastic, Acidophillum, streptomyces*)

(Ch. 7)

Bacteria like spp. and *Pseudomonas* have the ability to destroy the pyridines and other chemicals. Hence, these bacteria are used to clear the oil spills. These are called bacteria. It has been observed that species like *Vibrio*, can decompose the PET. Similarly, species of fungi like have ability of decomposing rubber from garbage. Sulphuric acid is source of energy for some species of bacteria like Hence, these bacteria can control the soil pollution occurring due to acid rain. convert the salts of uranium into insoluble salts.

(5) (*genome, bacteria, virus, insulin, biotechnology, vaccines, pancreas*) (Ch. 8)

The hormone is used for treatment of diabetes. Earlier, it was collected from the of horses. However, nowadays, due to insulin can be prepared with the help of Human insulin gene has been inserted into of bacteria. Various are also produced in the same way.



Q. 4 : 2/3/5 Marks each

Q. Read the given paragraph or information and answer the questions below :

Our atmosphere is getting hotter, more turbulent and more unpredictable because of the 'boiling and churning' effect caused by the heat-trapping greenhouse gases within the upper layers of atmosphere. With increase of carbon, methane, or other greenhouse gas levels in the atmosphere, our local weather and global climate is further agitated, heated and 'boiled'. The greenhouse gases such as water vapour, carbon dioxide, methane, nitrous oxide and ozone are causing great impact on life on earth. Due to global warming, the glaciers are melting, causing sea level rise, coastal submergence and frequent natural disasters. We have to rethink while taking the development projects and help to protect the existing life forms on the earth before its too late because climate change can exterminate us.

Questions :

- (1) Why is there increased global warming?
- (2) Which are main greenhouse gases that affect climate?
- (3) What can we do to reverse effects of climate change?

Q. 2 [B] : 2 Marks each

- (1) In which way is science of heredity useful these days? **(Ch. 1)**
- (2) Explain with suitable examples importance of anatomical evidences in evolution **(Ch. 1)**
- (3) What is cellular respiration? What are its two types? **(Ch. 2)**
- (4) How are proteins obtained? What are the components of proteins? **(Ch. 2)**
- (5) Explain asexual reproduction in plants. **(Ch. 3)**
- (6) What are the problems associated with power generation by thermal power plant? **(Ch. 5)**

- (7) What is meant by green energy? Which energy sources can be called green energy sources and why? Give examples. **(Ch. 5)**
- (8) Write limitations in an electric generator using wind energy.
(Ch. 5) (March '22)
- (9) Write the characteristics of chordates. **(Ch. 6)**

Q. 3/Q. 4 : 3/5 Marks each

- (1) How are the hereditary changes responsible for evolution?
(Ch. 1)
- (2) Where and in which forms are the amino acids formed after digestion of food used in the body? **(Ch. 2)**
- (3) How energy is formed from oxidation of carbohydrates, fats and proteins? **(Ch. 2)**
- (4) How do all the life processes contribute to the growth and development of the body? **(Ch. 2)**
- (5) Explain sexual reproduction in plants. **(Ch. 3)**
- (6) How can we get the required amount of energy by connecting solar panels? **(Ch. 5)**
- (7) Which fuels can be obtained by Microbial processes? Why is it necessary to increase the use of such fuels? **(Ch. 7)**
- (8) What is DNA fingerprinting? Explain it in brief, where is this technique used? Give any two examples. **(Ch. 8)**
- (9) Which changes occur in persons continuously using the internet and mobile phones? **(Ch. 9)**
- (10) Explain the role of district disaster control unit after occurrence of any disasters. **(Ch. 10)**
- (11) Which are the three aspects of disaster that are important for common citizens? **(Ch. 10)**
-



Time : 2.00 Hours] [Answers on QR Code] [Max. Marks : 40**Note :** (i) All questions are compulsory.

(ii) Use of calculator is not allowed.

(iii) The numbers to the right of the questions indicate full marks.

(iv) In case of MCQs [Q. No. 1 (A)], only the first attempt will be evaluated and will be given credit.

(v) For every MCQ, the correct alternative (A), (B), (C) or (D) with sub-question number is to be written as an answer.

For e.g. (i) (A) (ii) (B) (iii) (C)

(vi) Scientifically correct, labelled diagrams should be drawn wherever necessary.

Q. 1 A Choose the correct alternatives : 5

(i) Our muscle cells perform type of respiration during exercise.

(A) Aerobic

(B) Anaerobic

(C) Fermentation

(D) Krebs Cycle

(ii) Which of the following animals can regenerate its broken body part?

(A) Cockroach

(B) Frog

(C) Sparrow

(D) Starfish

(iii) Government of India has encouraged the for improving the productivity by launching NKM-16.

(A) Apiculture

(B) Cultivation of medicinal plants

(C) Pisciculture

(D) Fruit processing

(iv) At the district level is responsible for disaster management and implementation of rehabilitation schemes.

- (A) Collector (B) Tahsildar
(C) Education Officer (D) Chief Executive Officer

(v) of West Bengal is reserved for tigers.

- (A) Manas National Park
(B) Western Ghat
(C) Molai Jungle
(D) Sundarban National Park

9. 1 B Answer the following questions :

5

(i) State whether the following statement is True or False :

Evolution is the gradual change occurring in living organisms over a long duration.

(ii) Write the name of any *one* sexual disease.

(iii) Select odd man out :

Gutkha, Yoga, Alcoholic drinks, Cigarette.

(iv) Fill in the box with proper word :



(v) Match the correct pair :

Column 'A'	Column 'B'
Xylitol	(a) To impart acidity (b) To impart sweetness (c) Pigment.

9. 2 A Give scientific reasons : (Any two)

4

(i) Some of the organs in human body are most valuable.

(ii) Hydroelectric energy, solar energy and wind energy are called renewable energies.

(iii) Fibres are one of the important nutrients.



9. 2 B Answer the following questions : (Any three)

6

(i) Observe the given animals and answer the questions :



Fig. A



Fig. B

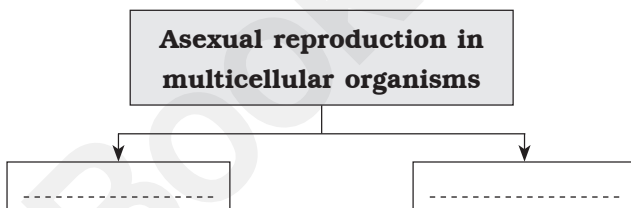
(a) Identify the animals shown in above figure.

(b) State the type of evidence of evolution related to above animals.

(ii) Complete the table :

Sr. No.	Microbe used	Name of beverage
1.	<i>Lactobacillus brevis</i>	Coffee
2.	<i>Candida, Hansenula, Pichia, Saccharomyces</i>	-----
3.	<i>Saccharomyces cerevisiae</i>	Wine
4.	<i>Saccharomyces cerevisiae</i>	-----

(iii) Fill in the box with proper word :



(iv) Give *two* examples of each :

(a) Physical problems arising due to excessive use of mobile phones.

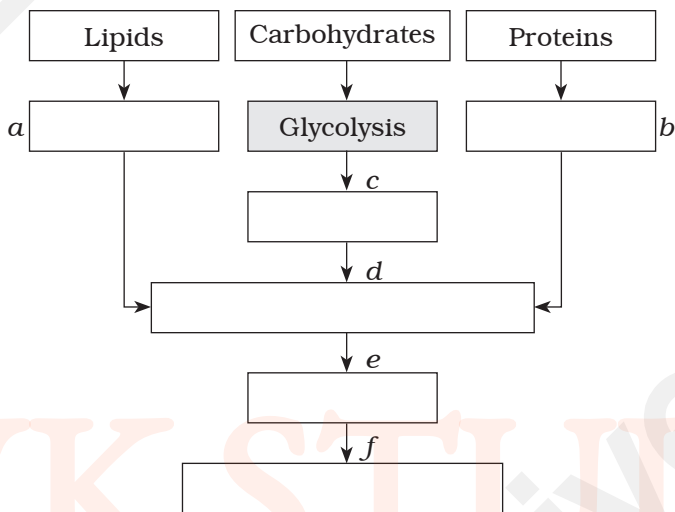
(b) Activities under the jurisdiction of cyber crime laws.

(v) Write any *two* objectives of disaster management.

Q. 3 Answer the following questions : (Any five)

15

- (i) Distinguish between Sexual reproduction and Asexual reproduction.
- (ii) Complete the chart :

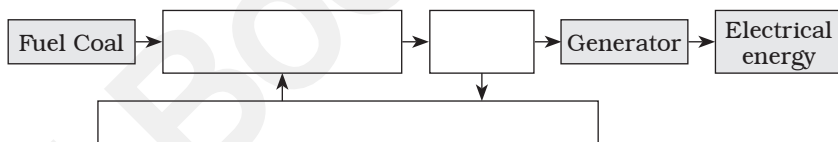


- (iii) Answer the following questions :

- (a) Define vestigial organs.
- (b) Write names of any *two* vestigial organs in human body.
- (c) Write the names of any *two* vestigial organs which are non-functional in human beings but functional in other organisms.

- (iv) Complete the following flow chart :

- (a) Generation of electrical energy using thermal energy :



- (b) Energy transformation in thermal power plant :



- (v) Complete the paragraph with the help of words given in the bracket :

(Lactobacillus cremoris, protease, cow, vegetarian, microbiological, rennet, Geobacter, Saccharomyces)

Cheese is produced on large scale from the abundantly available milk all over the world. First chemical and tests of milk are performed. Some colours and microbes like Lactobacillus lactis, and Streptococcus thermophilus are mixed with milk. It imparts sourness to the milk. After this, to impart the dense texture, whey needs to be removed.

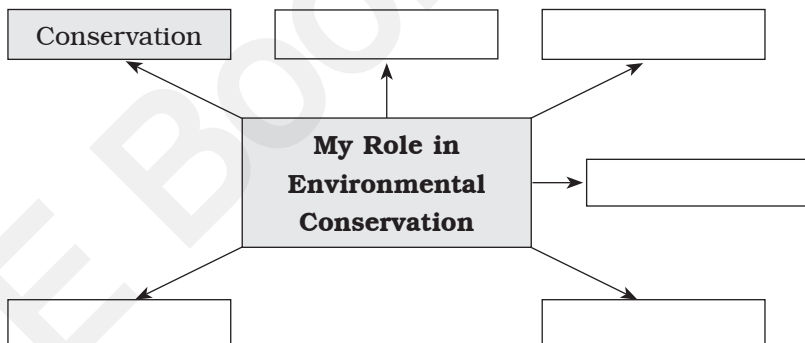
An enzyme obtained from alimentary canal of cattle was being traditionally used earlier. However an enzyme obtained from fungi is used at present to produce cheese.

- (vi) Write a short note on the importance of medicinal plants.
 (vii) Explain the importance of good communication with others.
 (viii) Identify the type of disaster and write its effect :
 (a) Terrorism (b) Forest fire (c) Theft.

Q. 4 Answer the following questions : (Any one)

5

- (i) Complete the following flow chart explaining my role in environmental conservation :



- (ii) Observe the following picture. Write the answers of the following questions :



Fig. A



Fig. B

- (a) Identify the class of animals shown in above picture.
- (b) Which similarities are shown in above animals from the adaptation point of view?
- (c) Why these animals are included in different classes inspite of their similarities?
- _____

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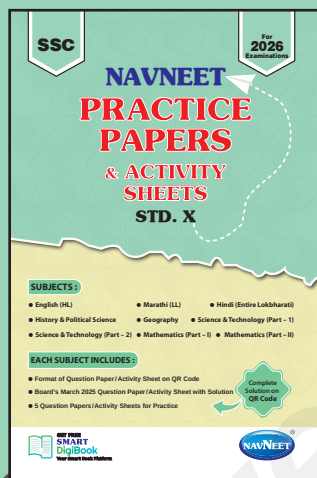


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